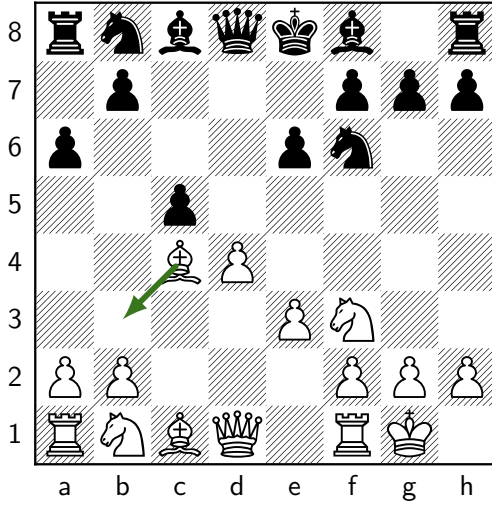


Melanie-ANALOGY — Queen’s Gambit Accepted

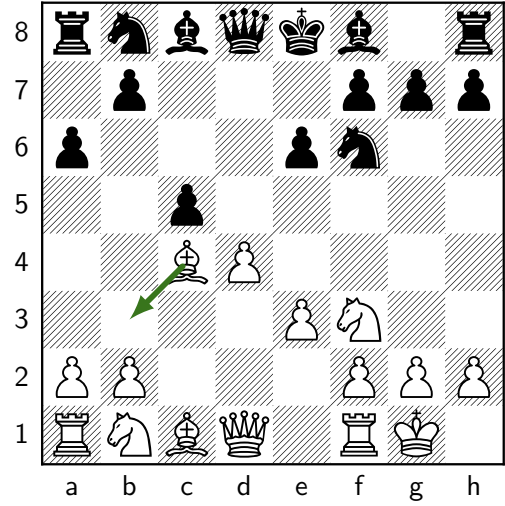
Melanie (Deep_Vision)

Abstract

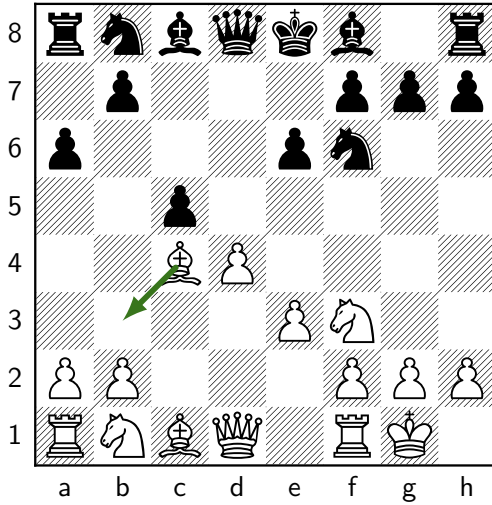
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



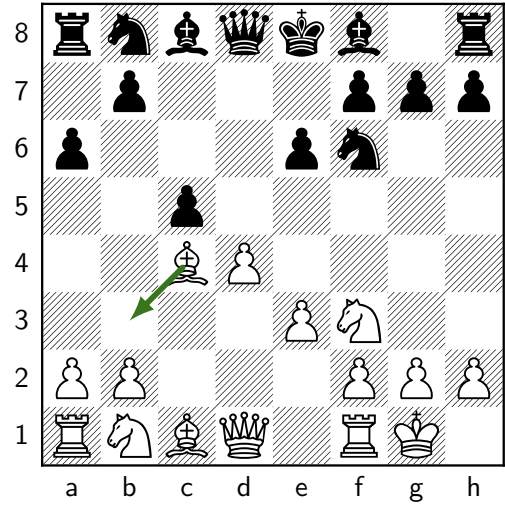
(a)



(b)

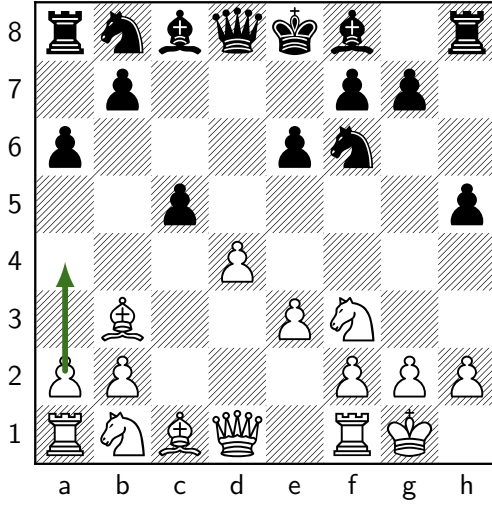


(c)

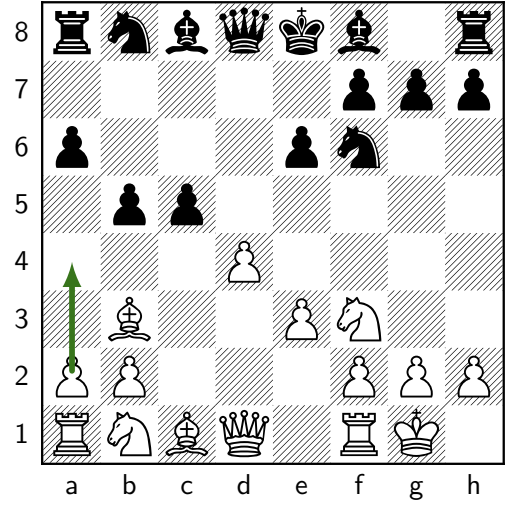


(d)

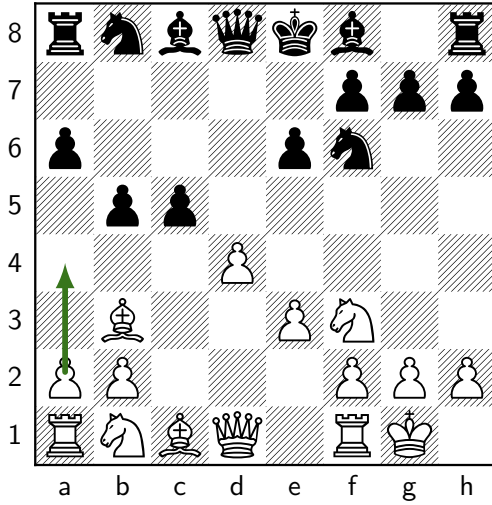
Figure 1: **Move 7, White: Bb3.** (a) Our position; Melanie plays Bb3 (c4b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1112` (ply 12) [structural distance $d=0$, QAP obj=166]: the master played Bb3; under μ our Bc4 plays the role of that game’s Bc4; transferred move c4b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1113` (ply 12) [structural distance $d=0$, QAP obj=166]: the master played Bb3; under μ our Bc4 plays the role of that game’s Bc4; transferred move c4b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1717` (ply 12) [structural distance $d=0$, QAP obj=166]: the master played Bb3; under μ our Bc4 plays the role of that game’s Bc4; transferred move c4b3 (**endorses this move**). Synthesis: 25 retrieved analogs transferred to Bb3 (summed QAP objective 4151, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



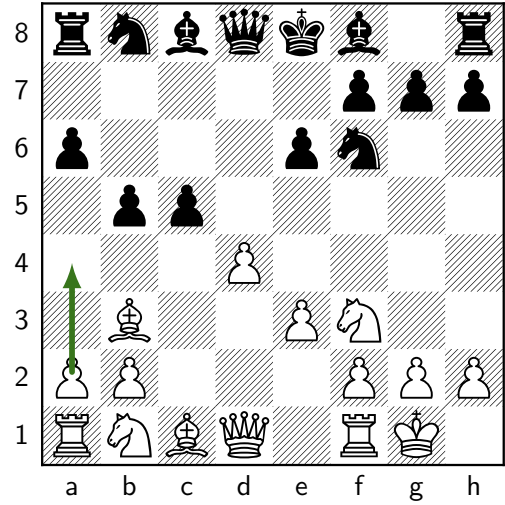
(a)



(b)

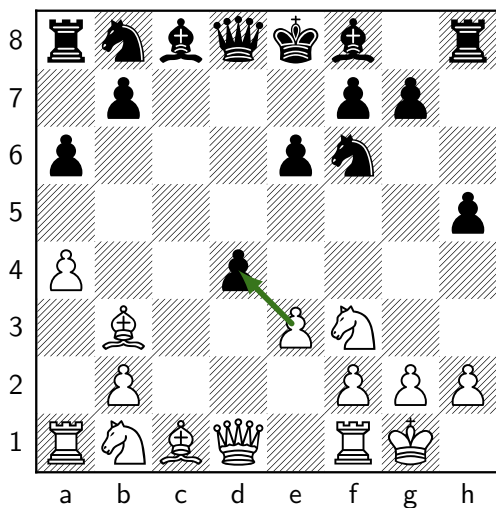


(c)

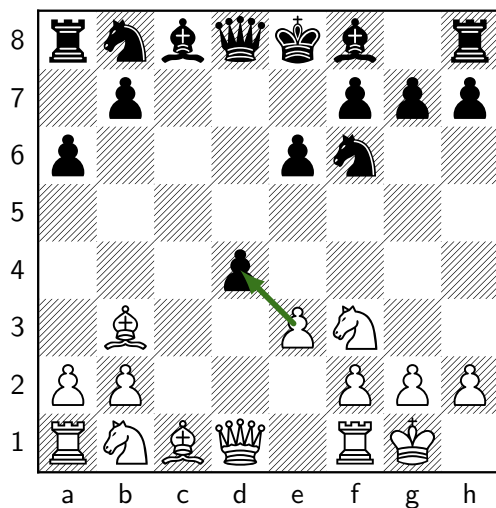


(d)

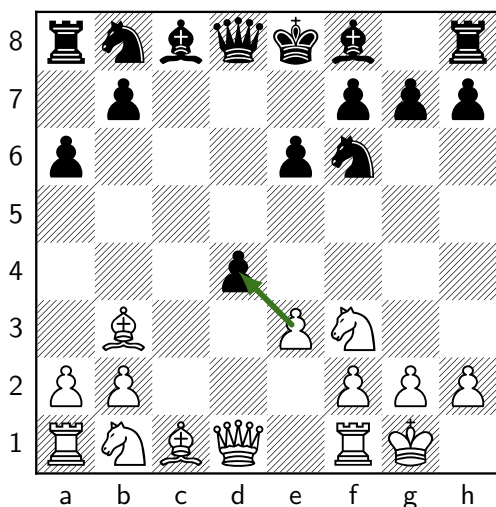
Figure 2: **Move 8, White: a4.** (a) Our position; Melanie plays **a4** (a2a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1717` (ply 14) [structural distance $d=1$, QAP obj=140]: the master played **a4**; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *Bc1* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move **a2a4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1718` (ply 14) [structural distance $d=1$, QAP obj=140]: the master played **a4**; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *Bc1* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move **a2a4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2816` (ply 14) [structural distance $d=1$, QAP obj=140]: the master played **a4**; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *Bc1* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move **a2a4** (**endorses this move**). Synthesis: 20 retrieved analogs transferred to **a4** (summed QAP objective 2802, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



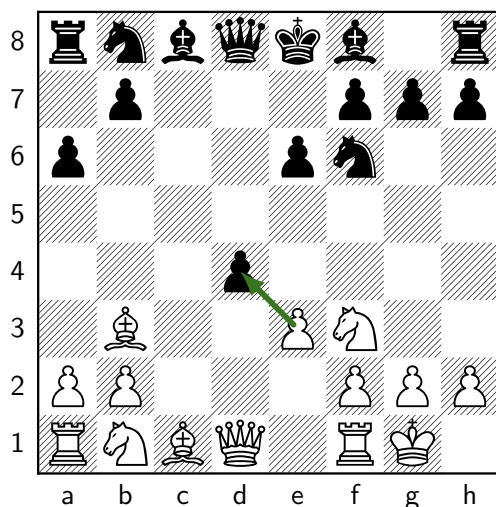
(a)



(b)

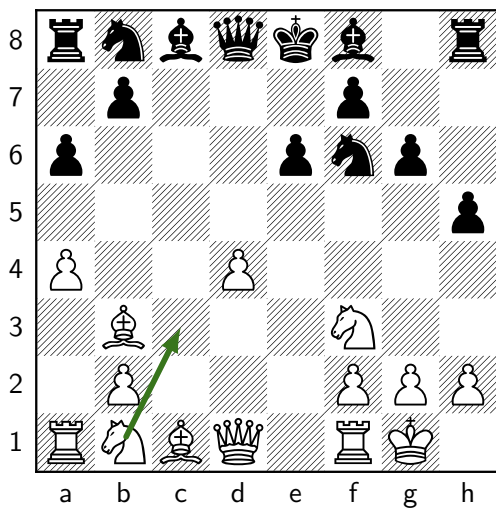


(c)

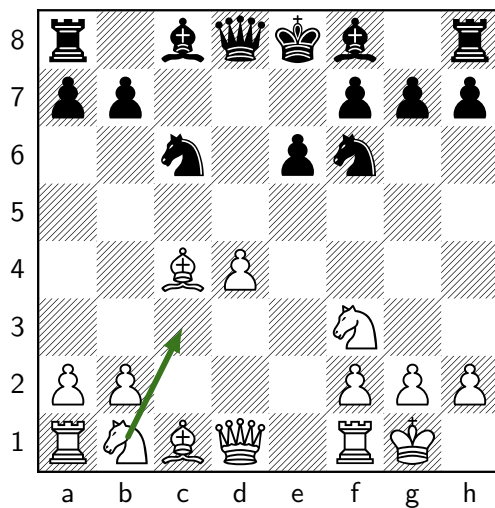


(d)

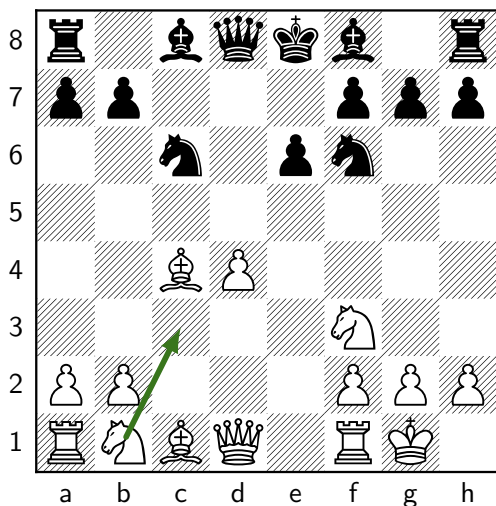
Figure 3: **Move 9, White: exd4.** (a) Our position; Melanie plays exd4 (e3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3657 (ply 14) [structural distance $d=5$, QAP obj=197]: the master played exd4; under μ our *Pe3* plays the role of that game’s *Pe3*; transferred move e3d4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3658 (ply 14) [structural distance $d=5$, QAP obj=197]: the master played exd4; under μ our *Pe3* plays the role of that game’s *Pe3*; transferred move e3d4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3429 (ply 14) [structural distance $d=5$, QAP obj=197]: the master played exd4; under μ our *Pe3* plays the role of that game’s *Pe3*; transferred move e3d4 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to exd4 (summed QAP objective 2929, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



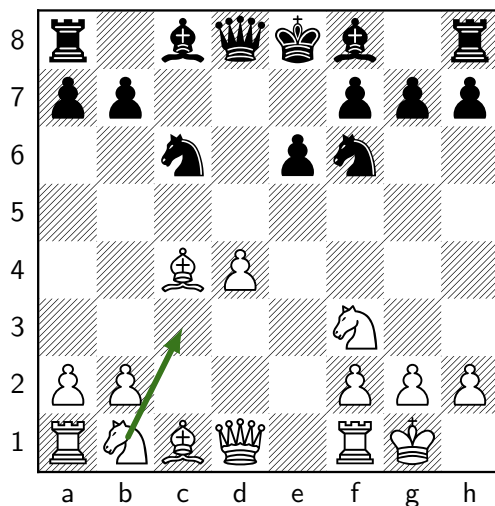
(a)



(b)

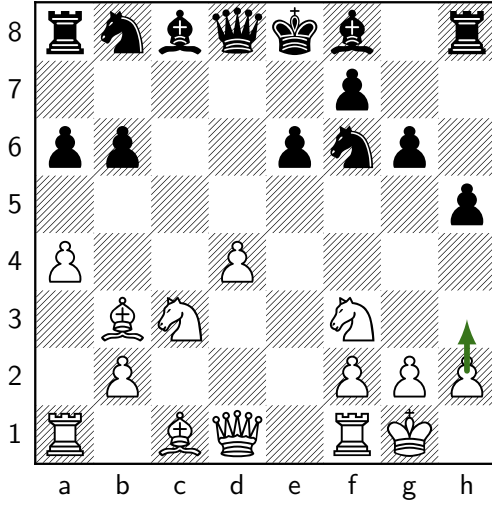


(c)

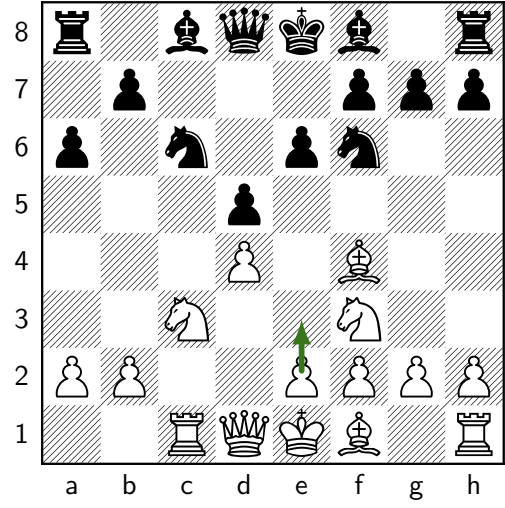


(d)

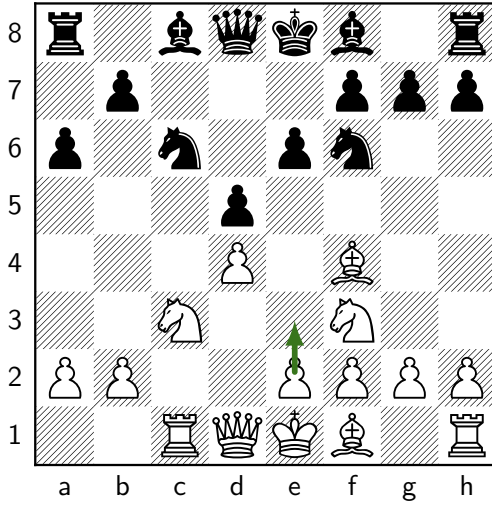
Figure 4: **Move 10, White: Nc3**. (a) Our position; Melanie plays Nc3 (b1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#297` (ply 16) [structural distance $d=38$, QAP obj=79]: the master played Nc3; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *pa6* as their *Ra1*, our *qd8* as their *nc6*; transferred move b1c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#298` (ply 16) [structural distance $d=38$, QAP obj=79]: the master played Nc3; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *pa6* as their *Ra1*, our *qd8* as their *nc6*; transferred move b1c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2920` (ply 14) [structural distance $d=38$, QAP obj=79]: the master played Nc3; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *pa6* as their *Ra1*, our *qd8* as their *nc6*; transferred move b1c3 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Nc3 (summed QAP objective 600, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



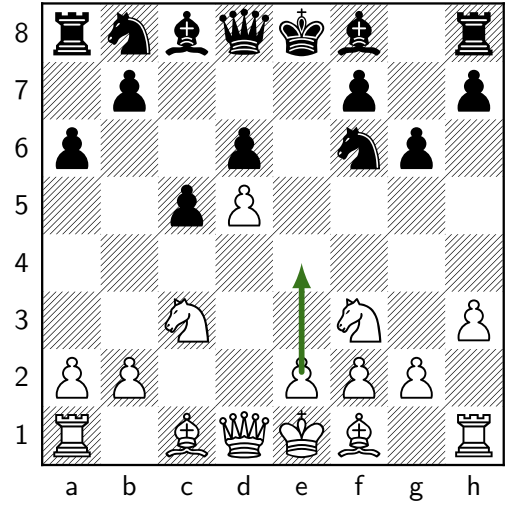
(a)



(b)

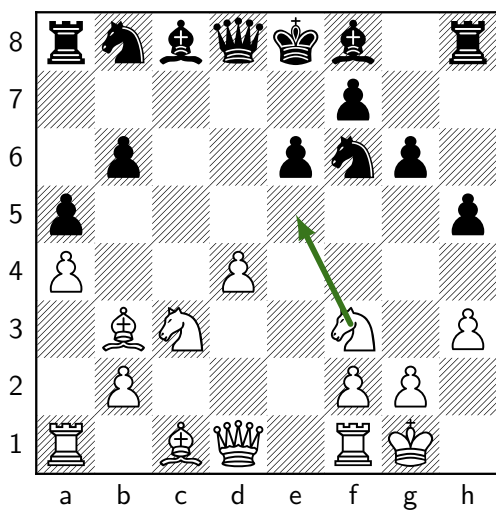


(c)

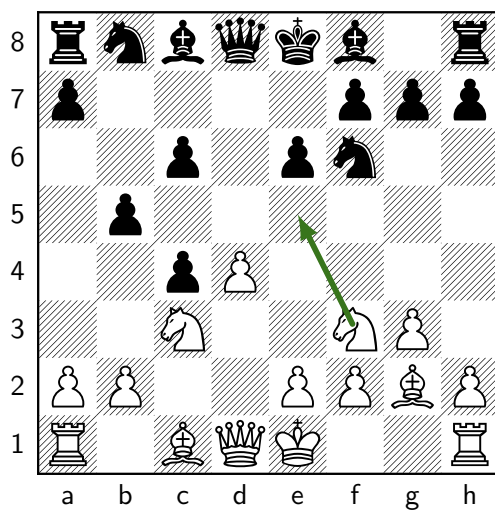


(d)

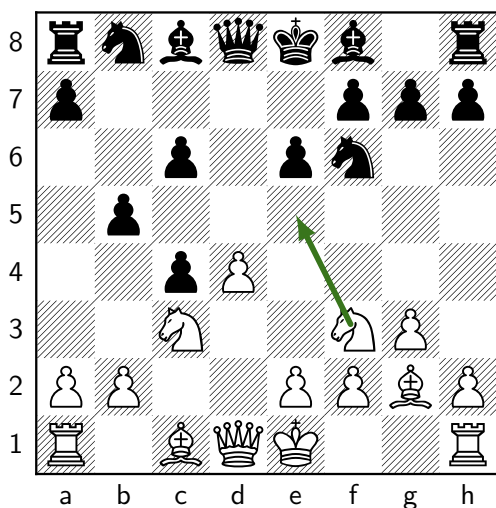
Figure 5: **Move 11, White: h3**. (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3973` (ply 14) [structural distance $d=50$, QAP obj=30]: the master played e3; under μ our $Ph2$ plays the role of that game’s $Pe2$; cross-type roles: our $Qd1$ as their $Rc1$, our $Rf1$ as their $Qd1$; transferred move h2h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3974` (ply 14) [structural distance $d=50$, QAP obj=30]: the master played e3; under μ our $Ph2$ plays the role of that game’s $Pe2$; cross-type roles: our $Qd1$ as their $Rc1$, our $Rf1$ as their $Qd1$; transferred move h2h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2686` (ply 14) [structural distance $d=47$, QAP obj=25]: the master played e4; under μ our $Ph2$ plays the role of that game’s $Pe2$; cross-type roles: our $pg6$ as their $Bc1$, our $Nf3$ as their $Qd1$; transferred move h2h3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to h3 (summed QAP objective 160, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



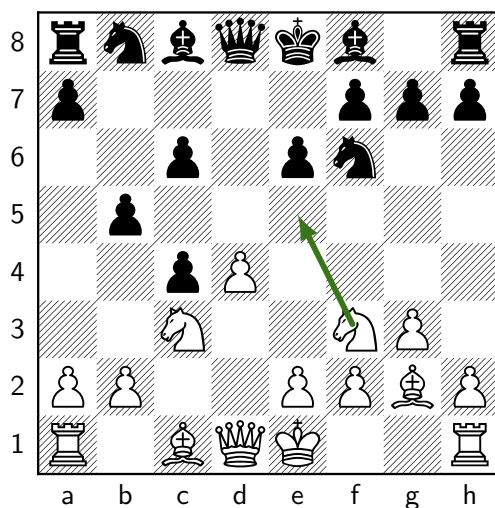
(a)



(b)

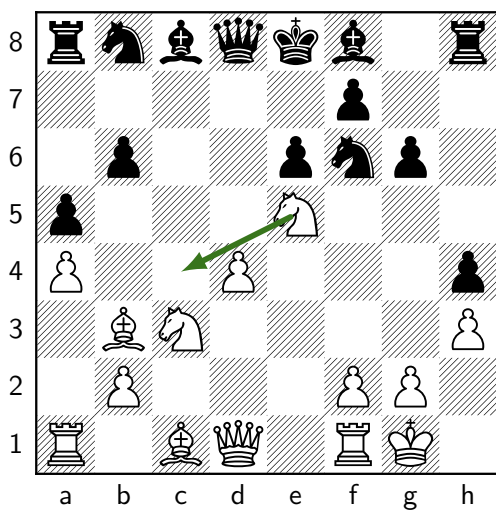


(c)

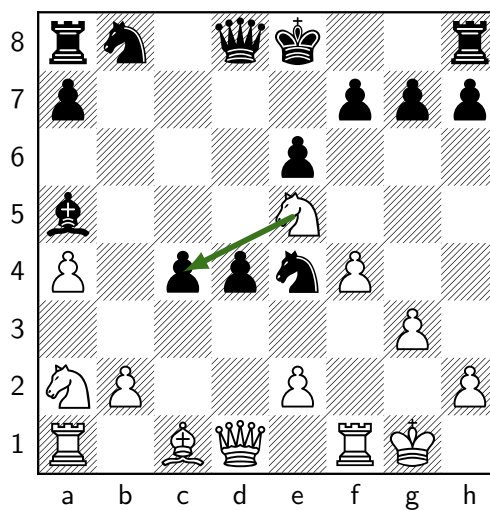


(d)

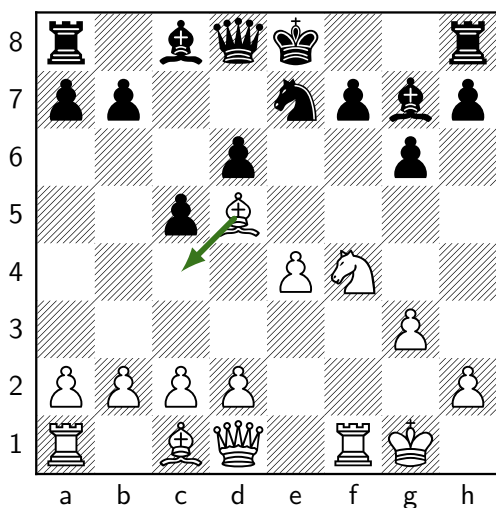
Figure 6: **Move 12, White: Ne5.** (a) Our position; Melanie plays Ne5 (f3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#428 (ply 12) [structural distance $d=68$, QAP obj=58]: the master played Ne5; under μ our Nf3 plays the role of that game's Nf3; cross-type roles: our Pa4 as their Bc1, our Rf1 as their Qd1; transferred move f3e5 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#429 (ply 12) [structural distance $d=68$, QAP obj=58]: the master played Ne5; under μ our Nf3 plays the role of that game's Nf3; cross-type roles: our Pa4 as their Bc1, our Rf1 as their Qd1; transferred move f3e5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#472 (ply 12) [structural distance $d=68$, QAP obj=58]: the master played Ne5; under μ our Nf3 plays the role of that game's Nf3; cross-type roles: our Pa4 as their Bc1, our Rf1 as their Qd1; transferred move f3e5 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Ne5 (summed QAP objective 819, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



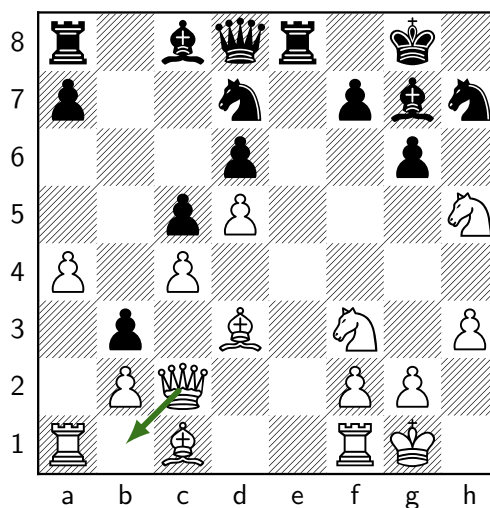
(a)



(b)

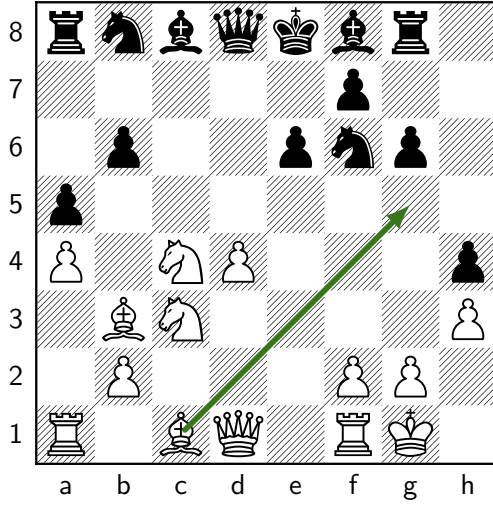


(c)

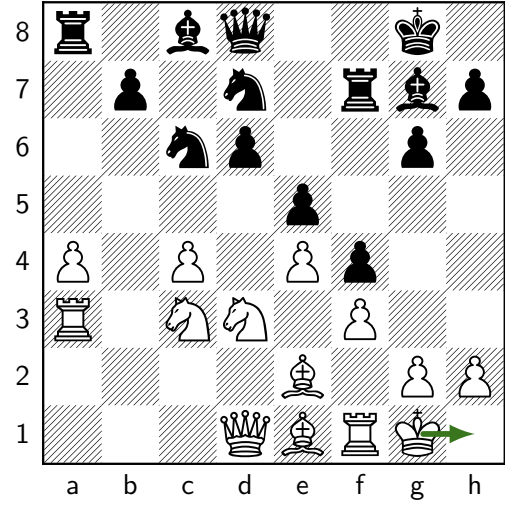


(d)

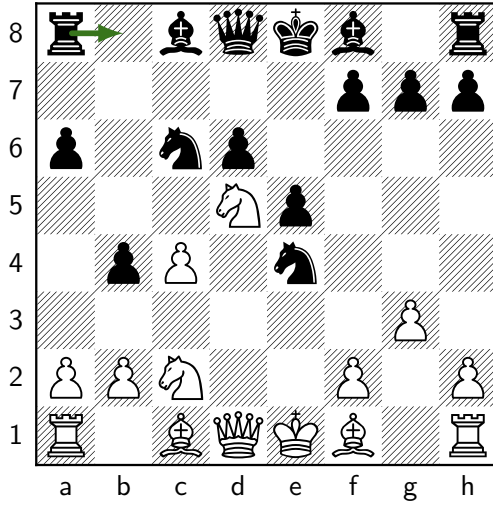
Figure 7: **Move 13, White: Nc4.** (a) Our position; Melanie plays Nc4 (e5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1968` (ply 28) [structural distance $d=80$, QAP obj=91]: the master played Nxc4; under μ our Ne5 plays the role of that game’s Ne5; cross-type roles: our Bc1 as their Ra1, our pb6 as their Bc1; transferred move e5c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2546` (ply 20) [structural distance $d=78$, QAP obj=27]: the master played Bc4; under μ our Ne5 plays the role of that game’s Bd5 (a **cross-type** correspondence); cross-type roles: our Nc3 as their Ra1, our Pg2 as their Bc1; transferred move e5c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3584` (ply 30) [structural distance $d=77$, QAP obj=26]: the master played Qb1; under μ our Ne5 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our nf6 as their Ra1, our qd8 as their Bd3; transferred move e5c4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nc4 (summed QAP objective 144, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



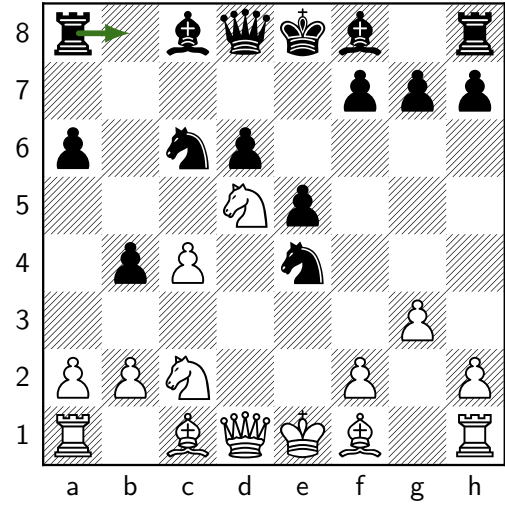
(a)



(b)

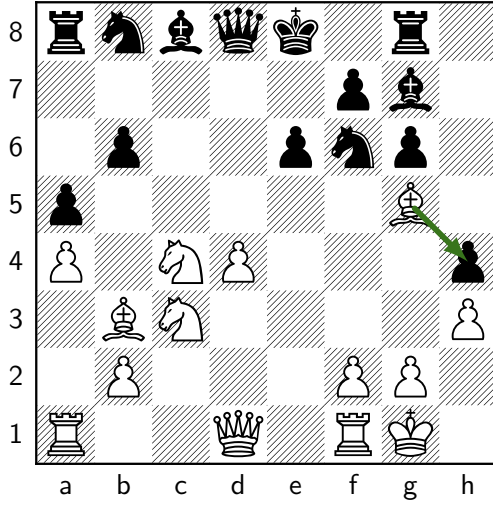


(c)

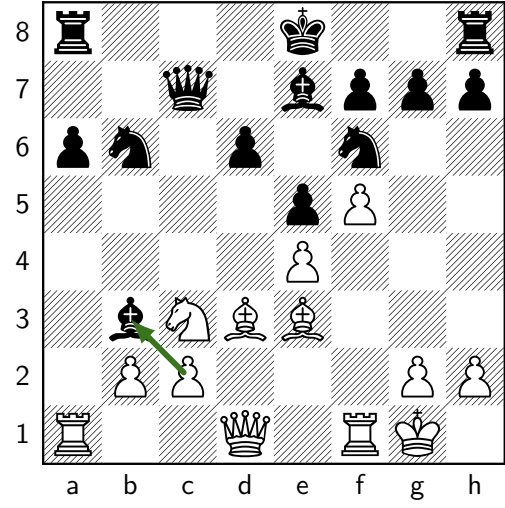


(d)

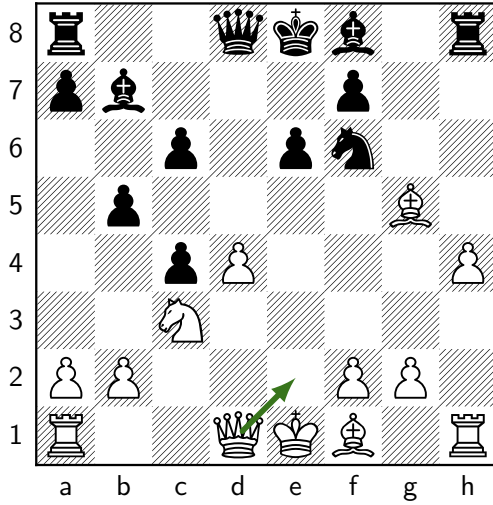
Figure 8: **Move 14, White: Bg5.** (a) Our position; Melanie plays Bg5 (c1g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1556` (ply 38) [structural distance $d=68$, QAP obj=93]: the master played Kh1; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our ph4 as their Qd1, our Pg2 as their Be1; transferred move g1h1. (c) Retrieved from `ccrl_top_p0000#2771` (ply 21) [structural distance $d=65$, QAP obj=85]: the master played Rb8; under μ our Nc3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Bc8, our Rf1 as their Qd8; transferred move c3e2. (d) Retrieved from `ccrl_top_p0003#1107` (ply 21) [structural distance $d=65$, QAP obj=85]: the master played Rb8; under μ our Nc3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Bc8, our Rf1 as their Qd8; transferred move c3e2. Synthesis: no retrieved analog transferred to Bg5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



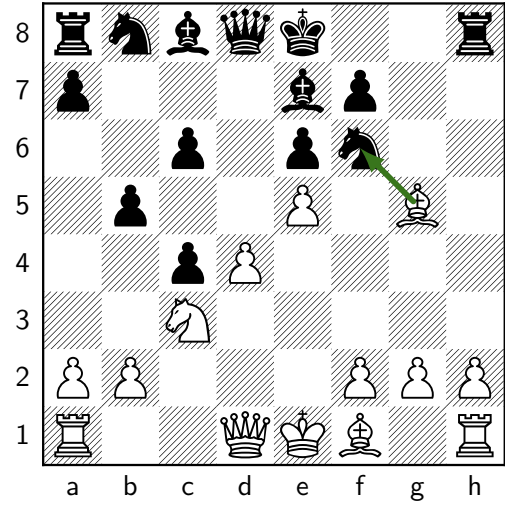
(a)



(b)

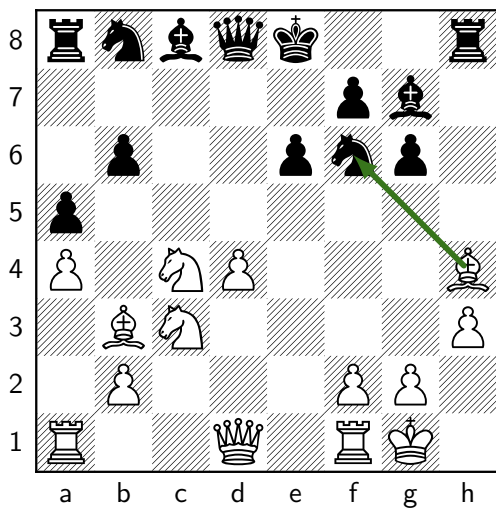


(c)

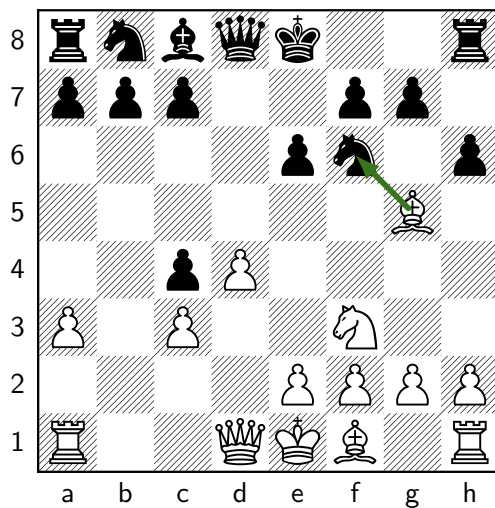


(d)

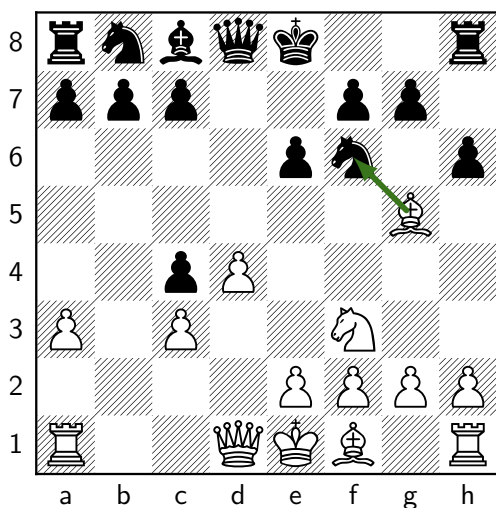
Figure 9: **Move 15, White: Bxh4**. (a) Our position; Melanie plays Bxh4 (g5h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#39` (ply 28) [structural distance $d=81$, QAP obj=29]: the master played `cxb3`; under μ our `Bg5` plays the role of that game’s `Pc2` (a **cross-type** correspondence); cross-type roles: our `Nc3` as their `Ra1`, our `nf6` as their `bb3`; transferred move `g5h4` (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#261` (ply 24) [structural distance $d=84$, QAP obj=65]: the master played `Qe2`; under μ our `Rf1` plays the role of that game’s `Qd1` (a **cross-type** correspondence); cross-type roles: our `bc8` as their `Ra1`, our `Pf2` as their `Bf1`; transferred move `f1e1`. (d) Retrieved from `ccr1_top_p0000#568` (ply 20) [structural distance $d=82$, QAP obj=53]: the master played `Bxf6`; under μ our `Bg5` plays the role of that game’s `Bg5`; cross-type roles: our `pb6` as their `Ra1`, our `Rf1` as their `Qd1`; transferred move `g5f6`. Synthesis: 1 retrieved analog transferred to Bxh4 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



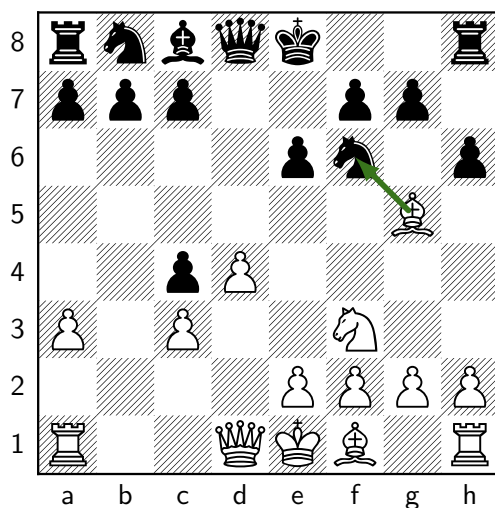
(a)



(b)

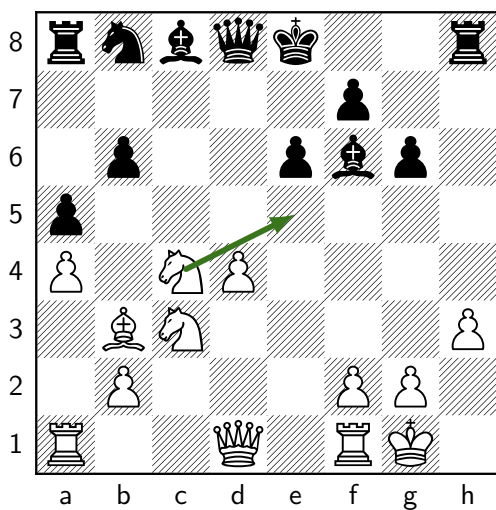


(c)

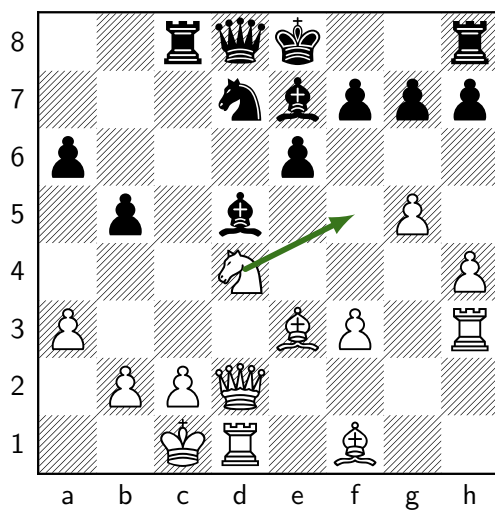


(d)

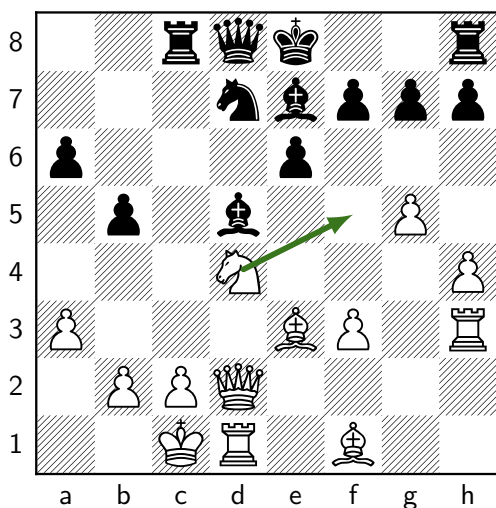
Figure 10: **Move 16, White: Bxf6.** (a) Our position; Melanie plays Bxf6 (h4f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3222 (ply 14) [structural distance $d=95$, QAP obj=71]: the master played Bxf6; under μ our Bh4 plays the role of that game’s Bg5; cross-type roles: our Pb2 as their Ra1, our Pg2 as their Bf1; transferred move h4f6 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#3223 (ply 14) [structural distance $d=95$, QAP obj=71]: the master played Bxf6; under μ our Bh4 plays the role of that game’s Bg5; cross-type roles: our Pb2 as their Ra1, our Pg2 as their Bf1; transferred move h4f6 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#360 (ply 14) [structural distance $d=95$, QAP obj=71]: the master played Bxf6; under μ our Bh4 plays the role of that game’s Bg5; cross-type roles: our Pb2 as their Ra1, our Pg2 as their Bf1; transferred move h4f6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bxf6 (summed QAP objective 565, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



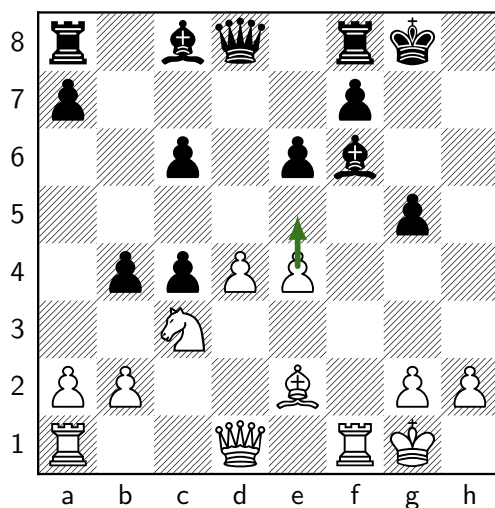
(a)



(b)

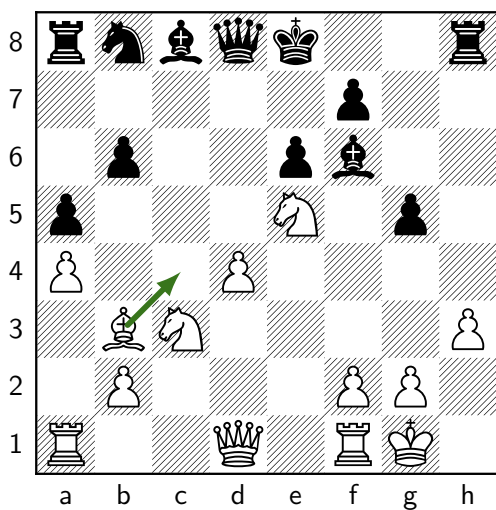


(c)

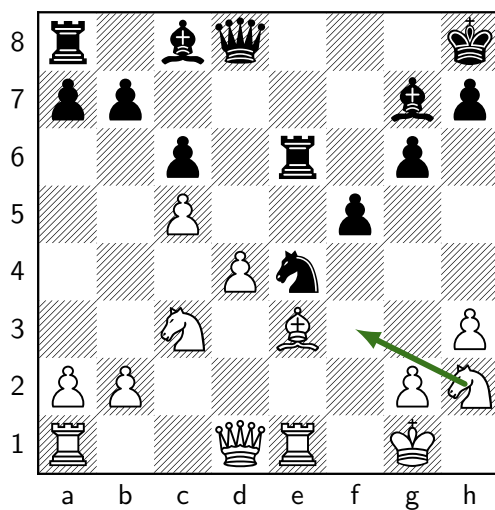


(d)

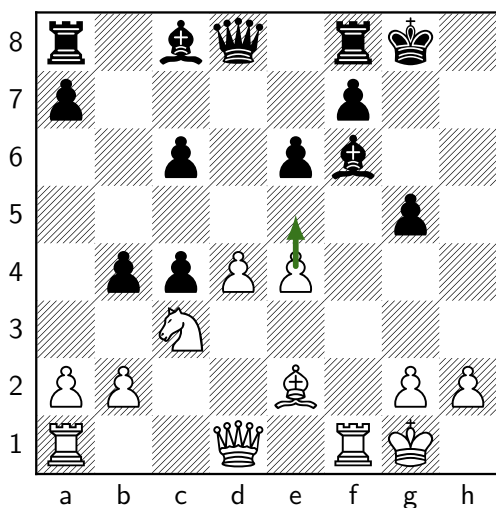
Figure 11: **Move 17, White: Ne5.** (a) Our position; Melanie plays Ne5 (c4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2864` (ply 32) [structural distance $d=74$, QAP obj=99]: the master played Nf5; under μ our Nc4 plays the role of that game's Nd4; cross-type roles: our Nc4 as their Bf1, our Nc3 as their Qd2; transferred move c4e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1129` (ply 32) [structural distance $d=74$, QAP obj=99]: the master played Nf5; under μ our Nc4 plays the role of that game's Nd4; cross-type roles: our Nc4 as their Bf1, our Nc3 as their Qd2; transferred move c4e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1145` (ply 30) [structural distance $d=82$, QAP obj=71]: the master played e5; under μ our Nc4 plays the role of that game's Pe4 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Ra1, our ra8 as their Pb2; transferred move c4e5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ne5 (summed QAP objective 520, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



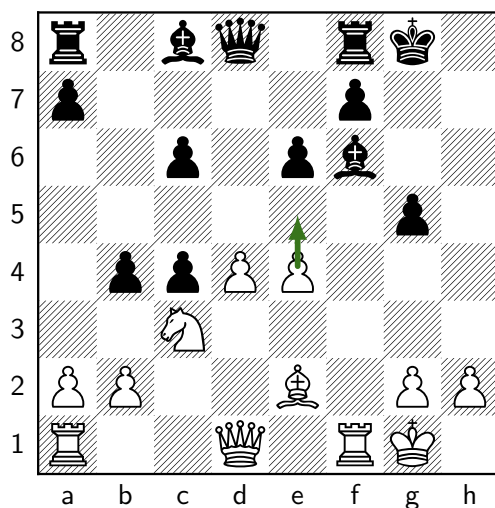
(a)



(b)

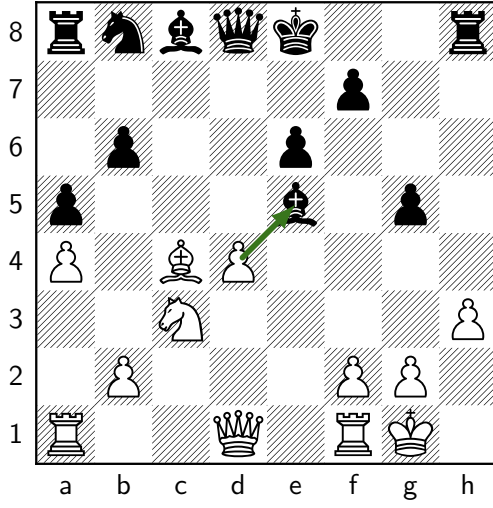


(c)

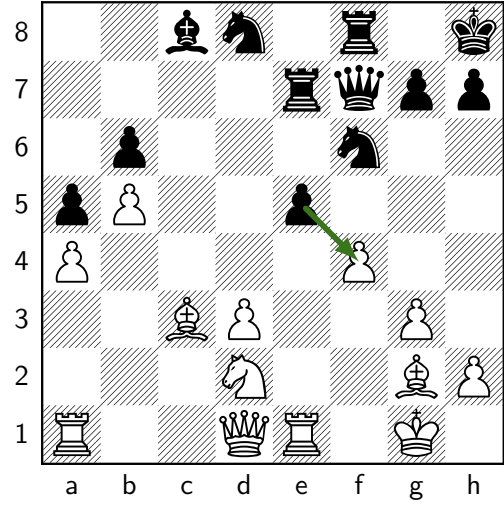


(d)

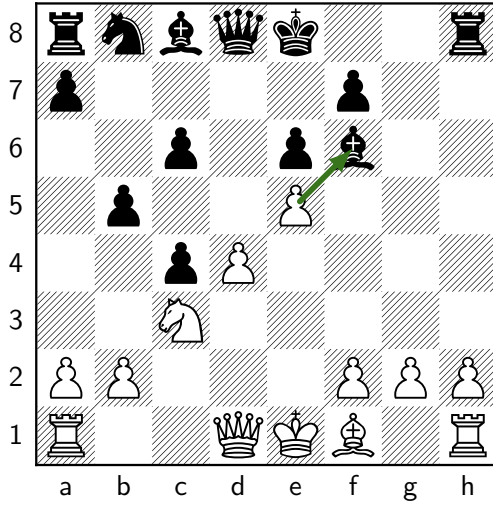
Figure 12: **Move 18, White: Bc4**. (a) Our position; Melanie plays Bc4 (b3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3491` (ply 40) [structural distance $d=84$, QAP obj=51]: the master played Nf3; under μ our Bb3 plays the role of that game’s Nh2 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ra1, our Ra1 as their Pa2; transferred move b3c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1145` (ply 30) [structural distance $d=78$, QAP obj=48]: the master played e5; under μ our Bb3 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our Pa4 as their Be2, our bf6 as their pb4; transferred move b3c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2206` (ply 30) [structural distance $d=78$, QAP obj=48]: the master played e5; under μ our Bb3 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our Pa4 as their Be2, our bf6 as their pb4; transferred move b3c4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bc4 (summed QAP objective 242, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



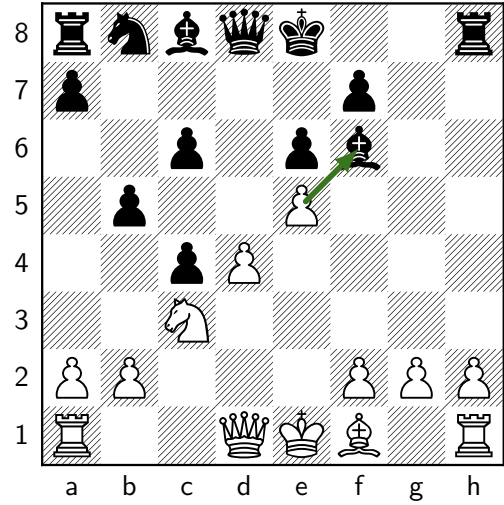
(a)



(b)

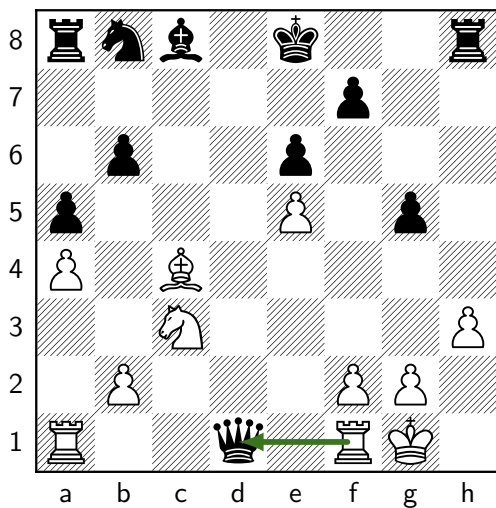


(c)

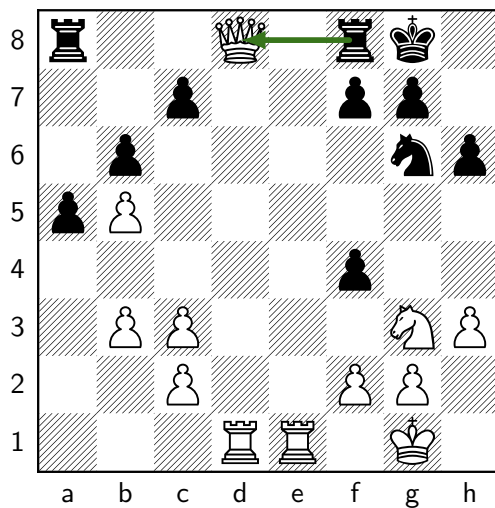


(d)

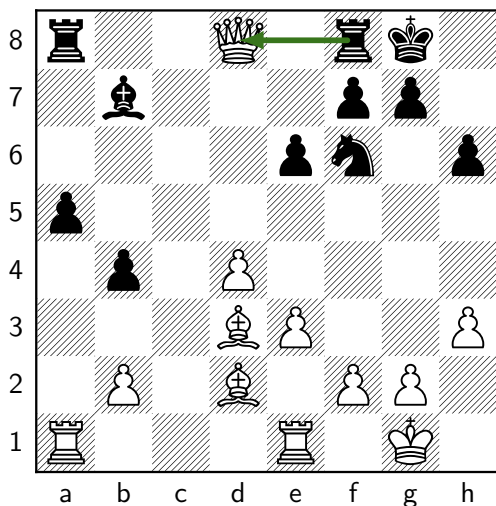
Figure 13: **Move 19, White: dxe5.** (a) Our position; Melanie plays dxe5 (d4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#562` (ply 43) [structural distance $d=71$, QAP obj=54]: the master played **exf4**; under μ our *Pd4* plays the role of that game’s *Pe4*; cross-type roles: our *Pa4* as their *Bc8*, our *Ra1* as their *Nd8*; transferred move **d4e5 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#568` (ply 22) [structural distance $d=60$, QAP obj=40]: the master played **exf6**; under μ our *Pd4* plays the role of that game’s *Pe5*; cross-type roles: our *Ph3* as their *Bf1*, our *Pf2* as their *Rh1*; transferred move **d4e5 (endorses this move)**. (d) Retrieved from `ccr1_top_p0003#2135` (ply 22) [structural distance $d=60$, QAP obj=40]: the master played **exf6**; under μ our *Pd4* plays the role of that game’s *Pe5*; cross-type roles: our *Ph3* as their *Bf1*, our *Pf2* as their *Rh1*; transferred move **d4e5 (endorses this move)**. Synthesis: 7 retrieved analogs transferred to dxe5 (summed QAP objective 248, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



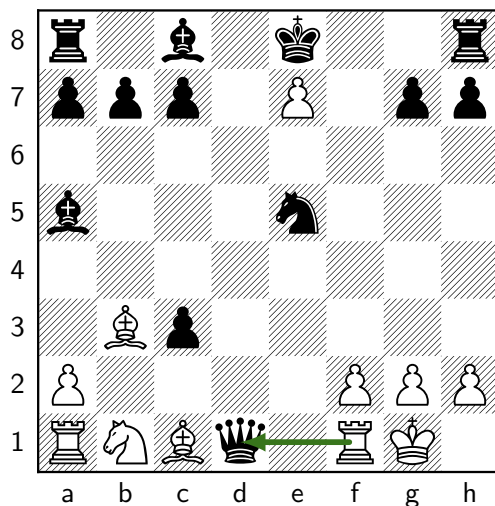
(a)



(b)

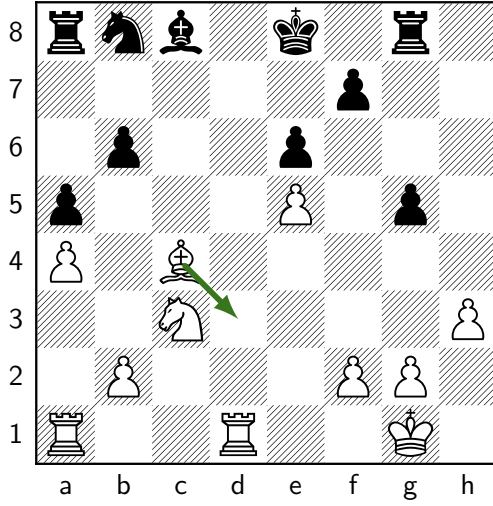


(c)

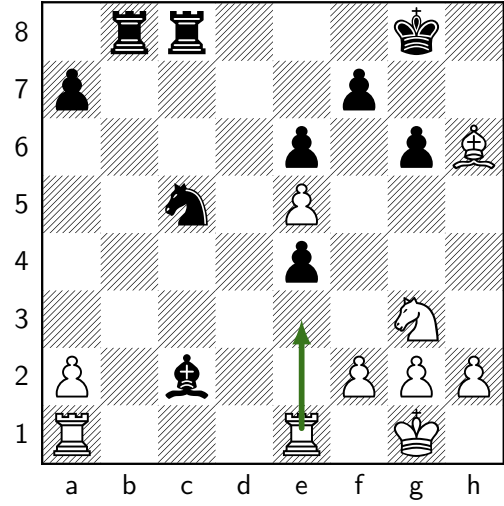


(d)

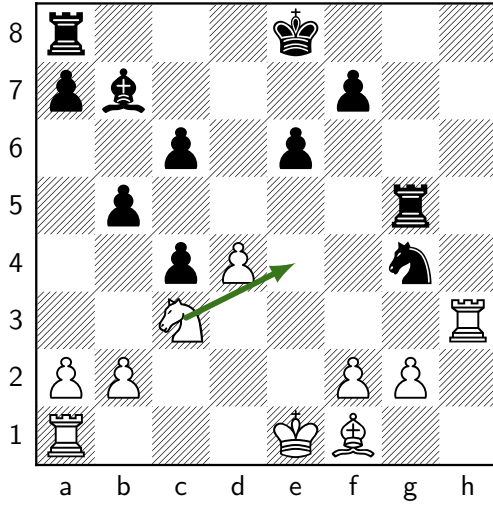
Figure 14: **Move 20, White: Rfxd1.** (a) Our position; Melanie plays Rfxd1 (f1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#493` (ply 45) [structural distance $d=82$, QAP obj=100]: the master played Rfxd8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Bc4* as their *Ng6*, our *Nc3* as their *Pf4*; transferred move **f1d1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0002#1555` (ply 41) [structural distance $d=70$, QAP obj=94]: the master played Rfxd8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Nc3* as their *Bb7*, our *Rf1* as their *Nf6*; transferred move **f1d1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#591` (ply 26) [structural distance $d=86$, QAP obj=93]: the master played Rxd1; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Ph3* as their *Ra1*, our *pb6* as their *Nb1*; transferred move **f1d1 (endorses this move)**. Synthesis: 36 retrieved analogs transferred to Rfxd1 (summed QAP objective 1819, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



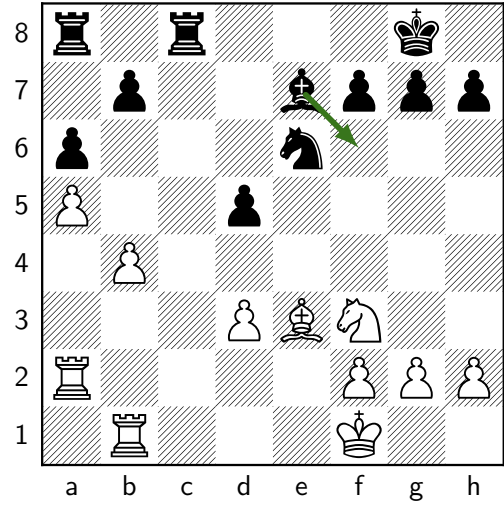
(a)



(b)

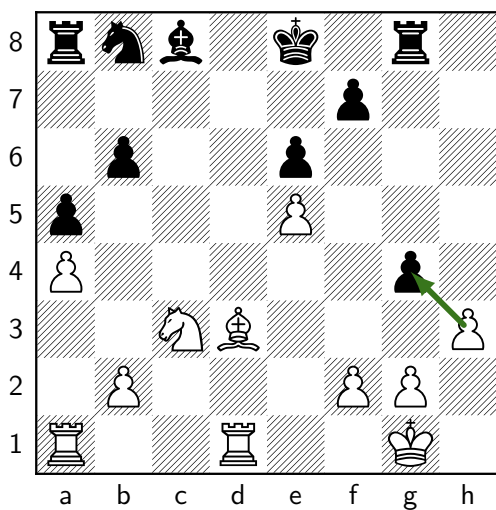


(c)

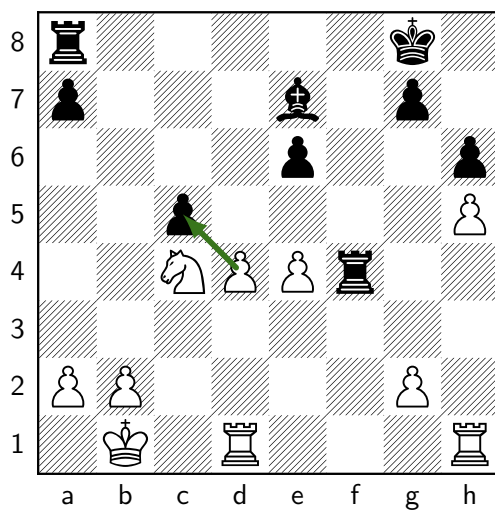


(d)

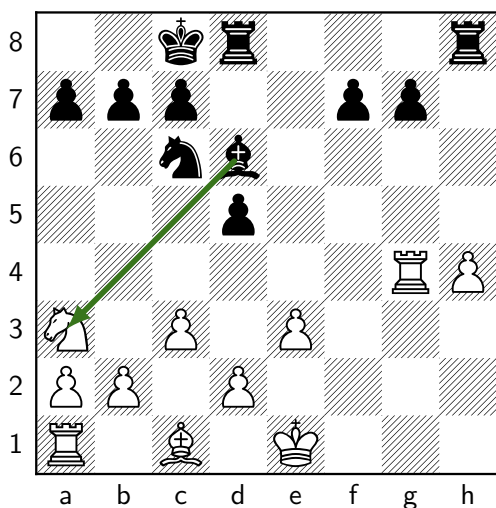
Figure 15: **Move 21, White: Bd3.** (a) Our position; Melanie plays Bd3 (c4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1873` (ply 46) [structural distance $d=49$, QAP obj=34]: the master played Re3; under μ our Bc4 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Ph3 as their Ra1, our nb8 as their Ph2; transferred move c4d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#261` (ply 36) [structural distance $d=47$, QAP obj=32]: the master played Ne4; under μ our Bc4 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Nc3 as their Bf1, our Ra1 as their Pb2; transferred move c4d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#744` (ply 45) [structural distance $d=39$, QAP obj=18]: the master played Bf6; under μ our Bc4 plays the role of that game’s Be2; cross-type roles: our Ph3 as their Ra8, our Ra1 as their Pb7; transferred move c4d3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bd3 (summed QAP objective 136, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



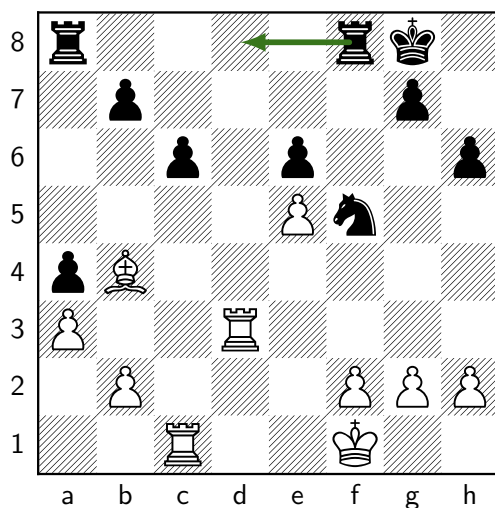
(a)



(b)

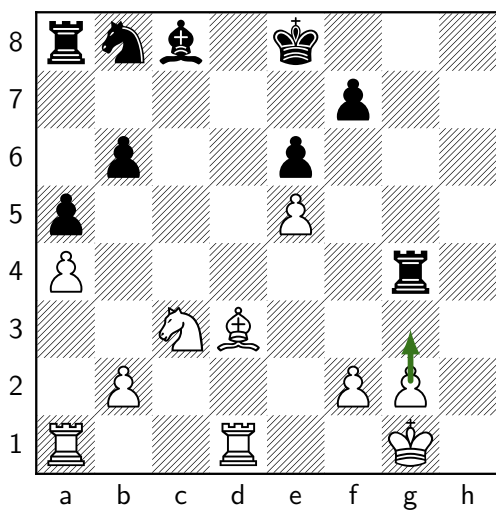


(c)

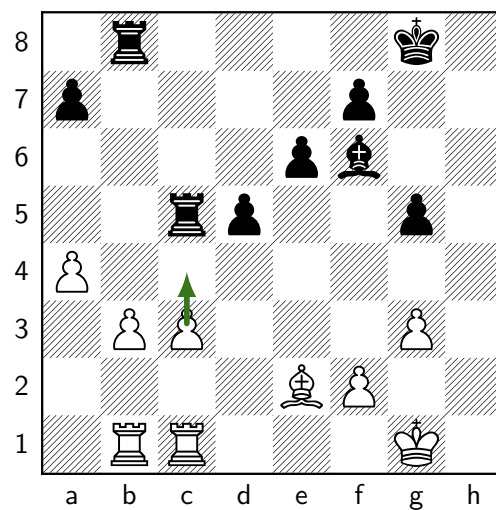


(d)

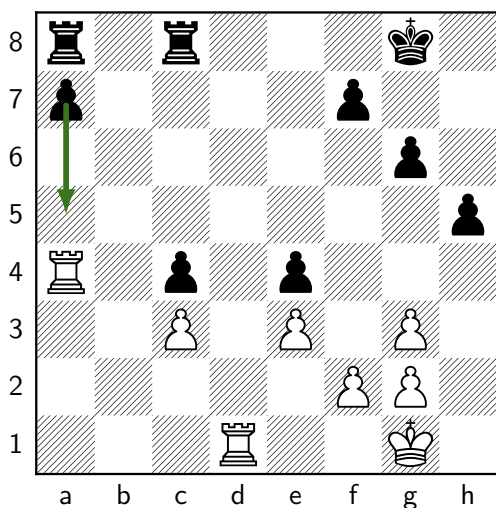
Figure 16: **Move 22, White: hxg4.** (a) Our position; Melanie plays **hxg4** (**h3g4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2476` (ply 50) [structural distance $d=52$, QAP obj=52]: the master played **dx5**; under μ our *Ph3* plays the role of that game’s *Pd4*; cross-type roles: our *pe6* as their *Nc4*, our *Ph3* as their *rf4*; transferred move **h3g4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2020` (ply 37) [structural distance $d=53$, QAP obj=20]: the master played **Bxa3**; under μ our *Ph3* plays the role of that game’s *Bd3* (a **cross-type** correspondence); cross-type roles: our *nb8* as their *Rh8*, our *Ra1* as their *Pf7*; transferred move **h3g4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1880` (ply 43) [structural distance $d=53$, QAP obj=18]: the master played **Rfd8**; under μ our *Ph3* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *Ra8*, our *Rd1* as their *Pb7*; transferred move **h3g4** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **hxg4** (summed QAP objective 160, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



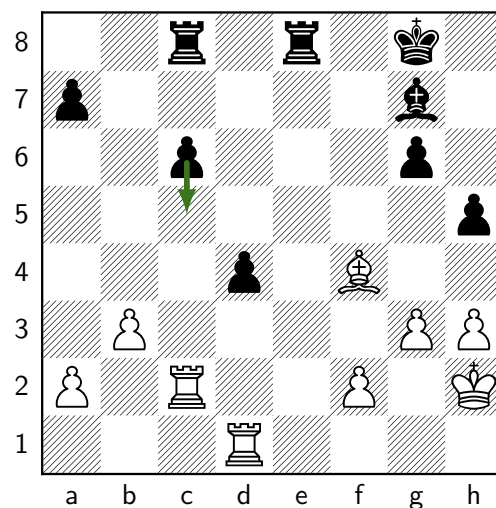
(a)



(b)

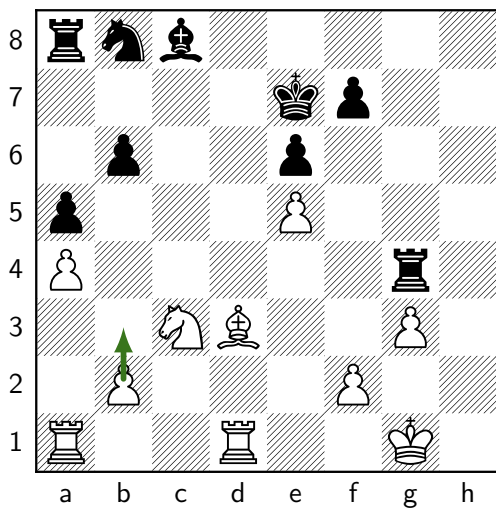


(c)

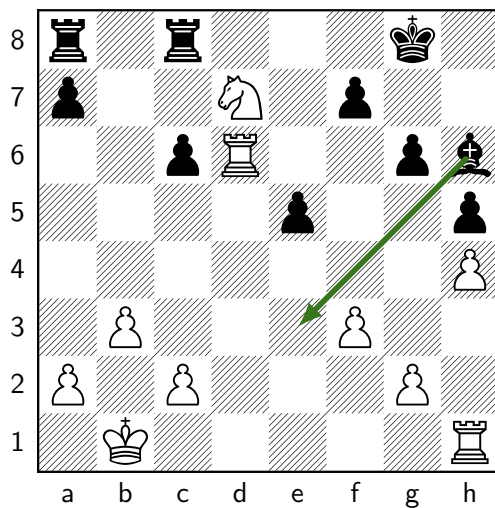


(d)

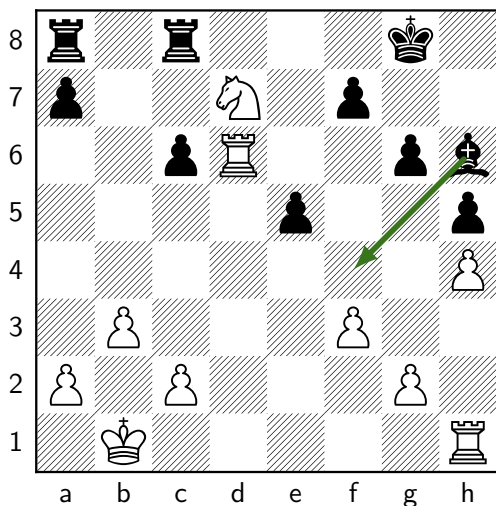
Figure 17: **Move 23, White: g3.** (a) Our position; Melanie plays g3 (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1533` (ply 60) [structural distance $d=43$, QAP obj=57]: the master played c4; under μ our $Pg2$ plays the role of that game’s $Pc3$; cross-type roles: our $Nc3$ as their $Rb1$, our $rg4$ as their $bf6$; transferred move g2g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2692` (ply 55) [structural distance $d=53$, QAP obj=45]: the master played a5; under μ our $Pg2$ plays the role of that game’s $Pa2$; cross-type roles: our $Nc3$ as their $Ra8$, our $bc8$ as their $pc3$; transferred move g2g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1413` (ply 55) [structural distance $d=51$, QAP obj=29]: the master played c5; under μ our $Pg2$ plays the role of that game’s $Pc3$; cross-type roles: our $Nc3$ as their $Rc8$, our $Ra1$ as their $Pa7$; transferred move g2g3 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to g3 (summed QAP objective 259, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



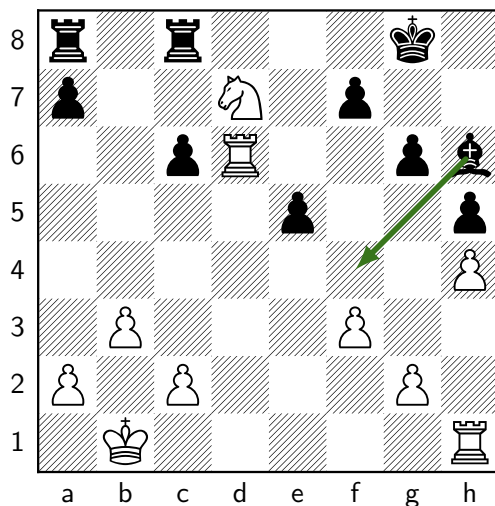
(a)



(b)

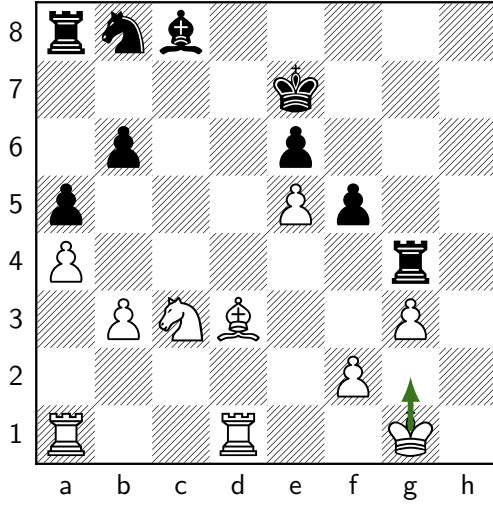


(c)

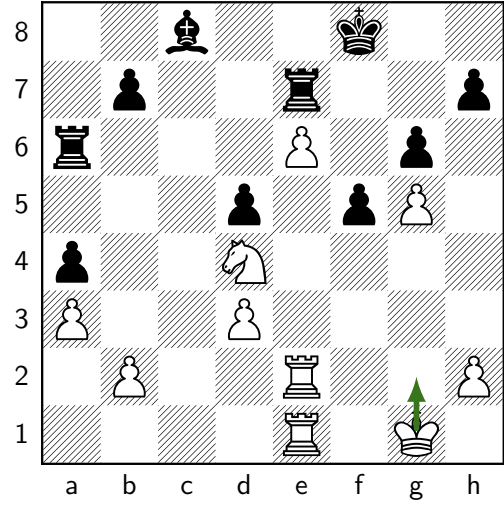


(d)

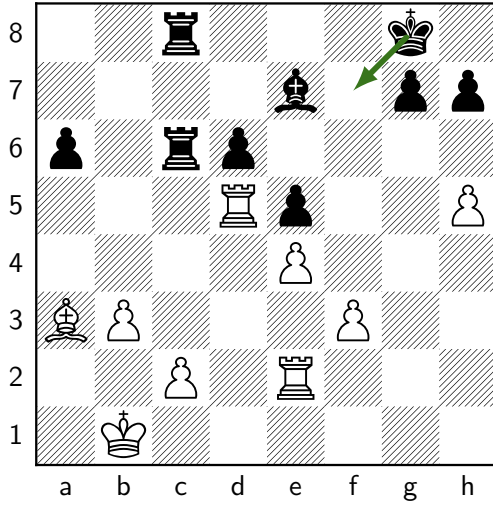
Figure 18: **Move 24, White: b3**. (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1374` (ply 47) [structural distance $d=44$, QAP obj=32]: the master played Be3; under μ our *Pb2* plays the role of that game’s *Bh3* (a **cross-type** correspondence); cross-type roles: our *Bd3* as their *Pa7*, our *rg4* as their *nd7*; transferred move b2b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1376` (ply 47) [structural distance $d=44$, QAP obj=32]: the master played Bf4; under μ our *Pb2* plays the role of that game’s *Bh3* (a **cross-type** correspondence); cross-type roles: our *Bd3* as their *Pa7*, our *rg4* as their *nd7*; transferred move b2b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#339` (ply 47) [structural distance $d=44$, QAP obj=32]: the master played Bf4; under μ our *Pb2* plays the role of that game’s *Bh3* (a **cross-type** correspondence); cross-type roles: our *Bd3* as their *Pa7*, our *rg4* as their *nd7*; transferred move b2b3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to b3 (summed QAP objective 162, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



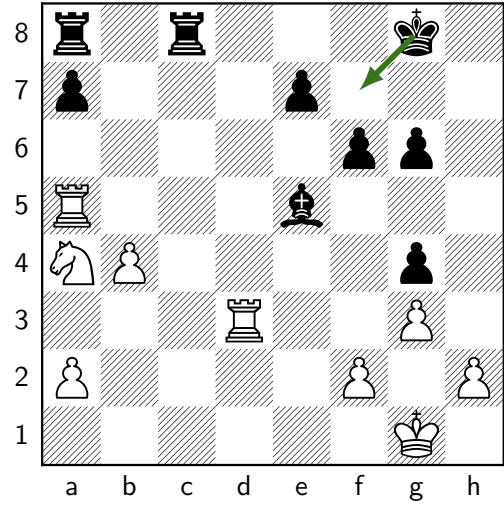
(a)



(b)

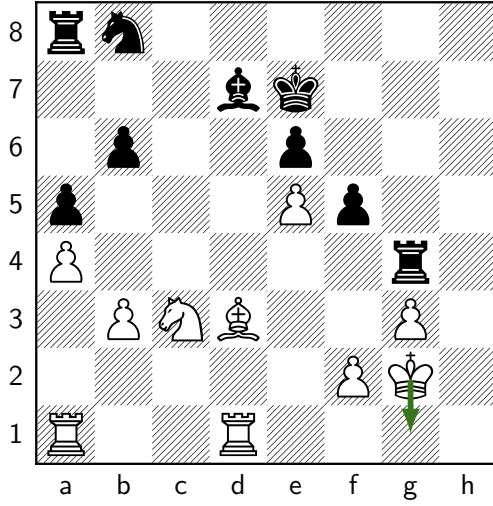


(c)

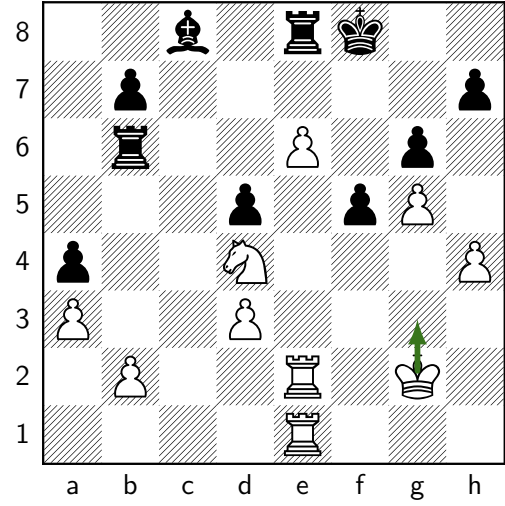


(d)

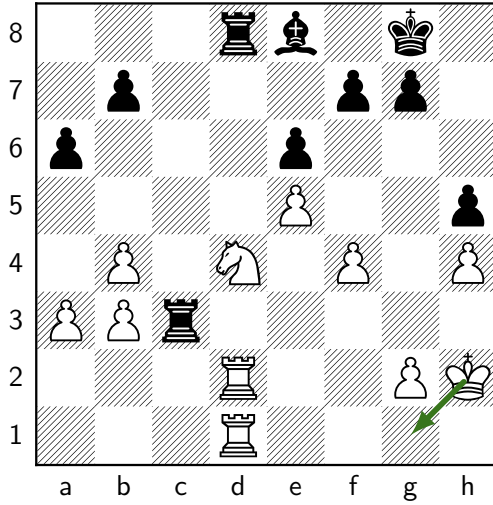
Figure 19: **Move 25, White: Kg2.** (a) Our position; Melanie plays Kg2 (**g1g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3721` (ply 62) [structural distance $d=43$, QAP obj=58]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Nc3 as their Re2, our Ra1 as their Ph2; transferred move g1g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1158` (ply 61) [structural distance $d=56$, QAP obj=50]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Nc3 as their Be7, our Bd3 as their Rc6; transferred move g1g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2310` (ply 51) [structural distance $d=59$, QAP obj=33]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Pf2 as their Ra8, our Pg3 as their Be5; transferred move g1g2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kg2 (summed QAP objective 216, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



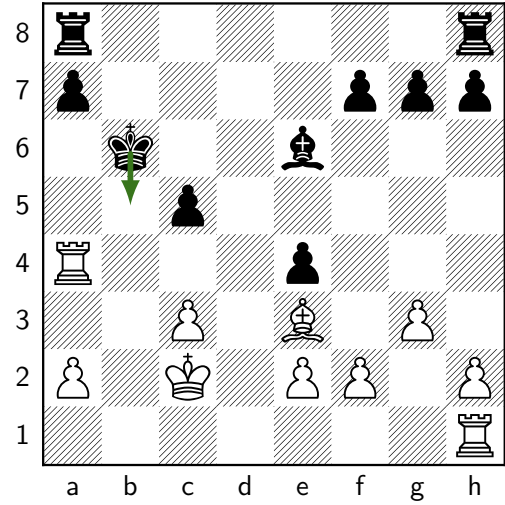
(a)



(b)

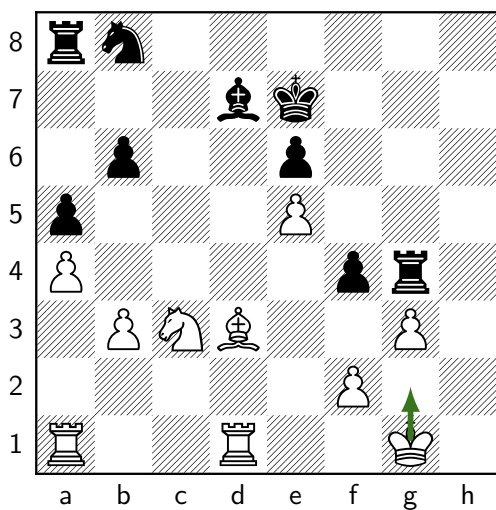


(c)

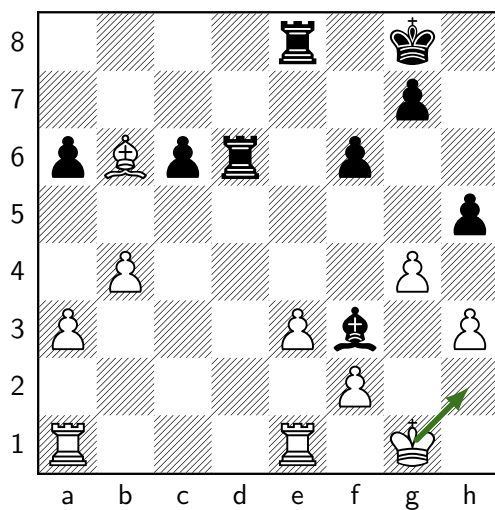


(d)

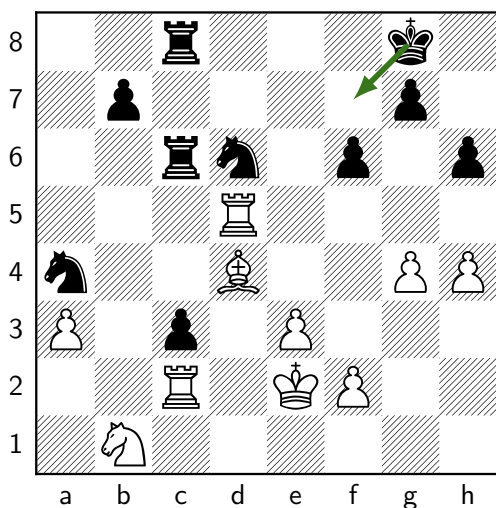
Figure 20: **Move 26, White: Kg1.** (a) Our position; Melanie plays Kg1 ($g2g1$). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3721` (ply 66) [structural distance $d=61$, QAP obj=47]: the master played Kg3; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Pe5 as their Re1, our Nc3 as their Re2; transferred move $g2g1$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3620` (ply 56) [structural distance $d=59$, QAP obj=40]: the master played Kg1; under μ our Kg2 plays the role of that game’s Kh2; cross-type roles: our nb8 as their Rd2, our Bd3 as their rc3; transferred move $g2g1$ (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2839` (ply 35) [structural distance $d=60$, QAP obj=36]: the master played Kb5; under μ our Kg2 plays the role of that game’s Kb3; cross-type roles: our Nc3 as their Ra8, our nb8 as their Pf7; transferred move $g2g1$ (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kg1 (summed QAP objective 249, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



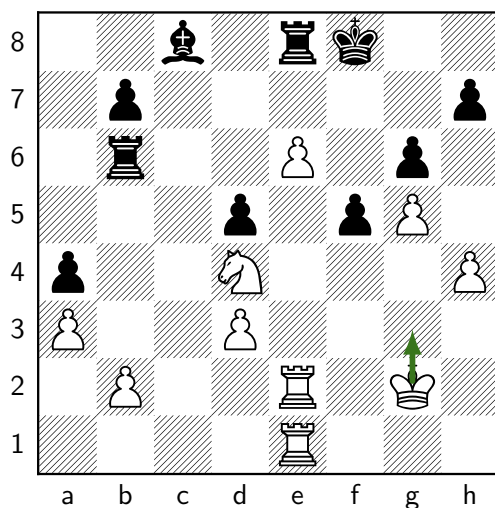
(a)



(b)

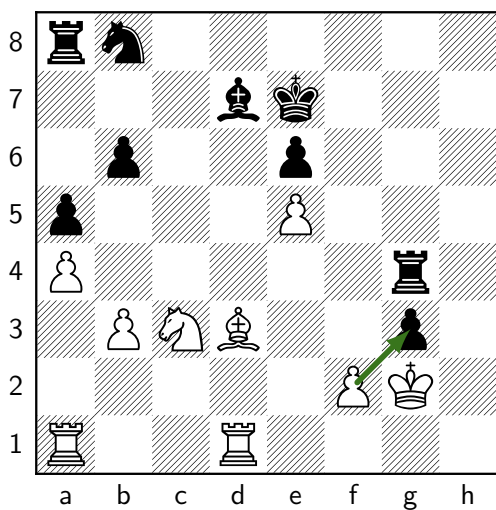


(c)

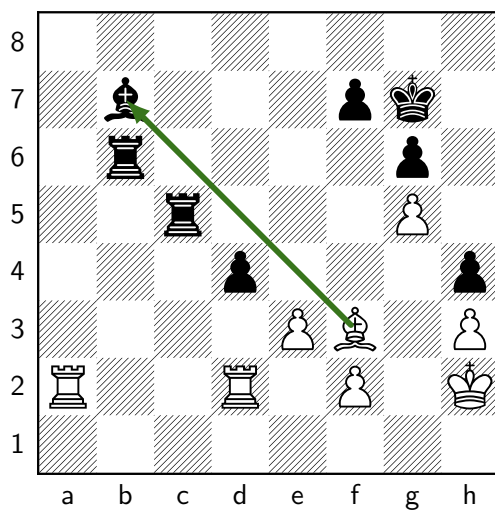


(d)

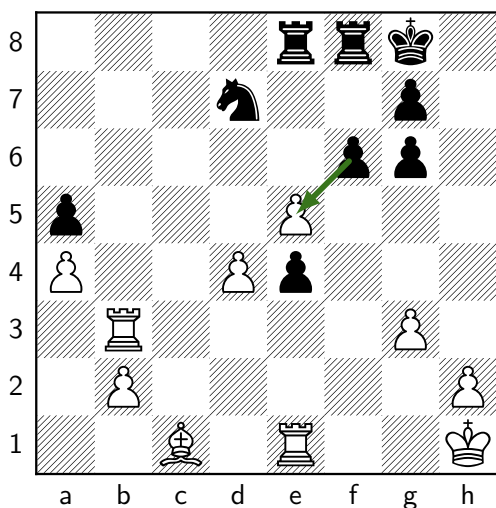
Figure 21: **Move 27, White: Kg2.** (a) Our position; Melanie plays Kg2 (**g1g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2877` (ply 50) [structural distance $d=56$, QAP obj=70]: the master played Kh2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Nc3 as their Ra1, our nb8 as their Pa3; transferred move g1g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1513` (ply 69) [structural distance $d=50$, QAP obj=58]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Nc3 as their Rc6, our Ra1 as their Nd6; transferred move g1g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3721` (ply 66) [structural distance $d=57$, QAP obj=54]: the master played Kg3; under μ our Kg1 plays the role of that game’s Kg2; cross-type roles: our Pf2 as their Re1, our Ra1 as their Pa3; transferred move g1g2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kg2 (summed QAP objective 215, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



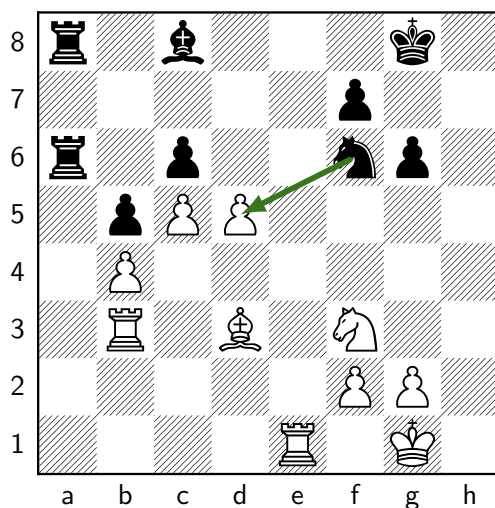
(a)



(b)

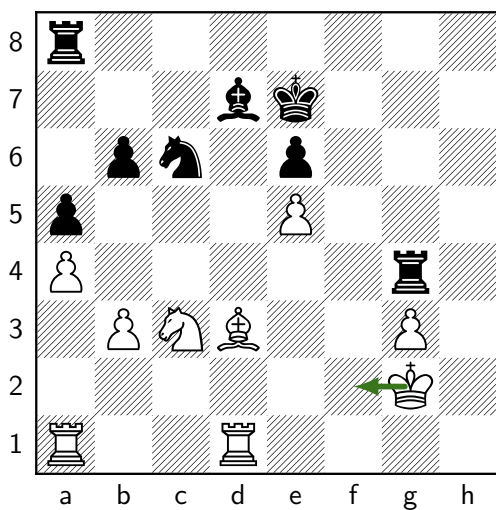


(c)

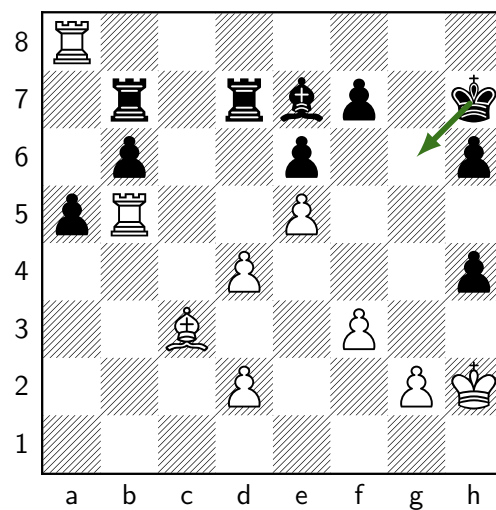


(d)

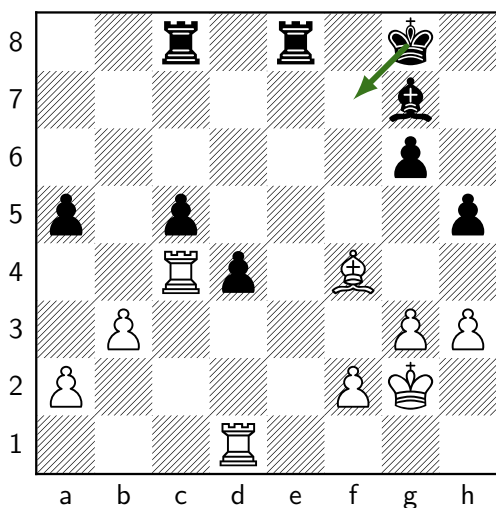
Figure 22: **Move 28, White: f2g3.** (a) Our position; Melanie plays f2g3 (f2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3062` (ply 66) [structural distance $d=52$, QAP obj=70]: the master played Bxb7; under μ our Pf2 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our Kg2 as their Rd2, our pg3 as their bb7; transferred move f2f4. (c) Retrieved from `ccr1_top_p0002#3941` (ply 45) [structural distance $d=57$, QAP obj=67]: the master played fxe5; under μ our Pf2 plays the role of that game’s Pf3; cross-type roles: our Kg2 as their Re8, our Kg2 as their Nd7; transferred move f2f3. (d) Retrieved from `ccr1_top_p0003#587` (ply 57) [structural distance $d=49$, QAP obj=56]: the master played Nxg5; under μ our Kg2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our ra8 as their Bc8, our Bd3 as their Ra6; transferred move g2g1. Synthesis: no retrieved analog transferred to f2g3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



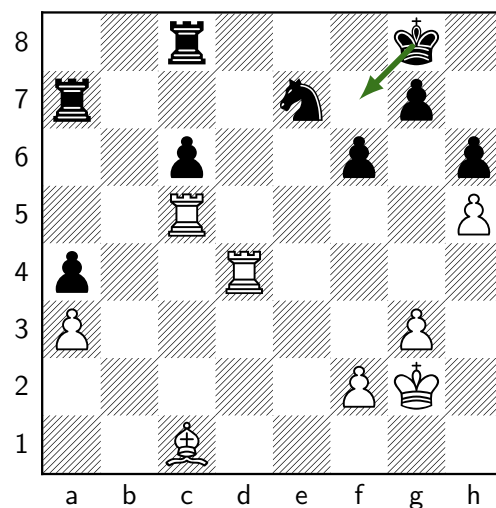
(a)



(b)

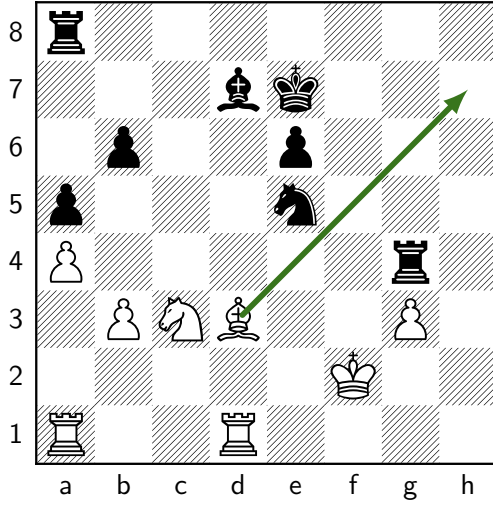


(c)

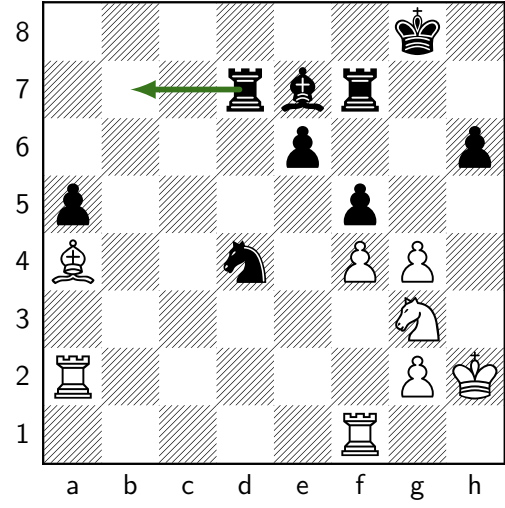


(d)

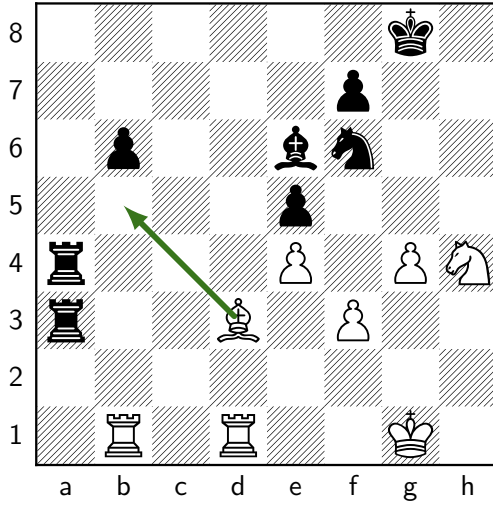
Figure 23: **Move 29, White: Kf2.** (a) Our position; Melanie plays Kf2 (g2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#714` (ply 73) [structural distance $d=70$, QAP obj=67]: the master played Kg6; under μ our Kg2 plays the role of that game’s Kh2; cross-type roles: our bd7 as their Rb7, our nc6 as their Rd7; transferred move g2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1413` (ply 59) [structural distance $d=67$, QAP obj=57]: the master played Kf7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Nc3 as their Rc8, our Bd3 as their Re8; transferred move g2f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1947` (ply 69) [structural distance $d=66$, QAP obj=47]: the master played Kf7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Nc3 as their Rc8, our Bd3 as their Ra7; transferred move g2f2 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kf2 (summed QAP objective 274, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



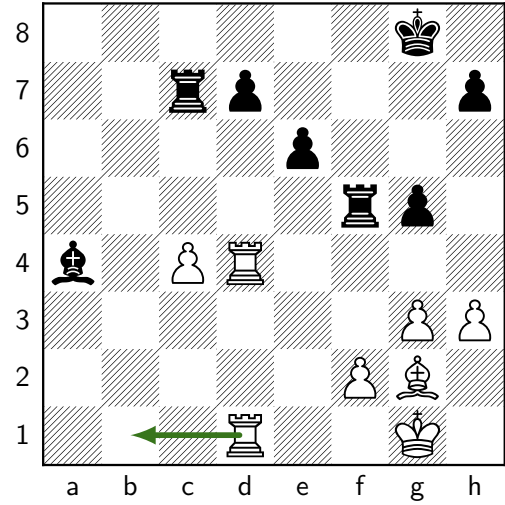
(a)



(b)

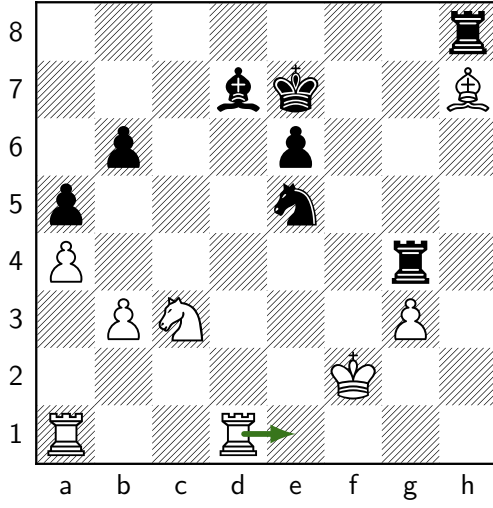


(c)

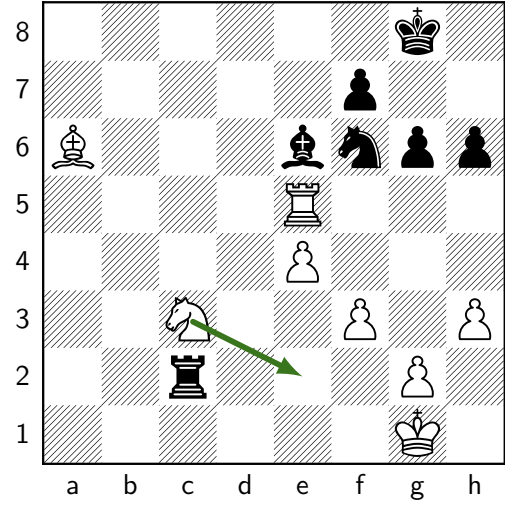


(d)

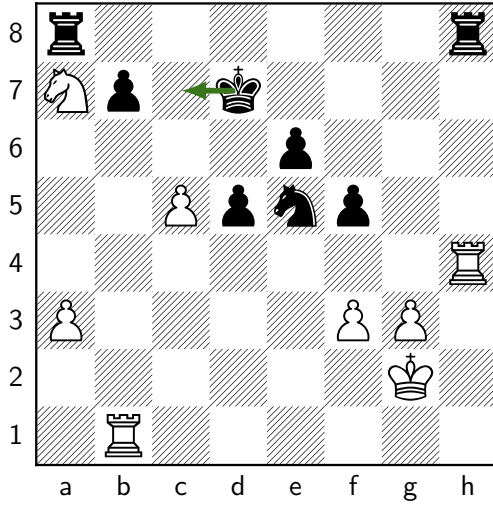
Figure 24: **Move 30, White: Bh7 (d3h7)**. (a) Our position; Melanie plays Bh7 (d3h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#370` (ply 75) [structural distance $d=52$, QAP obj=32]: the master played Rb7; under μ our Bd3 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our Rd1 as their Be7, our Nc3 as their Rf7; transferred move d3h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#723` (ply 66) [structural distance $d=58$, QAP obj=28]: the master played Bb5; under μ our Bd3 plays the role of that game’s Bd3; cross-type roles: our ne5 as their ra3, our pa5 as their Nh4; transferred move d3h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3472` (ply 60) [structural distance $d=64$, QAP obj=20]: the master played Rb1; under μ our Bd3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our pb6 as their Bg2, our Nc3 as their Pg3; transferred move d3h7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bh7 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



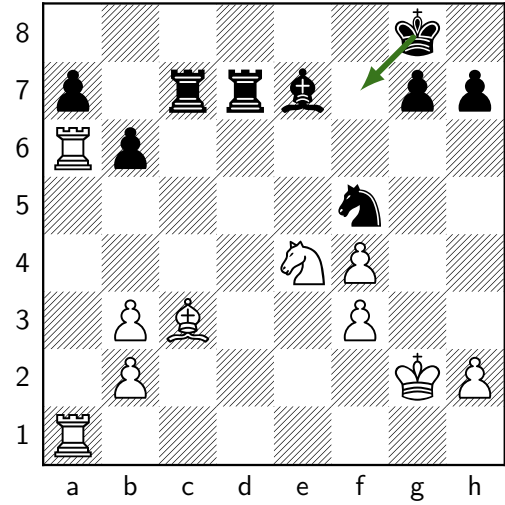
(a)



(b)

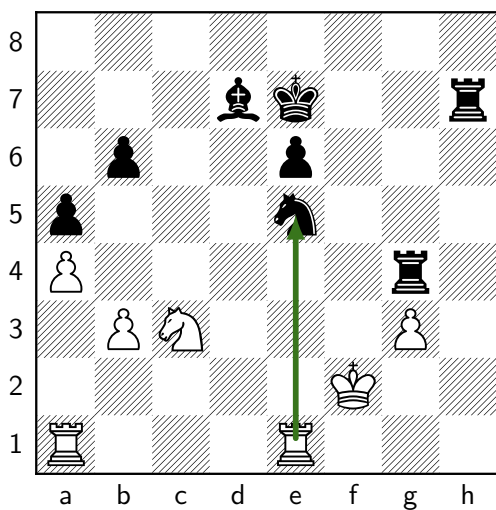


(c)

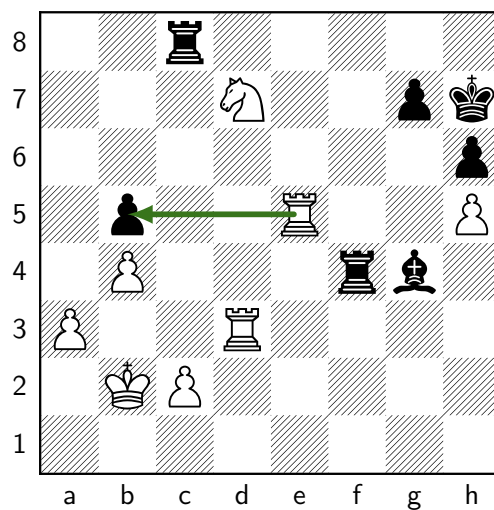


(d)

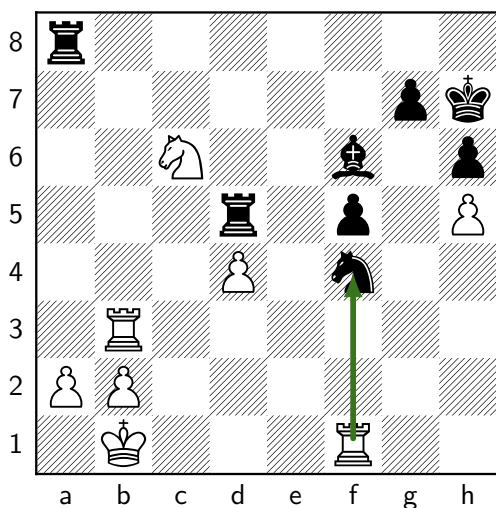
Figure 25: **Move 31, White: Re1.** (a) Our position; Melanie plays Re1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1108` (ply 66) [structural distance $d=65$, QAP obj=37]: the master played Ne2; under μ our Bh7 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Nc3 as their Pe4, our Pb3 as their Ba6; transferred move h7c2. (c) Retrieved from `ccrl_top_p0000#882` (ply 75) [structural distance $d=61$, QAP obj=27]: the master played Kc7; under μ our Kf2 plays the role of that game’s Kd2; cross-type roles: our Pg3 as their Rh8, our Bh7 as their na7; transferred move f2g2. (d) Retrieved from `ccrl_top_p0001#485` (ply 59) [structural distance $d=65$, QAP obj=26]: the master played Kf7; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our Nc3 as their Rd7, our Pb3 as their Be7; transferred move f2g2. Synthesis: no retrieved analog transferred to Re1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



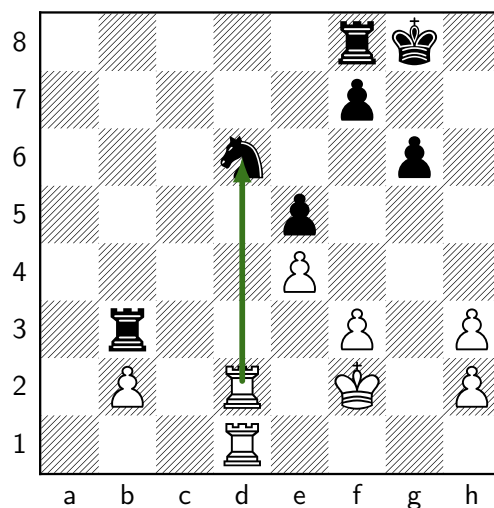
(a)



(b)

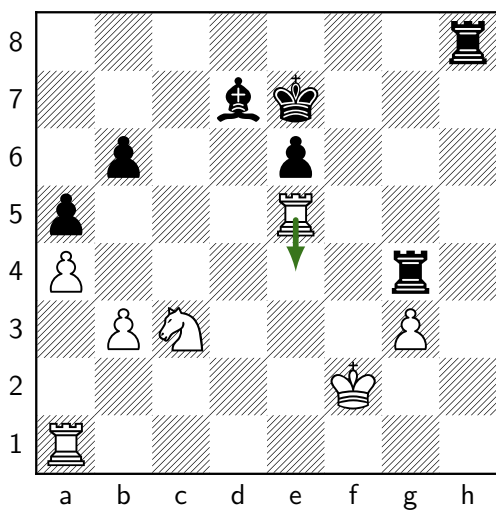


(c)

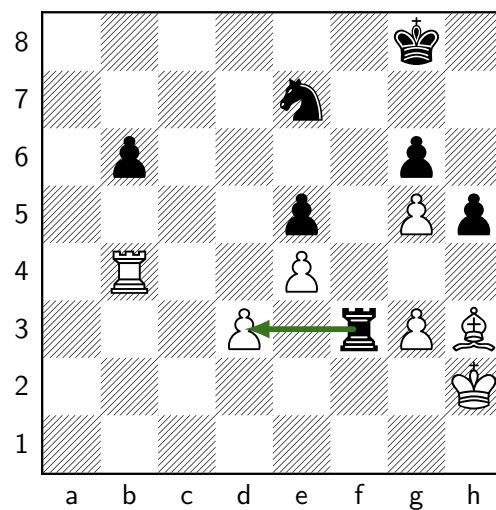


(d)

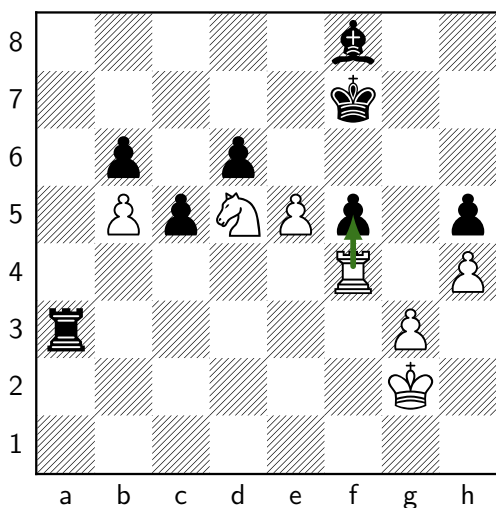
Figure 26: **Move 32, White: Rxe5**. (a) Our position; Melanie plays Rxe5 (e1e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#2650 (ply 76) [structural distance $d=43$, QAP obj=44]: the master played Rxb5; under μ our Re1 plays the role of that game's Re5; cross-type roles: our Nc3 as their Rd3, our bd7 as their rf4; transferred move e1e5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1699 (ply 68) [structural distance $d=43$, QAP obj=42]: the master played Rxf4; under μ our Re1 plays the role of that game's Rf1; cross-type roles: our Ra1 as their Pa2, our Nc3 as their Pb2; transferred move e1e5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#605 (ply 62) [structural distance $d=57$, QAP obj=37]: the master played Rxd6; under μ our Re1 plays the role of that game's Rd2; cross-type roles: our Nc3 as their Ph3; transferred move e1e5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rxe5 (summed QAP objective 178, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



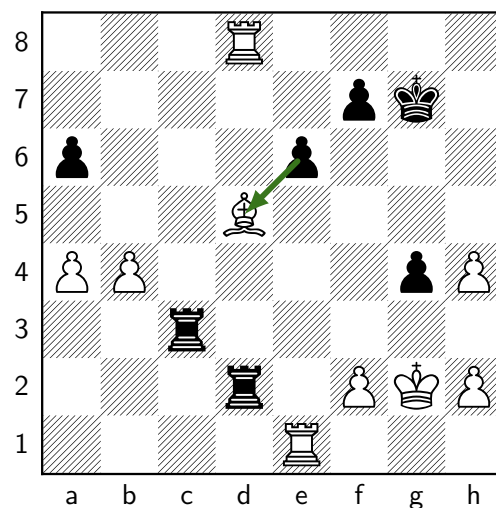
(a)



(b)

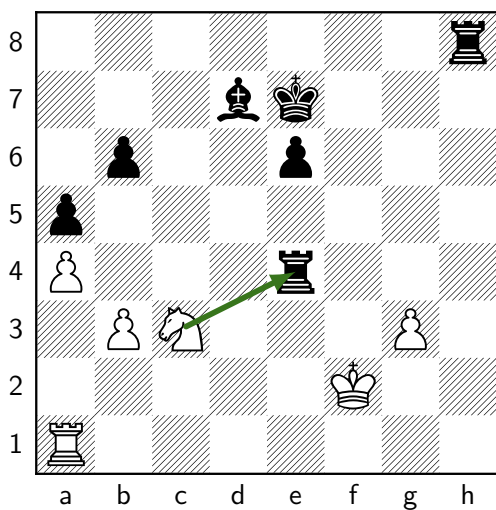


(c)

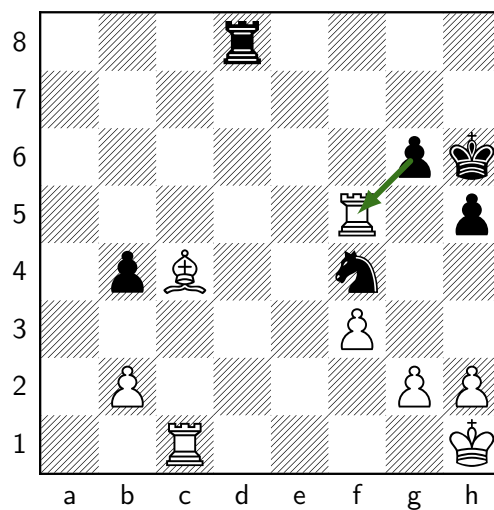


(d)

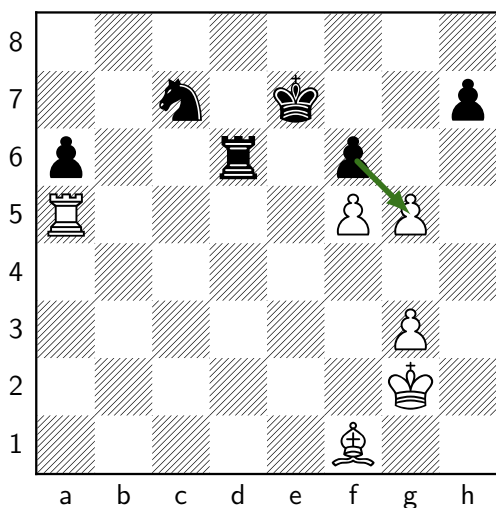
Figure 27: **Move 33, White: Re4.** (a) Our position; Melanie plays Re4 (e5e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2624` (ply 69) [structural distance $d=43$, QAP obj=65]: the master played Rxd3; under μ our Re5 plays the role of that game’s Rf6; cross-type roles: our Pa4 as their Ne7, our rh8 as their Pg6; transferred move e5e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2685` (ply 64) [structural distance $d=37$, QAP obj=64]: the master played Rxf5+; under μ our Re5 plays the role of that game’s Rf4; cross-type roles: our rh8 as their Ph4, our bd7 as their Nd5; transferred move e5e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2102` (ply 69) [structural distance $d=47$, QAP obj=39]: the master played exd5; under μ our Re5 plays the role of that game’s Pe3 (a **cross-type** correspondence); cross-type roles: our bd7 as their rd8, our pe6 as their bd5; transferred move e5e4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re4 (summed QAP objective 190, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



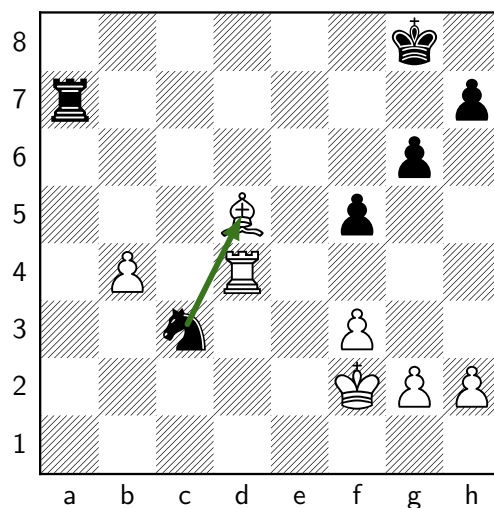
(a)



(b)

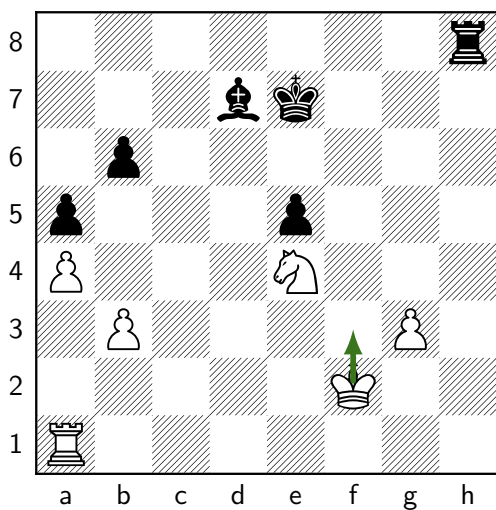


(c)

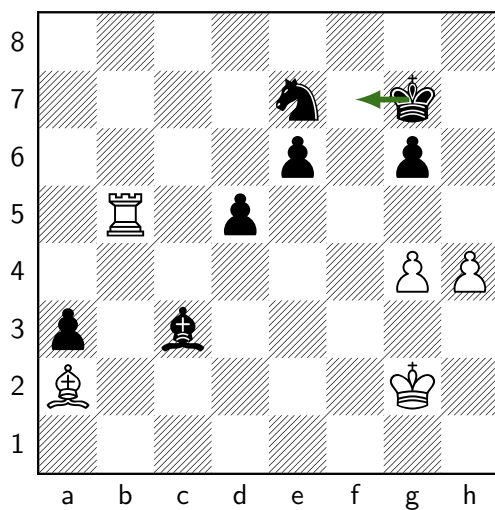


(d)

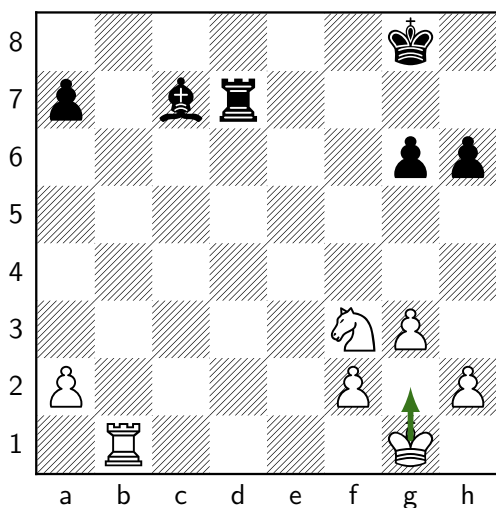
Figure 28: **Move 34, White: Nxe4**. (a) Our position; Melanie plays **Nxe4** (**c3e4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2958` (ply 67) [structural distance $d=34$, QAP obj=18]: the master played **gxf5**; under μ our **Nc3** plays the role of that game's **Pg3** (a **cross-type** correspondence); cross-type roles: our **Pb3** as their **Rd8**, our **pa5** as their **bc4**; transferred move **c3e4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#865` (ply 79) [structural distance $d=27$, QAP obj=11]: the master played **fxg5**; under μ our **Nc3** plays the role of that game's **Pf3** (a **cross-type** correspondence); cross-type roles: our **pe6** as their **Nc7**, our **bd7** as their **ra5**; transferred move **c3e4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2114` (ply 77) [structural distance $d=30$, QAP obj=4]: the master played **Nxd5**; under μ our **Nc3** plays the role of that game's **Nc6**; cross-type roles: our **pa5** as their **Ra7**, our **re4** as their **Pg6**; transferred move **c3e4** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Nxe4** (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



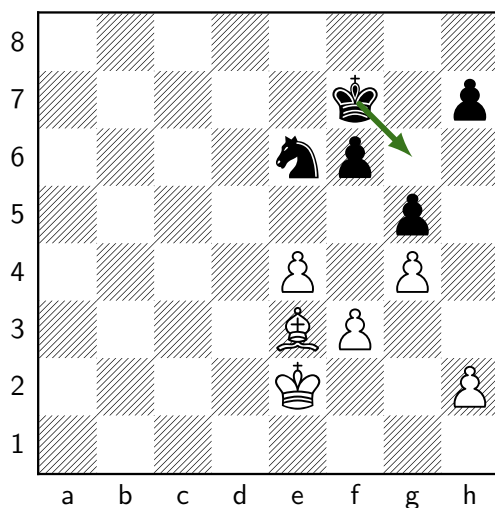
(a)



(b)

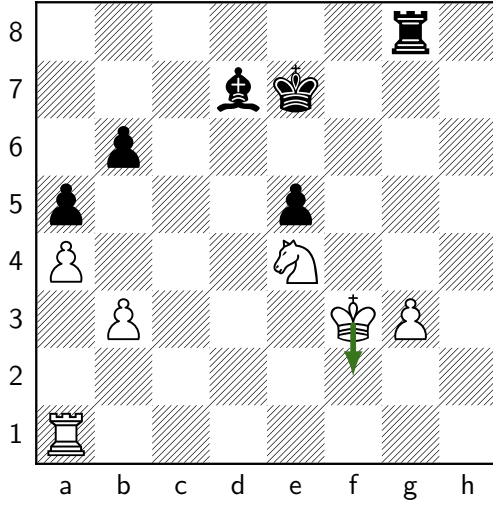


(c)

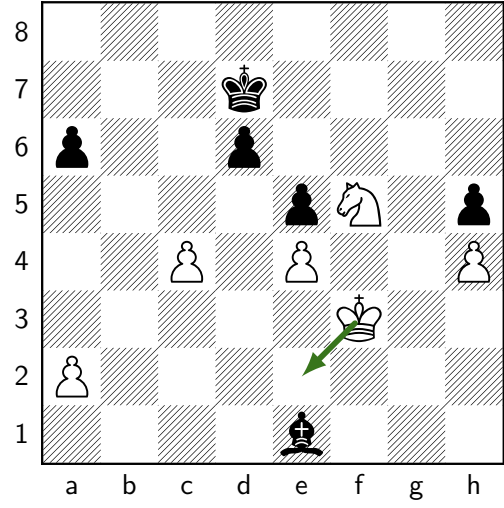


(d)

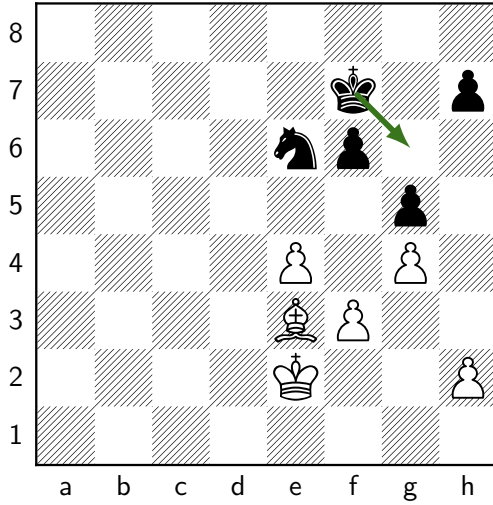
Figure 29: **Move 35, White: Kf3.** (a) Our position; Melanie plays Kf3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#274` (ply 83) [structural distance $d=23$, QAP obj=25]: the master played Kf7; under μ our Kf2 plays the role of that game’s Kg2; cross-type roles: our Ra1 as their Ne7, our bd7 as their rb5; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3435` (ply 62) [structural distance $d=27$, QAP obj=19]: the master played Kg2; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our Ra1 as their Pa2, our rh8 as their pa7; transferred move f2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2889` (ply 83) [structural distance $d=27$, QAP obj=18]: the master played Kg6; under μ our Kf2 plays the role of that game’s Kf2; cross-type roles: our Ra1 as their Ph7, our Pb3 as their Ne6; transferred move f2f3 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kf3 (summed QAP objective 176, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



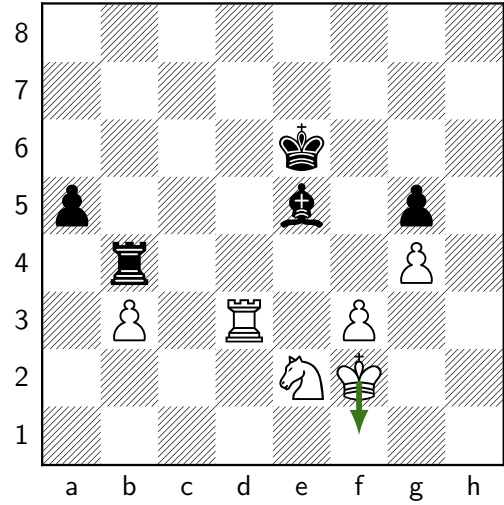
(a)



(b)

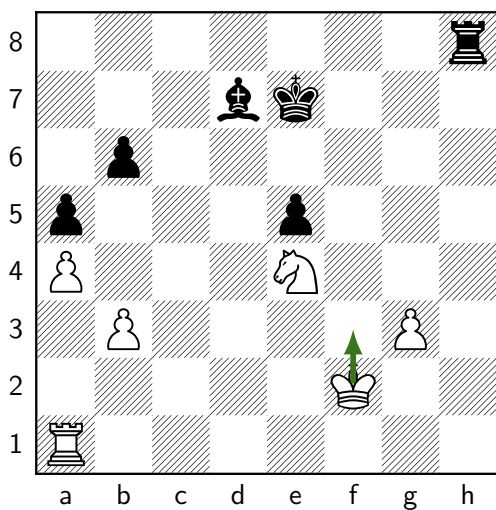


(c)

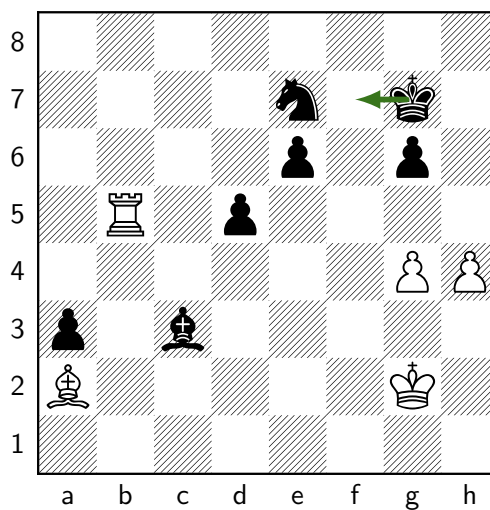


(d)

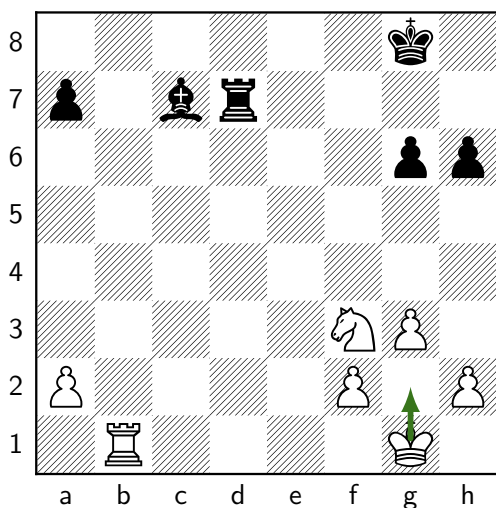
Figure 30: **Move 36, White: Kf2.** (a) Our position; Melanie plays Kf2 (f3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1159` (ply 80) [structural distance $d=25$, QAP obj=23]: the master played K_e2; under μ our K_f3 plays the role of that game’s K_f3; cross-type roles: our *rg*8 as their *be*1, our *Ra*1 as their *Pa*2; transferred move f3f2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2889` (ply 83) [structural distance $d=25$, QAP obj=21]: the master played K_g6; under μ our K_f3 plays the role of that game’s K_f2; cross-type roles: our *Ra*1 as their *Ph*7; transferred move f3f2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1160` (ply 82) [structural distance $d=23$, QAP obj=10]: the master played K_f1; under μ our K_f3 plays the role of that game’s K_f2; cross-type roles: our *bd*7 as their *rb*4, our *pb*6 as their *be*5; transferred move f3f2 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kf2 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



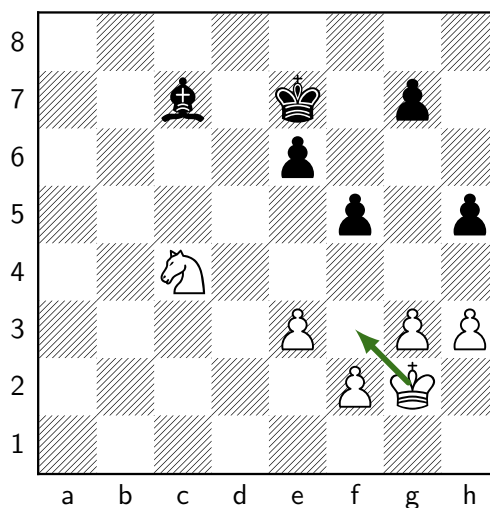
(a)



(b)

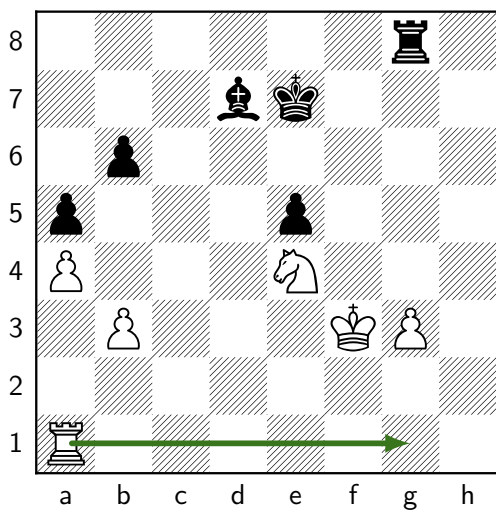


(c)

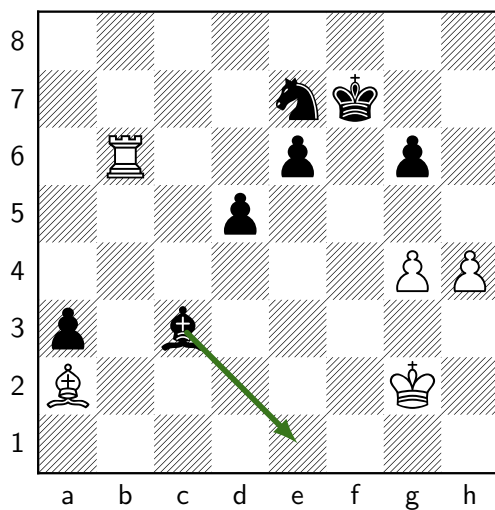


(d)

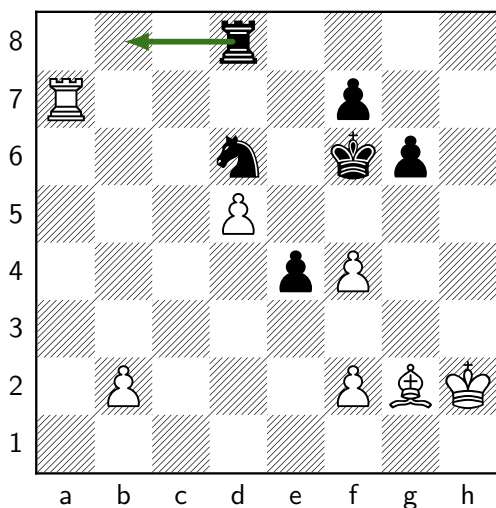
Figure 31: **Move 37, White: Kf3.** (a) Our position; Melanie plays Kf3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#274` (ply 83) [structural distance $d=23$, QAP obj=25]: the master played Kf7; under μ our Kf2 plays the role of that game’s Kg2; cross-type roles: our Ra1 as their Ne7, our bd7 as their rb5; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3435` (ply 62) [structural distance $d=27$, QAP obj=19]: the master played Kg2; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our Ra1 as their Pa2, our rh8 as their pa7; transferred move f2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3546` (ply 76) [structural distance $d=27$, QAP obj=18]: the master played Kf3; under μ our Kf2 plays the role of that game’s Kg2; cross-type roles: our Ne4 as their Ph3, our rh8 as their Nc4; transferred move f2f3 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kf3 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



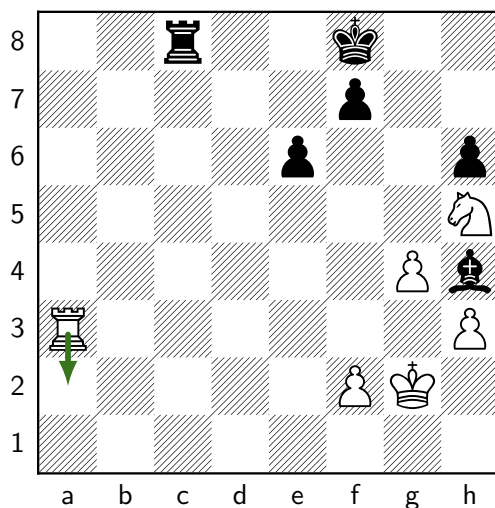
(a)



(b)

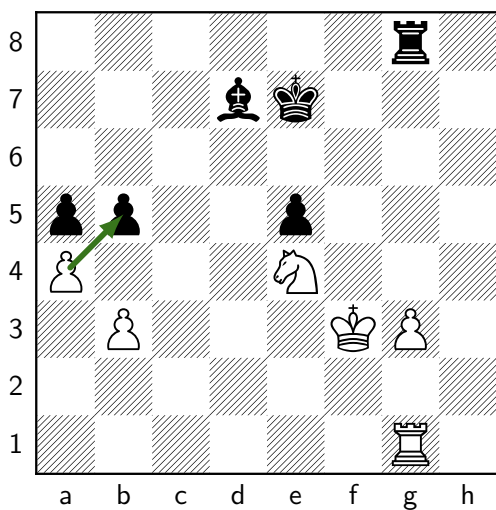


(c)

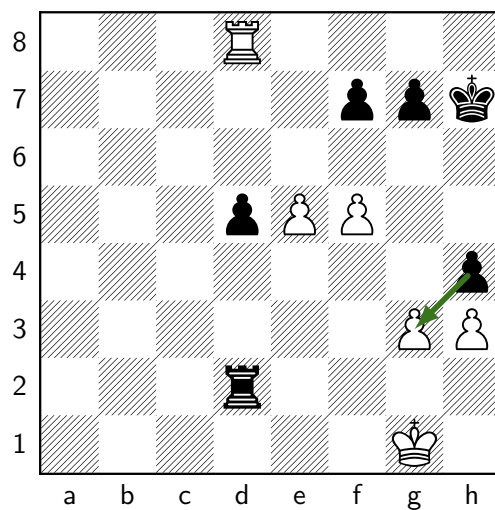


(d)

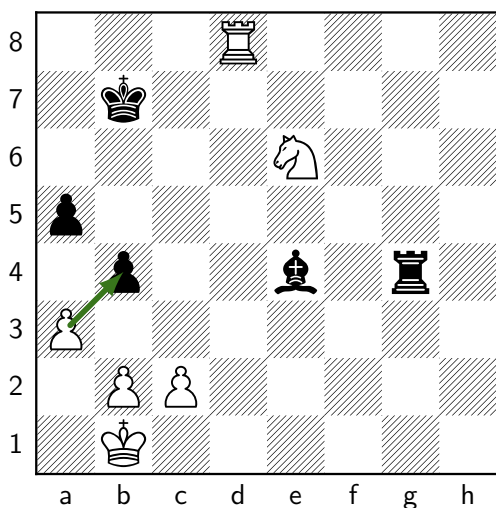
Figure 32: **Move 38, White: Rg1.** (a) Our position; Melanie plays Rg1 (a1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#274 (ply 85) [structural distance $d=25$, QAP obj=35]: the master played Be1; under μ our Ra1 plays the role of that game's Bc6 (a **cross-type** correspondence); transferred move a1g1 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#3367 (ply 81) [structural distance $d=17$, QAP obj=26]: the master played Rb8; under μ our Ra1 plays the role of that game's Rd1; transferred move a1g1 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2910 (ply 78) [structural distance $d=23$, QAP obj=15]: the master played Ra2; under μ our Ra1 plays the role of that game's Ra3; transferred move a1g1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rg1 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



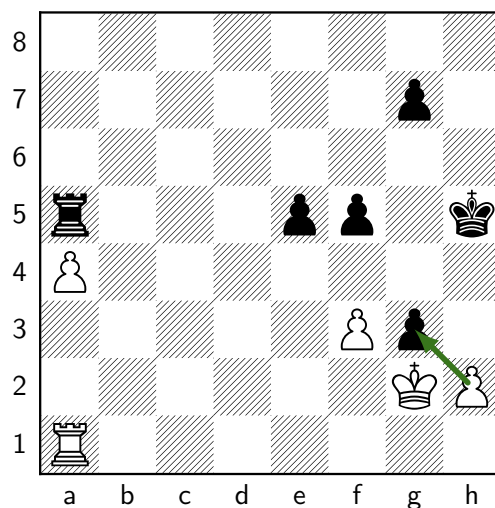
(a)



(b)

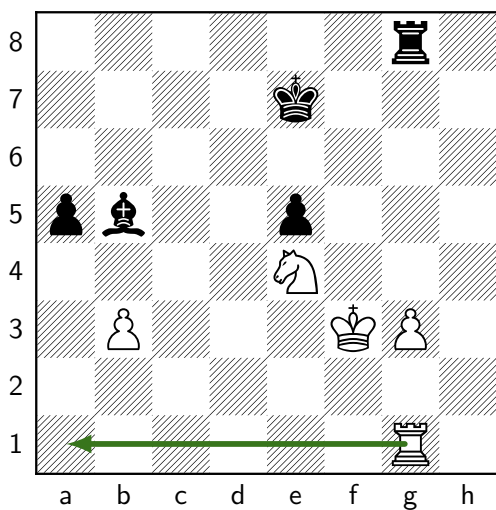


(c)

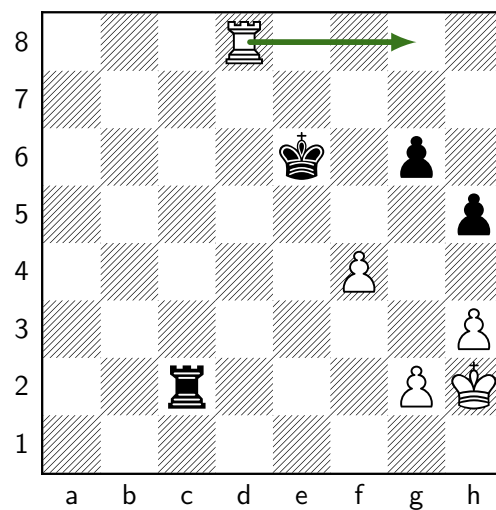


(d)

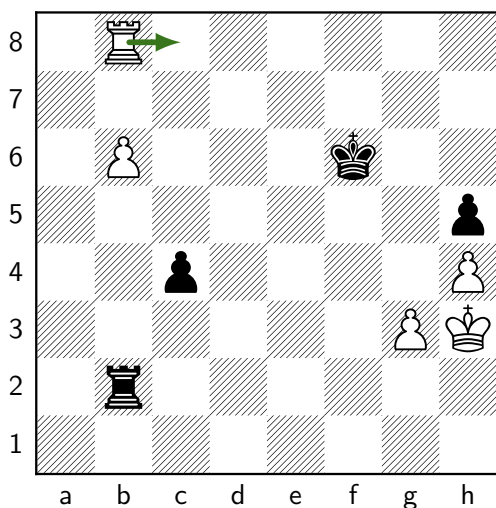
Figure 33: **Move 39, White: axb5.** (a) Our position; Melanie plays axb5 (a4b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#1273 (ply 79) [structural distance $d=28$, QAP obj=53]: the master played hxg3; under μ our Pa4 plays the role of that game’s Ph5; cross-type roles: our Ne4 as their Pg7, our bd7 as their pf5; transferred move a4b5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1315 (ply 70) [structural distance $d=27$, QAP obj=52]: the master played axb4; under μ our Pa4 plays the role of that game’s Pa3; cross-type roles: our rg8 as their be4, our pe5 as their rg4; transferred move a4b5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3534 (ply 88) [structural distance $d=28$, QAP obj=47]: the master played hxg3; under μ our Pa4 plays the role of that game’s Ph2; cross-type roles: our bd7 as their Ra1, our Ne4 as their Pf3; transferred move a4b5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to axb5 (summed QAP objective 396, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



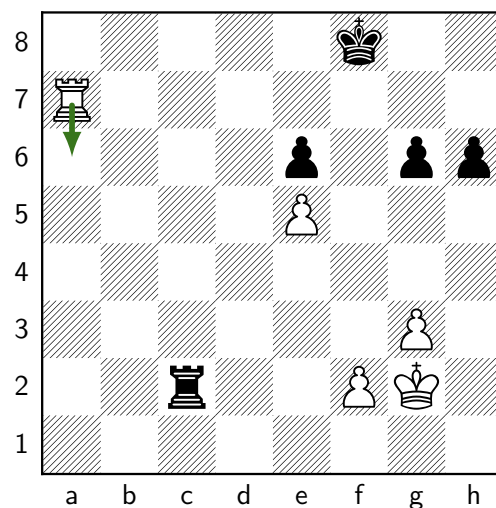
(a)



(b)

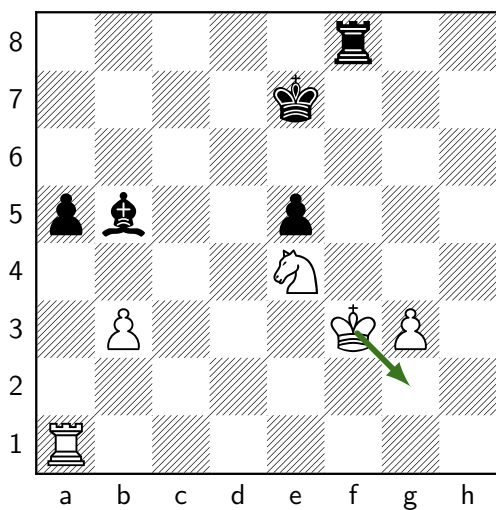


(c)

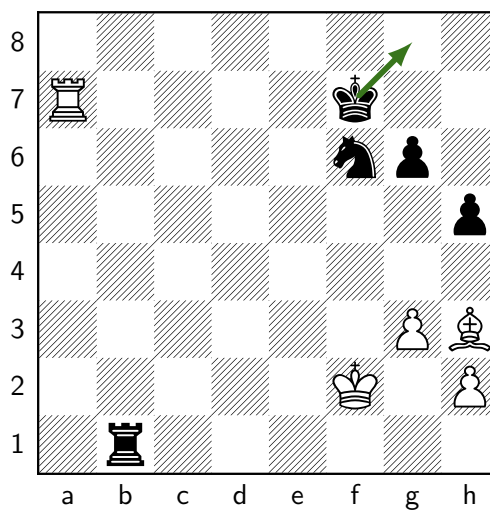


(d)

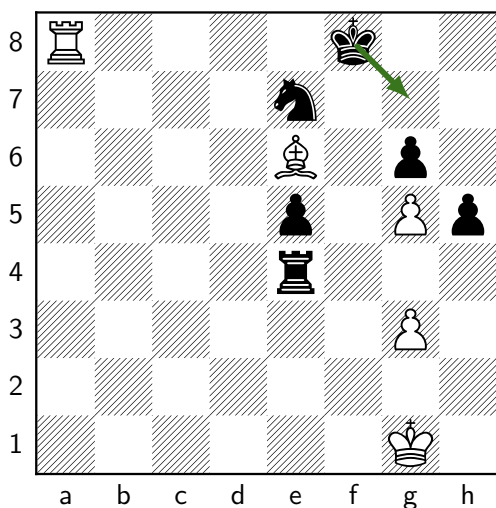
Figure 34: **Move 40, White: Ra1 (g1a1)**. (a) Our position; Melanie plays Ra1 (g1a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#133` (ply 66) [structural distance $d=19$, QAP obj=35]: the master played Rg8; under μ our Rg1 plays the role of that game’s Rd8; cross-type roles: our Ne4 as their Ph3; transferred move g1a1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2685` (ply 80) [structural distance $d=15$, QAP obj=34]: the master played Rc8; under μ our Rg1 plays the role of that game’s Rb8; cross-type roles: our bb5 as their Pg3, our Ne4 as their Ph4; transferred move g1a1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2965` (ply 80) [structural distance $d=20$, QAP obj=27]: the master played Ra6; under μ our Rg1 plays the role of that game’s Ra7; cross-type roles: our Ne4 as their Pg3, our bb5 as their pg6; transferred move g1a1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ra1 (summed QAP objective 145, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



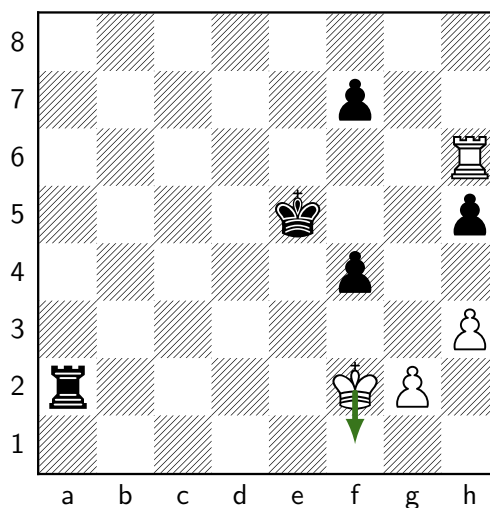
(a)



(b)

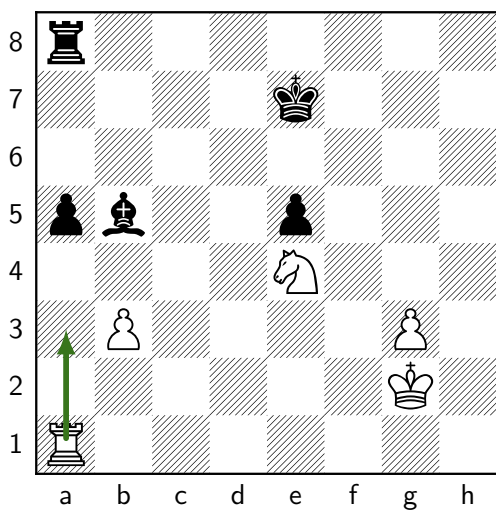


(c)

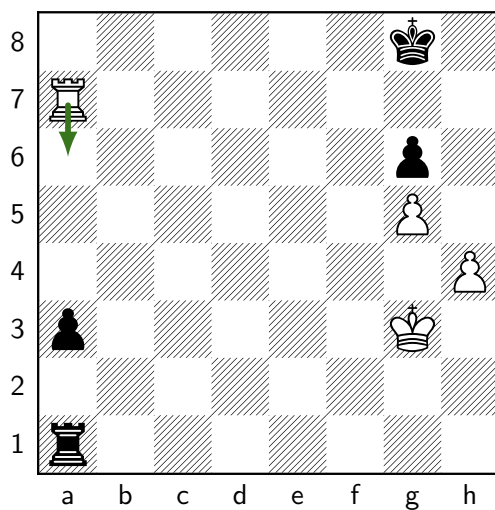


(d)

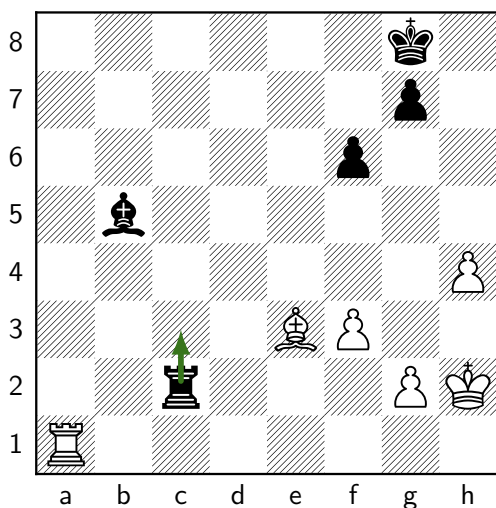
Figure 35: **Move 41, White: Kg2.** (a) Our position; Melanie plays Kg2 (f3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#696` (ply 85) [structural distance $d=17$, QAP obj=73]: the master played Kg8; under μ our Kf3 plays the role of that game’s Kf2; cross-type roles: our pe5 as their bh3, our bb5 as their ph2; transferred move f3g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2624` (ply 79) [structural distance $d=21$, QAP obj=71]: the master played Kg7; under μ our Kf3 plays the role of that game’s Kf1; cross-type roles: our pe5 as their be6, our bb5 as their Pe5; transferred move f3g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#841` (ply 74) [structural distance $d=20$, QAP obj=69]: the master played Kf1; under μ our Kf3 plays the role of that game’s Kf2; cross-type roles: our bb5 as their ph5; transferred move f3g2 (**endorses this move**). Synthesis: 22 retrieved analogs transferred to Kg2 (summed QAP objective 1077, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



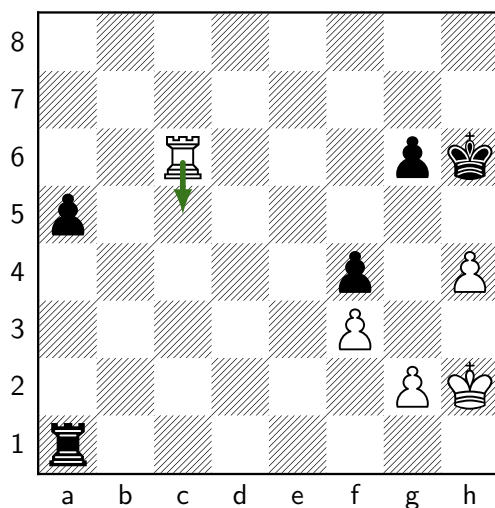
(a)



(b)

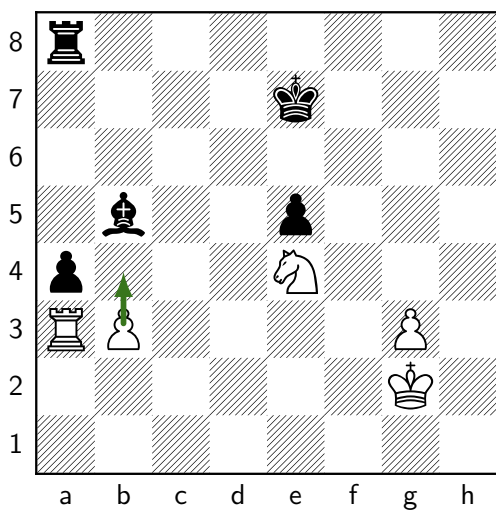


(c)

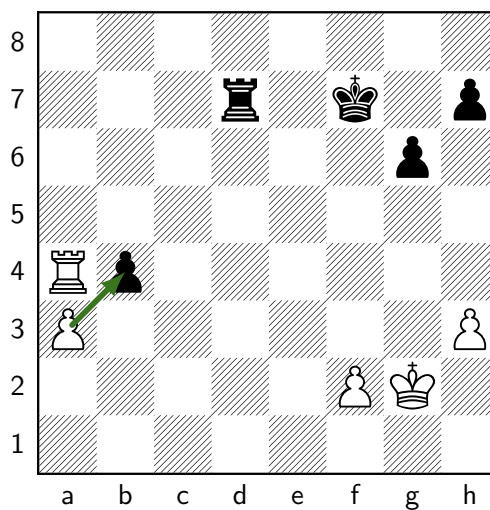


(d)

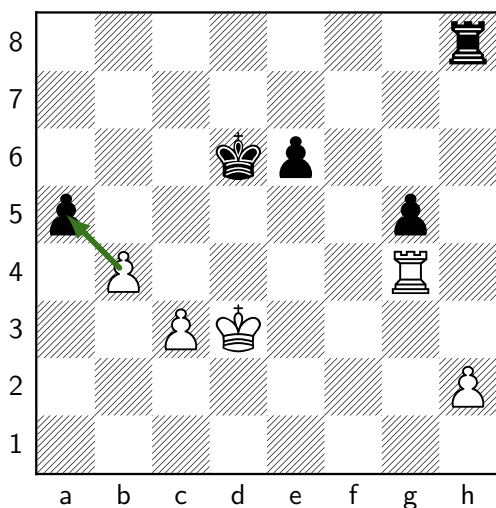
Figure 36: **Move 42, White: Ra3.** (a) Our position; Melanie plays Ra3 (a1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3059` (ply 88) [structural distance $d=13$, QAP obj=44]: the master played Ra6; under μ our Ra1 plays the role of that game’s Ra7; cross-type roles: our Ne4 as their Pg5; transferred move a1a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1054` (ply 99) [structural distance $d=16$, QAP obj=41]: the master played Rc3; under μ our Ra1 plays the role of that game’s Rc7; cross-type roles: our Pb3 as their Bb5, our Ne4 as their ph4; transferred move a1a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1024` (ply 96) [structural distance $d=16$, QAP obj=40]: the master played Rc5; under μ our Ra1 plays the role of that game’s Rc6; cross-type roles: our Ne4 as their pf4, our bb5 as their pa5; transferred move a1a3 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Ra3 (summed QAP objective 456, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



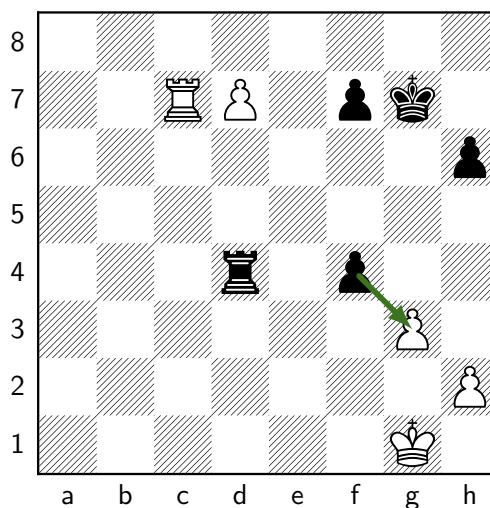
(a)



(b)

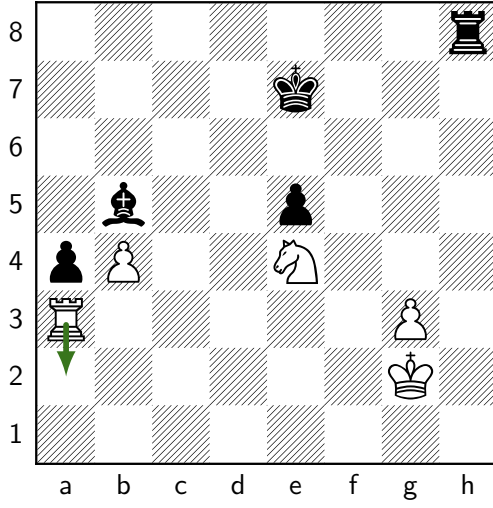


(c)

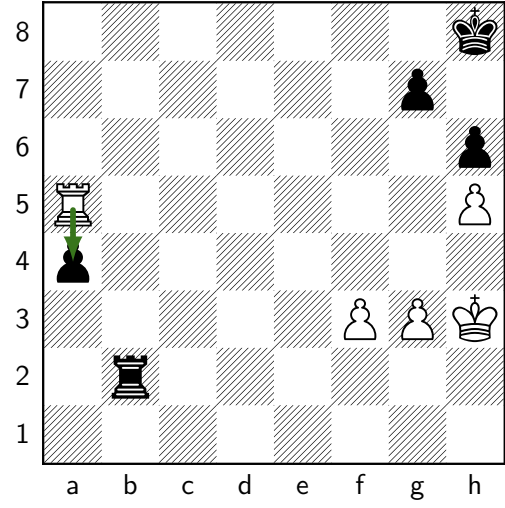


(d)

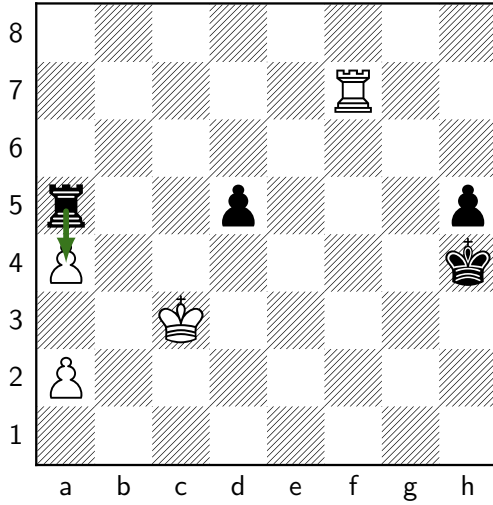
Figure 37: **Move 43, White: b4.** (a) Our position; Melanie plays b4 (b3b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#82` (ply 76) [structural distance $d=25$, QAP obj=83]: the master played `axb4`; under μ our *Pb3* plays the role of that game’s *Pa3*; cross-type roles: our *ra8* as their *Ph3*, our *Ne4* as their *pg6*; transferred move `b3b4` (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2587` (ply 78) [structural distance $d=23$, QAP obj=52]: the master played `bxa5`; under μ our *Pb3* plays the role of that game’s *Pb4*; cross-type roles: our *bb5* as their *pe6*, our *Ne4* as their *rh8*; transferred move `b3b4` (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3468` (ply 89) [structural distance $d=21$, QAP obj=45]: the master played `fxg3`; under μ our *Pb3* plays the role of that game’s *Pf5*; cross-type roles: our *bb5* as their *rc7*, our *Ne4* as their *Ph6*; transferred move `b3b4` (**endorses this move**). Synthesis: 10 retrieved analogs transferred to b4 (summed QAP objective 404, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



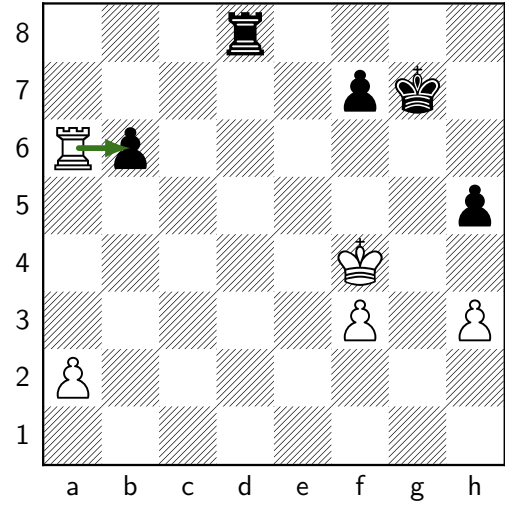
(a)



(b)

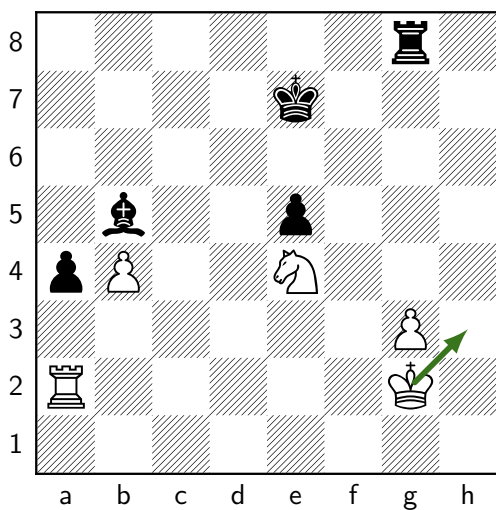


(c)

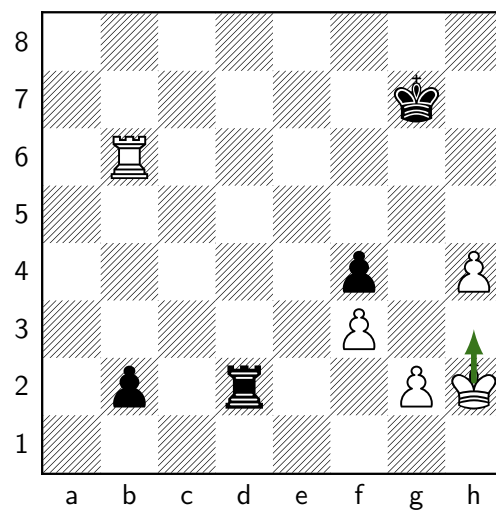


(d)

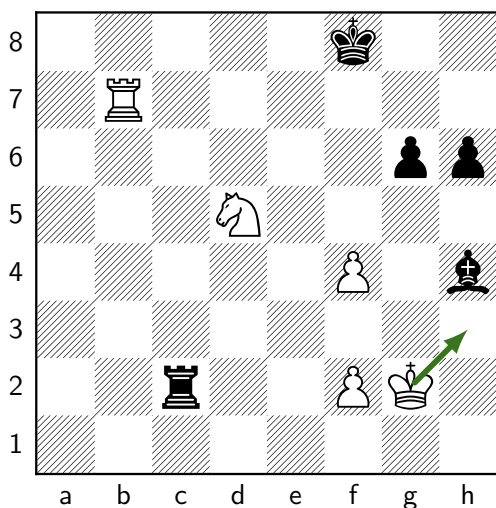
Figure 38: **Move 44, White: Ra2.** (a) Our position; Melanie plays Ra2 (a3a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3003` (ply 94) [structural distance $d=17$, QAP obj=20]: the master played `Rxa4`; under μ our *Ra3* plays the role of that game’s *Ra5*; cross-type roles: our *bb5* as their *ph6*, our *Ne4* as their *pg7*; transferred move `a3a2` (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1132` (ply 91) [structural distance $d=19$, QAP obj=20]: the master played `Rxa4`; under μ our *Ra3* plays the role of that game’s *Ra4*; cross-type roles: our *Pb4* as their *rf7*, our *Ne4* as their *Pd5*; transferred move `a3a2` (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3143` (ply 90) [structural distance $d=18$, QAP obj=20]: the master played `Rxb6`; under μ our *Ra3* plays the role of that game’s *Ra6*; cross-type roles: our *rh8* as their *Pf3*, our *Ne4* as their *Ph3*; transferred move `a3a2` (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ra2 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



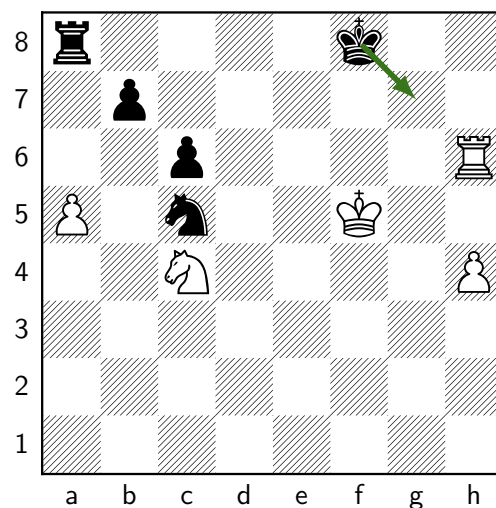
(a)



(b)

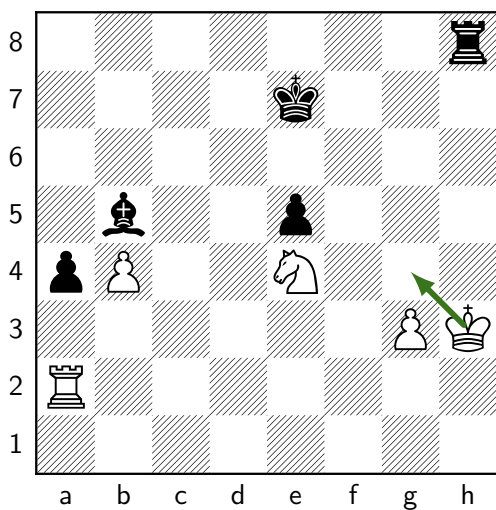


(c)

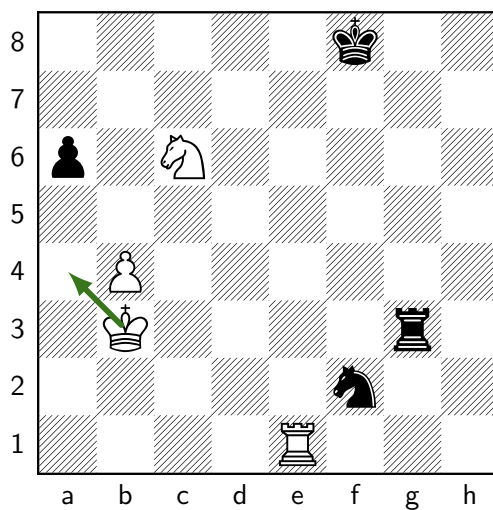


(d)

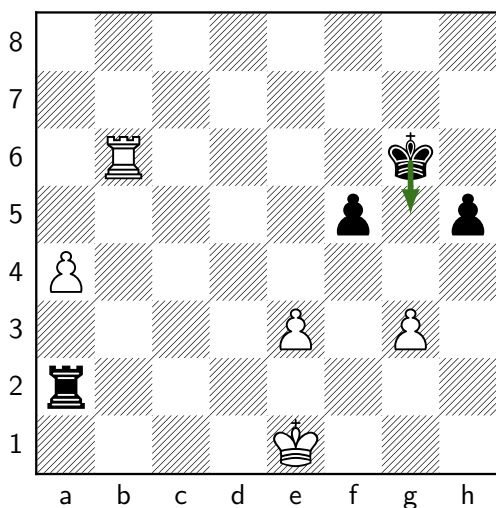
Figure 39: **Move 45, White: Kh3.** (a) Our position; Melanie plays Kh3 (**g2h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1056` (ply 92) [structural distance $d=19$, QAP obj=32]: the master played Kh3; under μ our Kg2 plays the role of that game’s Kh2; cross-type roles: our Ne4 as their Pf3; transferred move g2h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2995` (ply 88) [structural distance $d=16$, QAP obj=27]: the master played Kh3; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Ra2 as their bh4, our bb5 as their Nd5; transferred move g2h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1848` (ply 95) [structural distance $d=18$, QAP obj=25]: the master played Kg7; under μ our Kg2 plays the role of that game’s Kf1; cross-type roles: our Ne4 as their Pb7, our pe5 as their Nc5; transferred move g2h3 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kh3 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



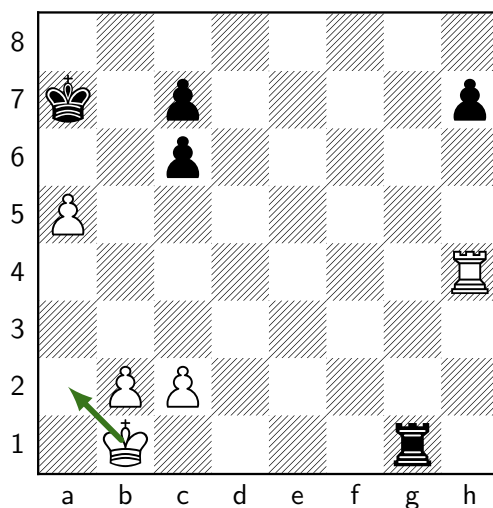
(a)



(b)

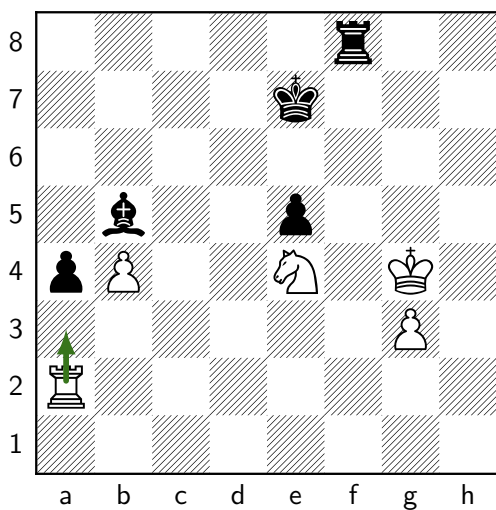


(c)

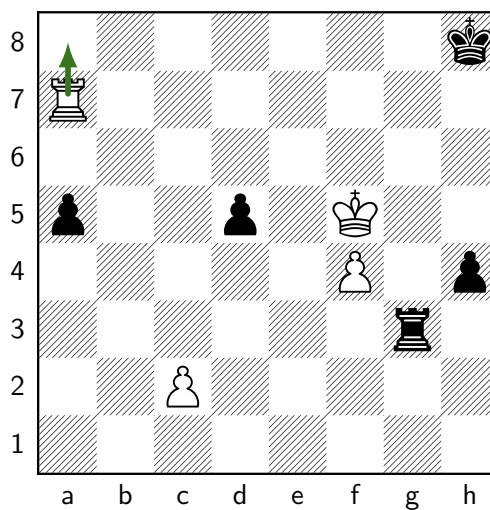


(d)

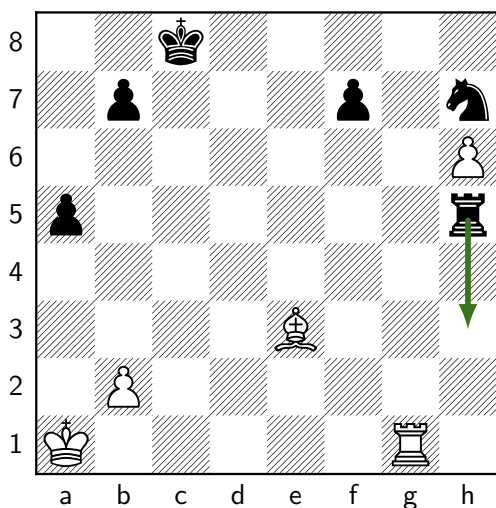
Figure 40: **Move 46, White: Kg4.** (a) Our position; Melanie plays Kg4 (h3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2807` (ply 82) [structural distance $d=17$, QAP obj=74]: the master played Ka4; under μ our Kh3 plays the role of that game’s Kb3; cross-type roles: our pa4 as their nf2; transferred move h3g4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#111` (ply 107) [structural distance $d=14$, QAP obj=73]: the master played Kg5; under μ our Kh3 plays the role of that game’s Kg3; cross-type roles: our Ra2 as their pa4, our Ne4 as their Ra2; transferred move h3g4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#449` (ply 86) [structural distance $d=14$, QAP obj=73]: the master played Ka2; under μ our Kh3 plays the role of that game’s Kb1; cross-type roles: our Ne4 as their Rh4, our Ra2 as their Pa5; transferred move h3g4 (**endorses this move**). Synthesis: 32 retrieved analogs transferred to Kg4 (summed QAP objective 1540, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



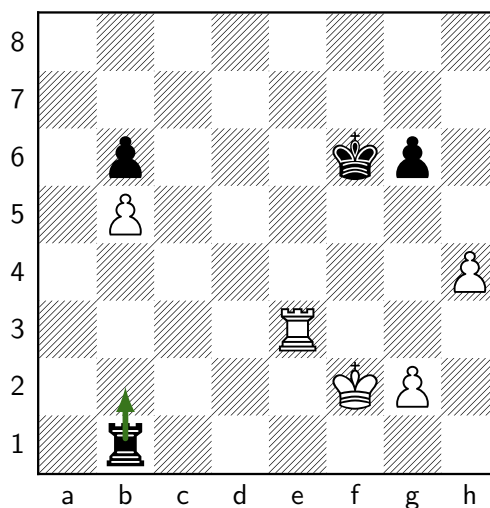
(a)



(b)

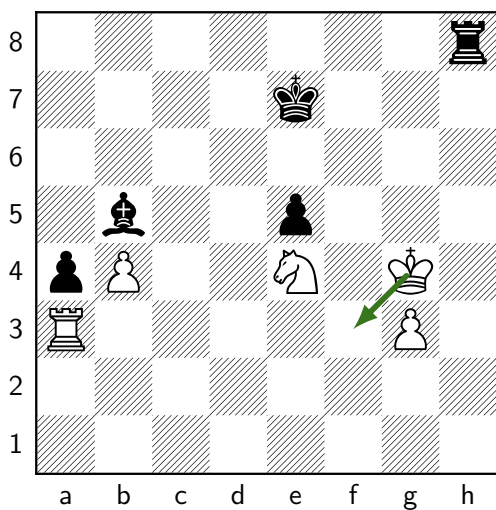


(c)

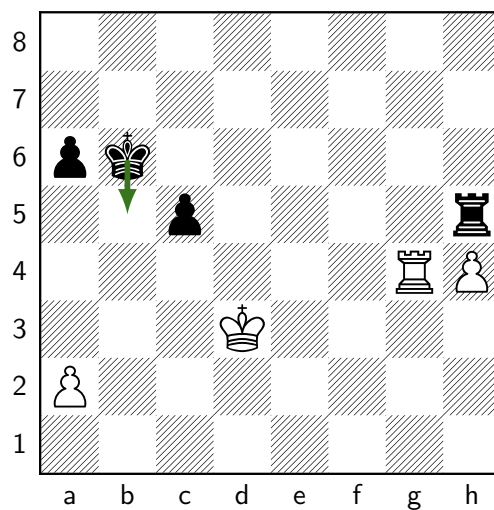


(d)

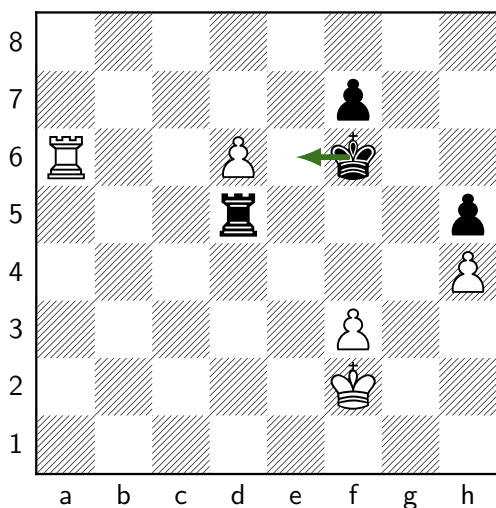
Figure 41: **Move 47, White: Ra3.** (a) Our position; Melanie plays Ra3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3948` (ply 92) [structural distance $d=20$, QAP obj=20]: the master played Ra8+; under μ our Ra2 plays the role of that game’s Ra7; cross-type roles: our Ne4 as their ph4; transferred move a2a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#236` (ply 97) [structural distance $d=19$, QAP obj=12]: the master played Rh3; under μ our Ra2 plays the role of that game’s Rh4; cross-type roles: our pe5 as their Nh7, our Ne4 as their pb2; transferred move a2a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1104` (ply 107) [structural distance $d=17$, QAP obj=12]: the master played Rb2+; under μ our Ra2 plays the role of that game’s Rb8; cross-type roles: our bb5 as their Pb6, our Ne4 as their pg2; transferred move a2a3 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Ra3 (summed QAP objective 83, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



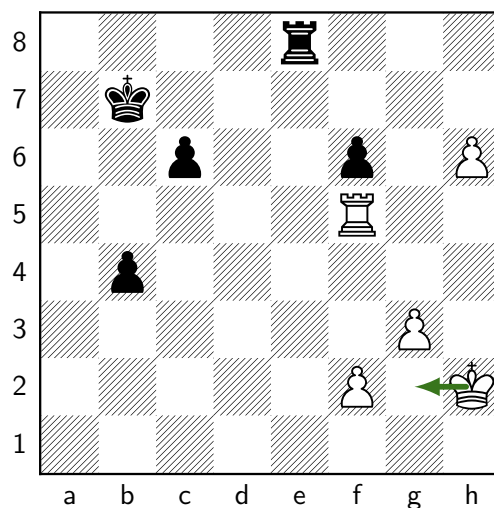
(a)



(b)

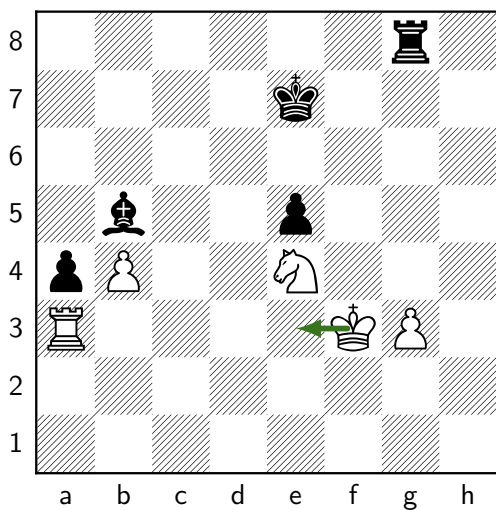


(c)

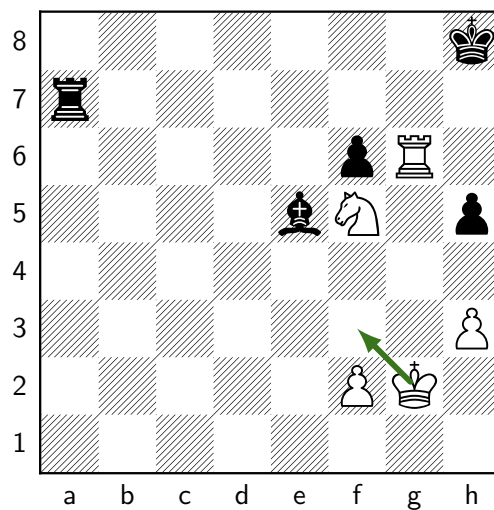


(d)

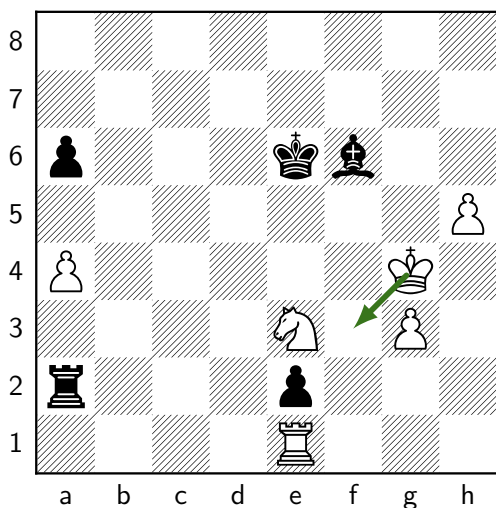
Figure 42: **Move 48, White: Kf3.** (a) Our position; Melanie plays Kf3 (**g4f3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3772` (ply 107) [structural distance $d=20$, QAP obj=23]: the master played Kb5; under μ our Kg4 plays the role of that game’s Kb3; cross-type roles: our bb5 as their rg4; transferred move g4f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#360` (ply 103) [structural distance $d=19$, QAP obj=22]: the master played Ke6; under μ our Kg4 plays the role of that game’s Kf3; cross-type roles: our bb5 as their ra6, our Ne4 as their Ph5; transferred move g4f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#186` (ply 90) [structural distance $d=17$, QAP obj=20]: the master played Kg2; under μ our Kg4 plays the role of that game’s Kh2; cross-type roles: our Ne4 as their pb4, our bb5 as their pc6; transferred move g4f3 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kf3 (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



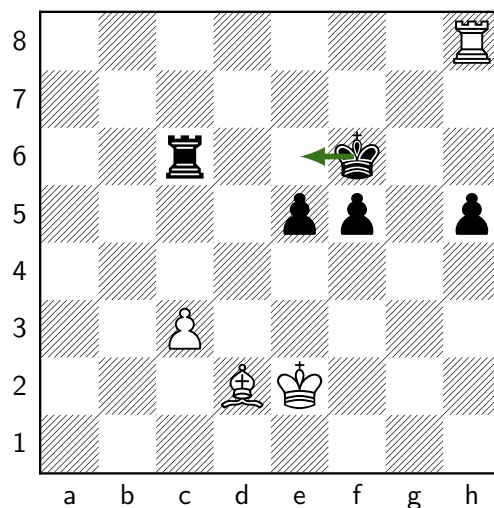
(a)



(b)

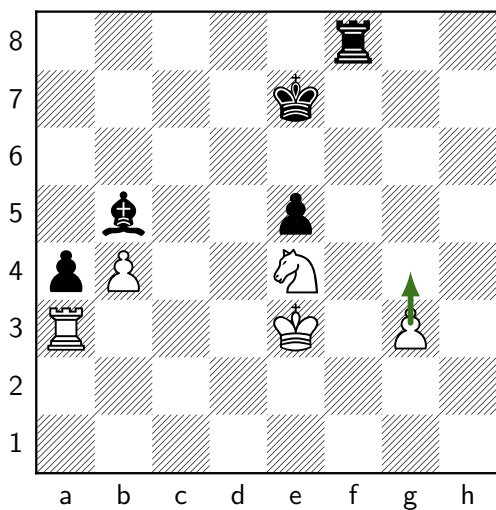


(c)

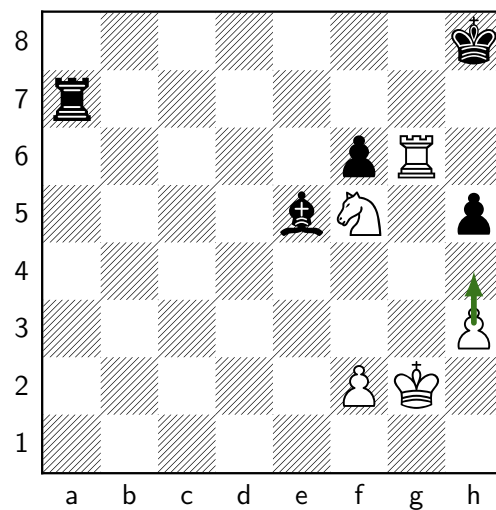


(d)

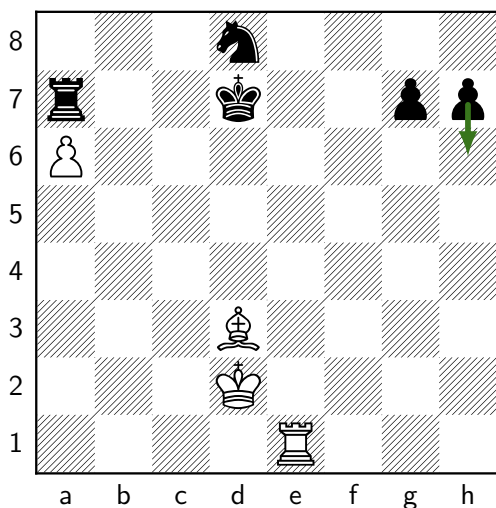
Figure 43: **Move 49, White: Ke3**. (a) Our position; Melanie plays Ke3 (f3e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#718` (ply 108) [structural distance $d=15$, QAP obj=24]: the master played Kf3; under μ our Kf3 plays the role of that game’s Kg2; transferred move f3e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#169` (ply 92) [structural distance $d=19$, QAP obj=20]: the master played Kf3; under μ our Kf3 plays the role of that game’s Kg4; cross-type roles: our bb5 as their ra2, our rg8 as their Ph5; transferred move f3e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1072` (ply 97) [structural distance $d=20$, QAP obj=9]: the master played Ke6; under μ our Kf3 plays the role of that game’s Kf3; cross-type roles: our Pg3 as their rh8, our Ne4 as their Pf5; transferred move f3e3 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Ke3 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



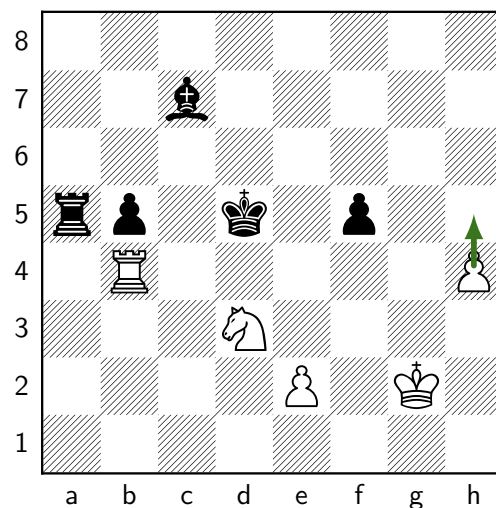
(a)



(b)

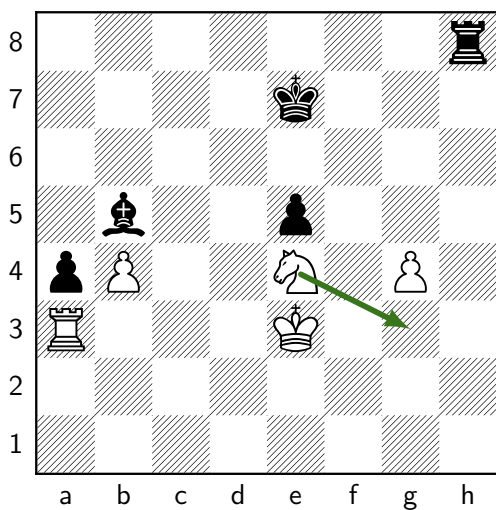


(c)

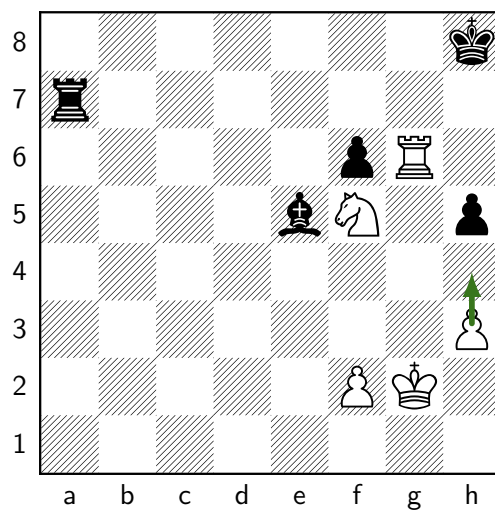


(d)

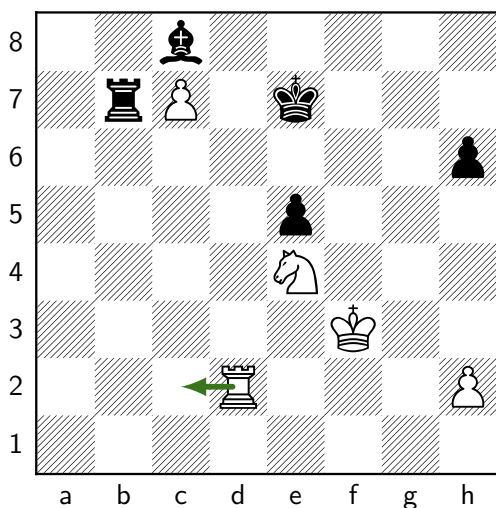
Figure 44: **Move 50, White: g4.** (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#718` (ply 112) [structural distance $d=19$, QAP obj=25]: the master played **h4**; under μ our *Pg3* plays the role of that game’s *Ph3*; transferred move **g3g4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2702` (ply 107) [structural distance $d=17$, QAP obj=23]: the master played **h6**; under μ our *Pg3* plays the role of that game’s *Ph2*; transferred move **g3g4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2796` (ply 112) [structural distance $d=19$, QAP obj=20]: the master played **h5**; under μ our *Pg3* plays the role of that game’s *Ph4*; cross-type roles: our *bb5* as their *ra5*, our *rf8* as their *bc7*; transferred move **g3g4** (**endorses this move**). Synthesis: 11 retrieved analogs transferred to **g4** (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



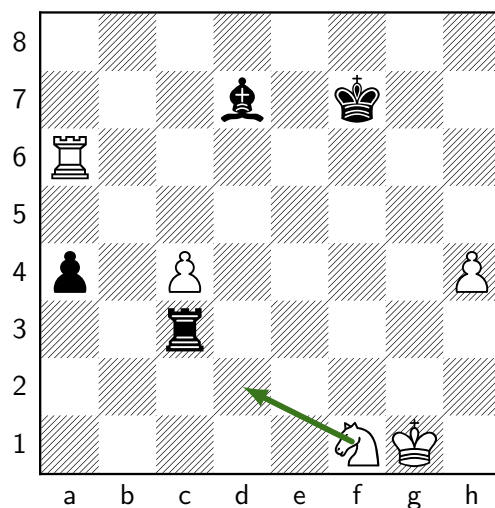
(a)



(b)

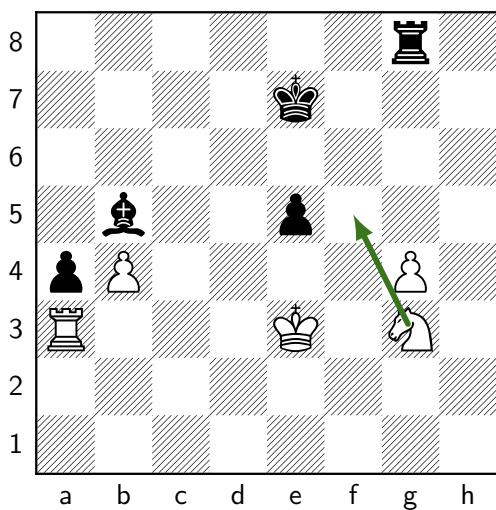


(c)

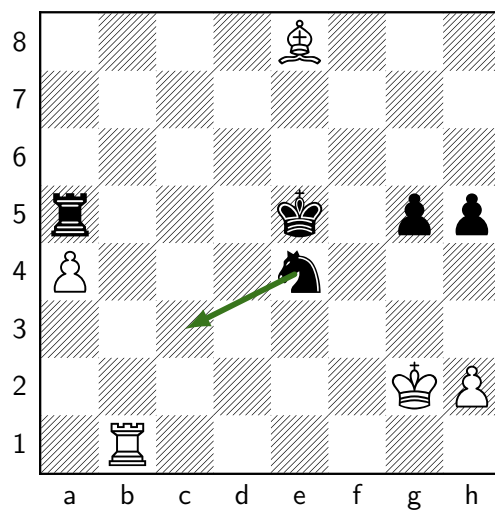


(d)

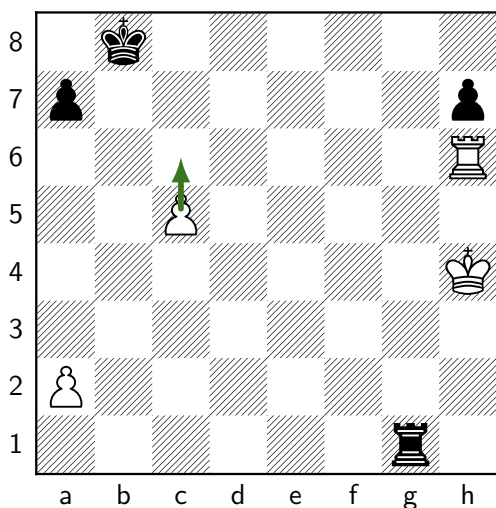
Figure 45: **Move 51, White: Ng3.** (a) Our position; Melanie plays Ng3 (e4g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#718` (ply 112) [structural distance $d=19$, QAP obj=24]: the master played h4; under μ our Ne4 plays the role of that game’s Ph3 (a **cross-type** correspondence); cross-type roles: our Pg4 as their Nf5; transferred move e4g3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1156` (ply 88) [structural distance $d=18$, QAP obj=19]: the master played Rc2; under μ our Ne4 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our rh8 as their Ne4; transferred move e4g3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3987` (ply 114) [structural distance $d=17$, QAP obj=19]: the master played Nd2; under μ our Ne4 plays the role of that game’s Nf1; transferred move e4g3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ng3 (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



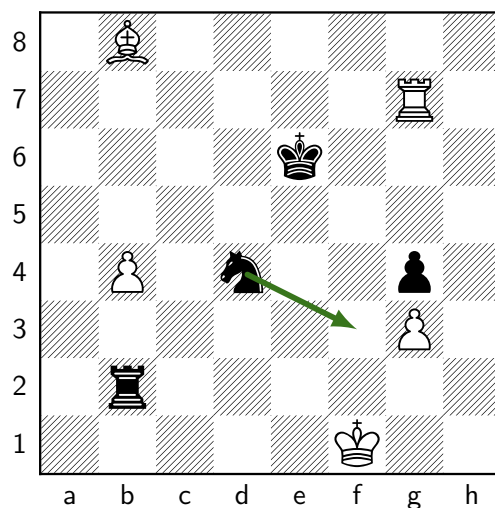
(a)



(b)

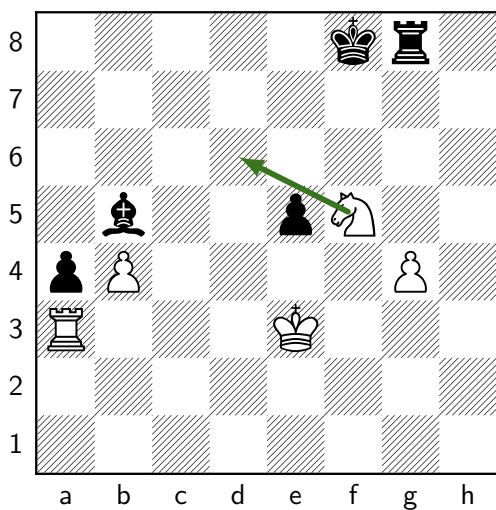


(c)

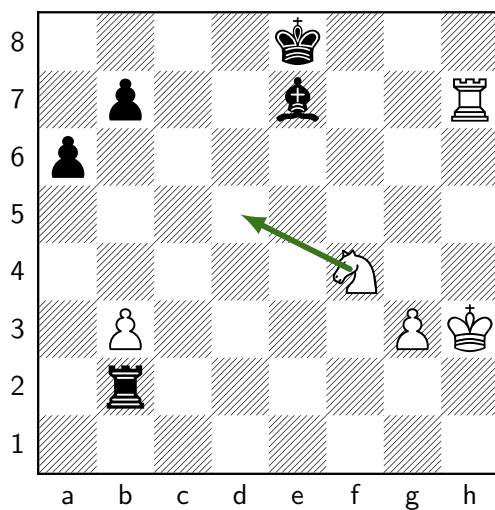


(d)

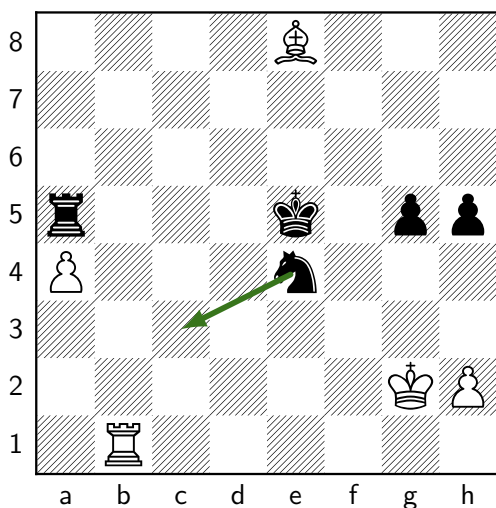
Figure 46: **Move 52, White: Nf5+.** (a) Our position; Melanie plays Nf5+ (g3f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2238` (ply 113) [structural distance $d=20$, QAP obj=27]: the master played Nc3; under μ our Ng3 plays the role of that game’s Ne5; cross-type roles: our rg8 as their be8, our bb5 as their rb1; transferred move g3f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2211` (ply 90) [structural distance $d=20$, QAP obj=21]: the master played c6; under μ our Ng3 plays the role of that game’s Pc5 (a **cross-type** correspondence); cross-type roles: our pe5 as their rg1, our bb5 as their pa7; transferred move g3f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2497` (ply 111) [structural distance $d=15$, QAP obj=21]: the master played Nf3; under μ our Ng3 plays the role of that game’s Nd5; cross-type roles: our pe5 as their bb8, our bb5 as their pg3; transferred move g3f5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nf5+ (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



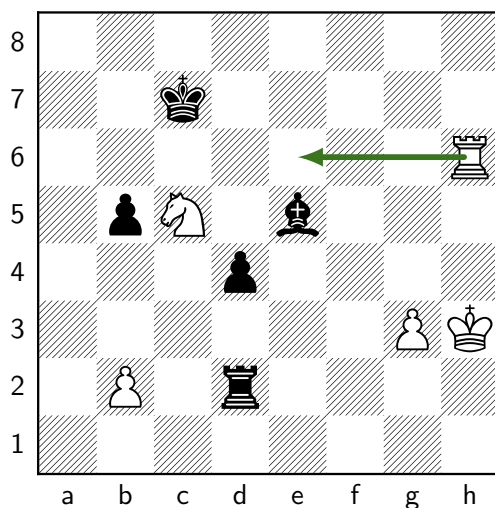
(a)



(b)

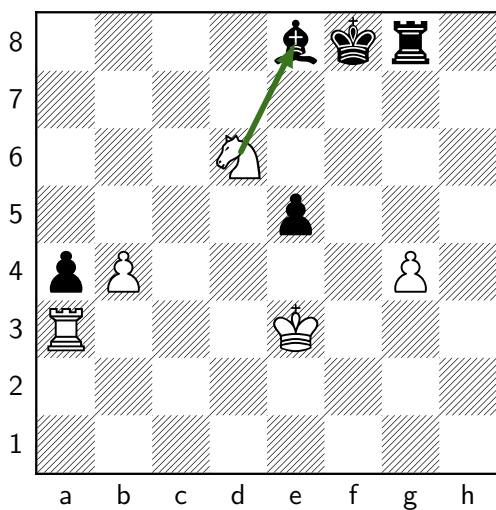


(c)

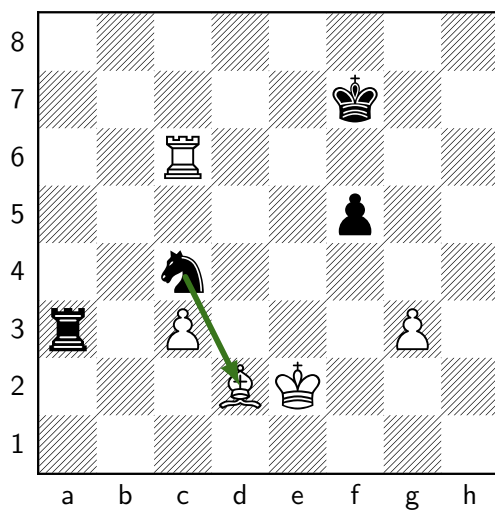


(d)

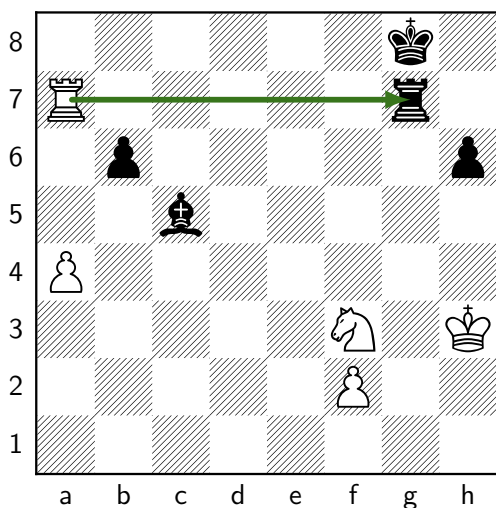
Figure 47: **Move 53, White: Nd6.** (a) Our position; Melanie plays Nd6 (f5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3756` (ply 106) [structural distance $d=25$, QAP obj=30]: the master played Nd5; under μ our Nf5 plays the role of that game’s Nf4; cross-type roles: our bb5 as their pa6, our pa4 as their be7; transferred move f5d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2238` (ply 113) [structural distance $d=23$, QAP obj=26]: the master played Nc3; under μ our Nf5 plays the role of that game’s Ne5; cross-type roles: our rg8 as their be8, our bb5 as their rb1; transferred move f5d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3233` (ply 90) [structural distance $d=24$, QAP obj=17]: the master played Re6; under μ our Nf5 plays the role of that game’s Rh6 (a **cross-type** correspondence); cross-type roles: our bb5 as their Nc5, our rg8 as their be5; transferred move f5d6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nd6 (summed QAP objective 96, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



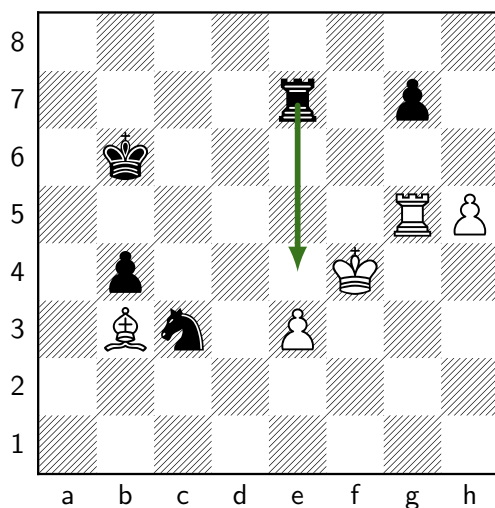
(a)



(b)



(c)



(d)

Figure 48: **Move 54, White: Nxe8**. (a) Our position; Melanie plays **Nxe8** (d6e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#859` (ply 105) [structural distance $d=25$, QAP obj=27]: the master played **Nxd2**; under μ our *Nd6* plays the role of that game’s *Nc5*; cross-type roles: our *pe5* as their *rc6*, our *rg8* as their *pg3*; transferred move **d6e8** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#971` (ply 90) [structural distance $d=21$, QAP obj=16]: the master played **Rxg7+**; under μ our *Nd6* plays the role of that game’s *Ra7* (a **cross-type** correspondence); cross-type roles: our *Pg4* as their *Nf3*, our *Ra3* as their *bc5*; transferred move **d6e8** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2050` (ply 115) [structural distance $d=18$, QAP obj=8]: the master played **Re4+**; under μ our *Nd6* plays the role of that game’s *Re2* (a **cross-type** correspondence); cross-type roles: our *Pg4* as their *Nc3*, our *rg8* as their *pe3*; transferred move **d6e8** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **Nxe8** (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

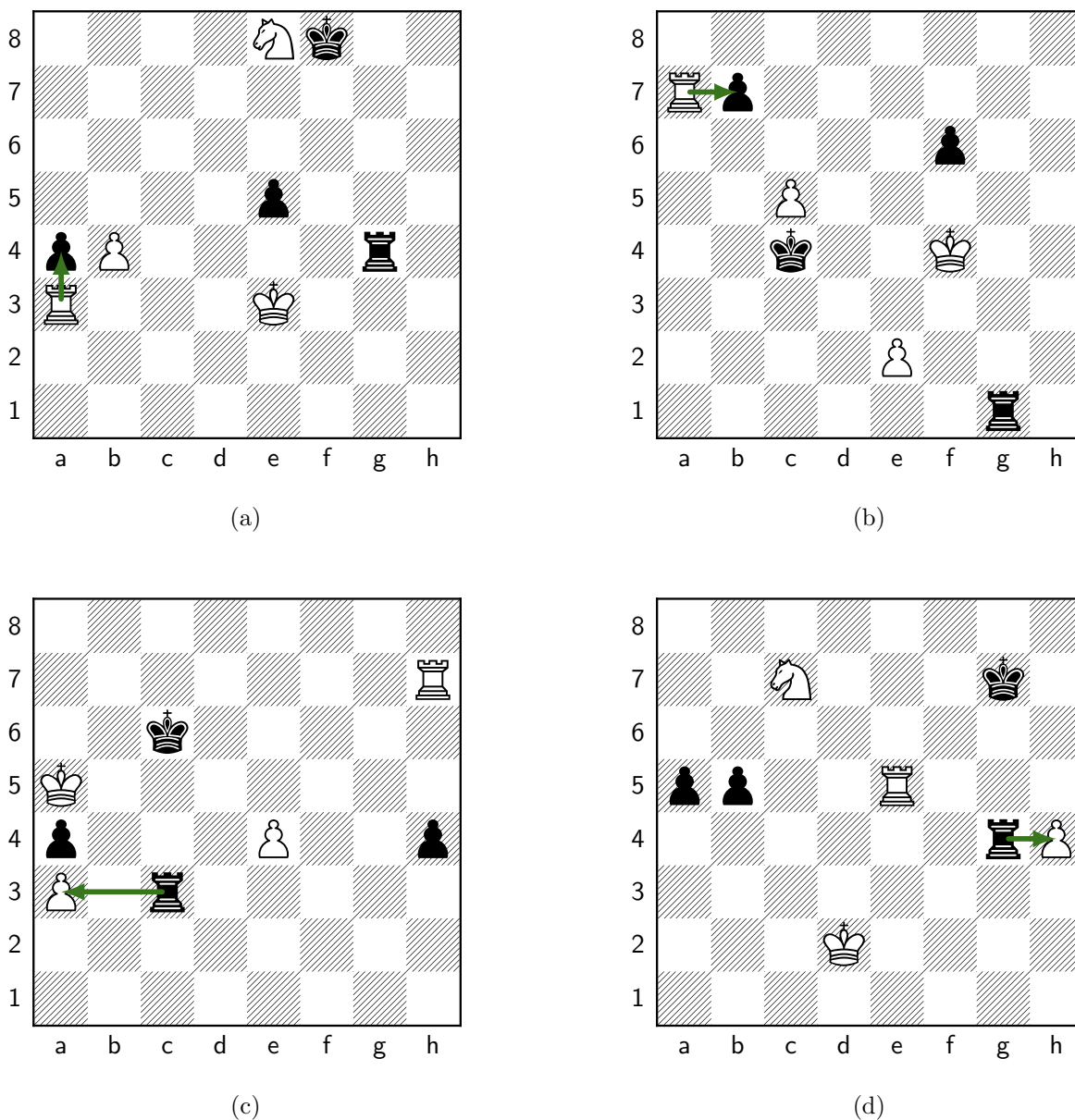
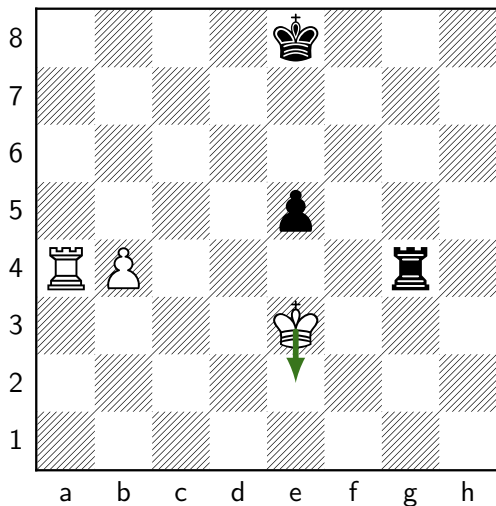
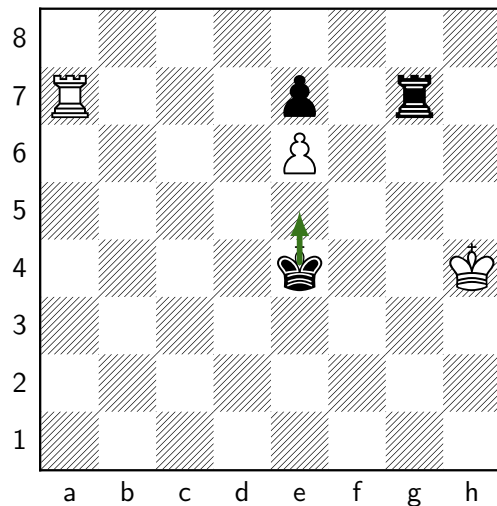


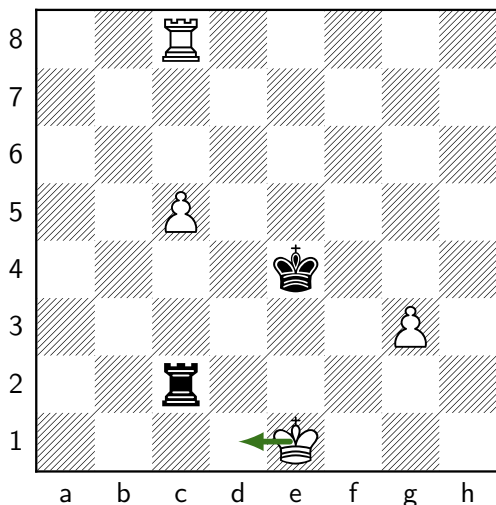
Figure 49: **Move 55, White: Rxa4**. (a) Our position; Melanie plays Rxa4 (a3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3180` (ply 100) [structural distance $d=11$, QAP obj=13]: the master played Rxb7; under μ our *Ra3* plays the role of that game’s *Ra7*; cross-type roles: our *Ne8* as their *Pc5*; transferred move a3a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1651` (ply 105) [structural distance $d=8$, QAP obj=9]: the master played Rxa3; under μ our *Ra3* plays the role of that game’s *Rc6*; cross-type roles: our *Ne8* as their *Pa4*; transferred move a3a4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2459` (ply 95) [structural distance $d=14$, QAP obj=9]: the master played Rxb4; under μ our *Ra3* plays the role of that game’s *Rg5*; cross-type roles: our *Pb4* as their *nc7*, our *Ne8* as their *Pb5*; transferred move a3a4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Rxa4 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



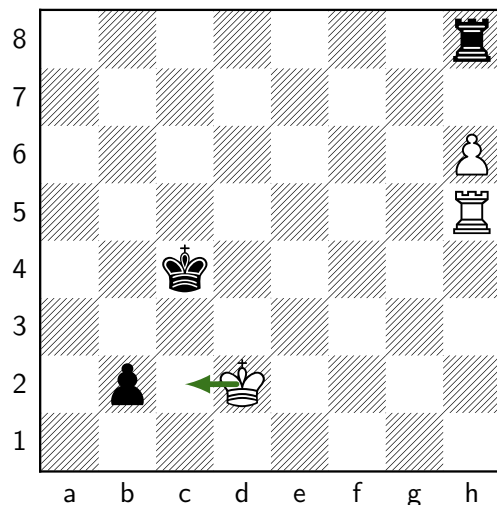
(a)



(b)



(c)



(d)

Figure 50: **Move 56, White: Ke2.** (a) Our position; Melanie plays Ke2 (e3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2066` (ply 115) [structural distance $d=5$, QAP obj=35]: the master played Ke5; under μ our Ke3 plays the role of that game's Ke5; transferred move e3e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3680` (ply 118) [structural distance $d=5$, QAP obj=28]: the master played Kd1; under μ our Ke3 plays the role of that game's Ke1; transferred move e3e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3999` (ply 104) [structural distance $d=2$, QAP obj=19]: the master played Kc2; under μ our Ke3 plays the role of that game's Kd2; transferred move e3e2 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to Ke2 (summed QAP objective 172, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

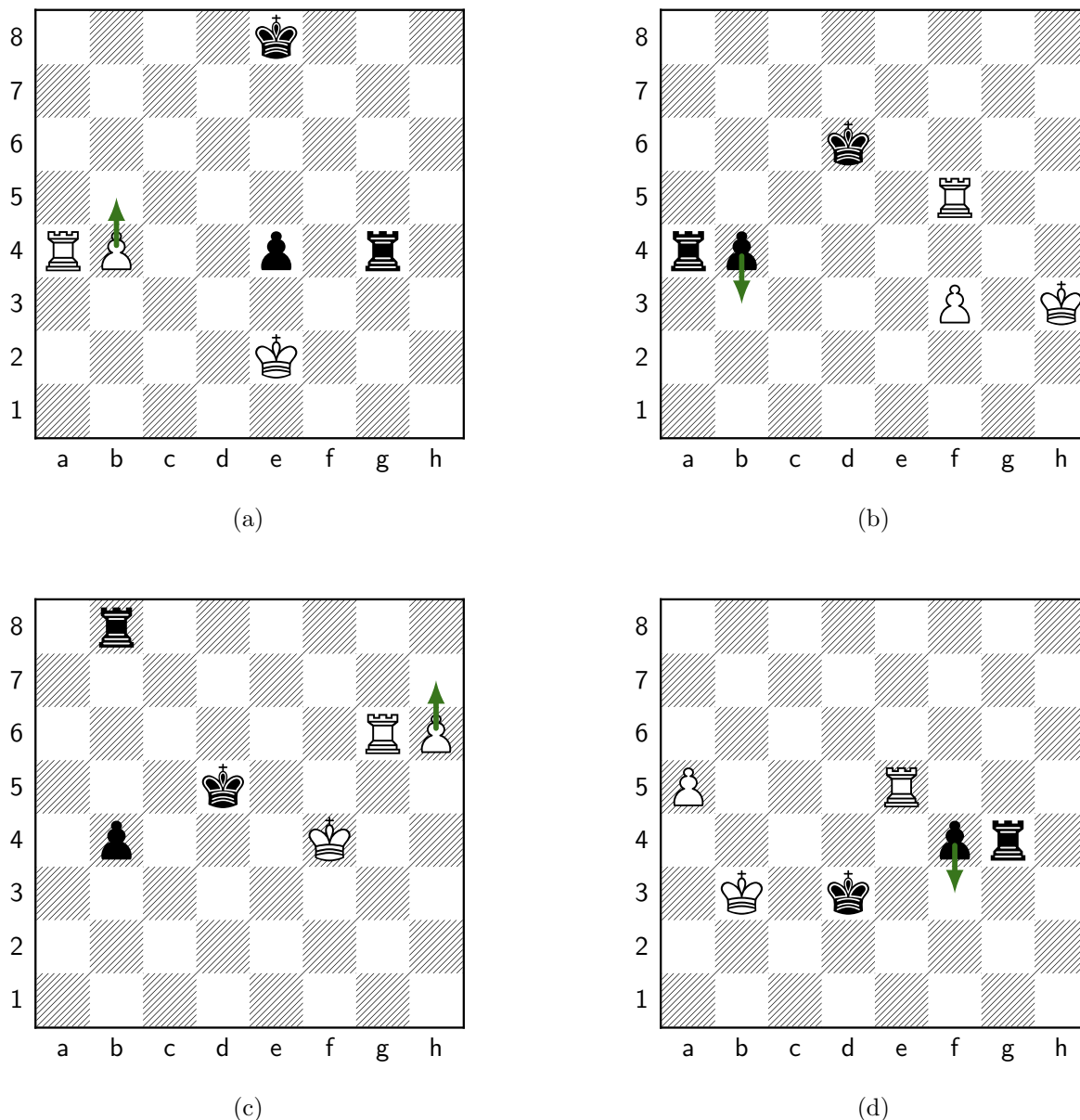
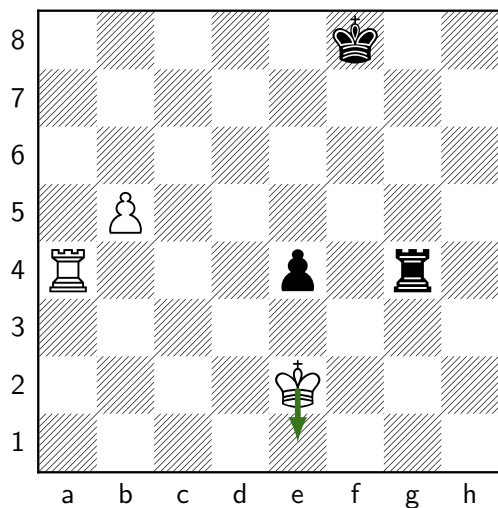
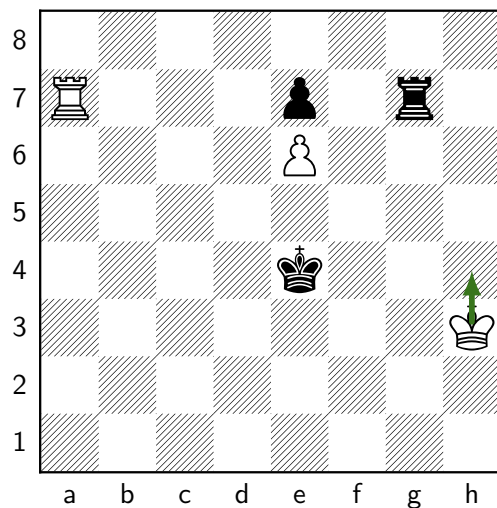


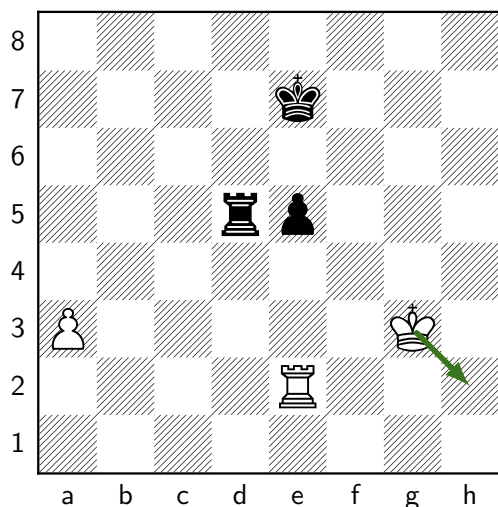
Figure 51: **Move 57, White: b5.** (a) Our position; Melanie plays **b5** (**b4b5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#72` (ply 109) [structural distance $d=2$, QAP obj=3]: the master played **b3**; under μ our Pb_4 plays the role of that game’s Pb_5 ; transferred move **b4b5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1769` (ply 104) [structural distance $d=2$, QAP obj=2]: the master played **h7**; under μ our Pb_4 plays the role of that game’s Ph_6 ; transferred move **b4b5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2934` (ply 129) [structural distance $d=2$, QAP obj=2]: the master played **f3**; under μ our Pb_4 plays the role of that game’s Pf_5 ; transferred move **b4b5** (**endorses this move**). Synthesis: 15 retrieved analogs transferred to **b5** (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



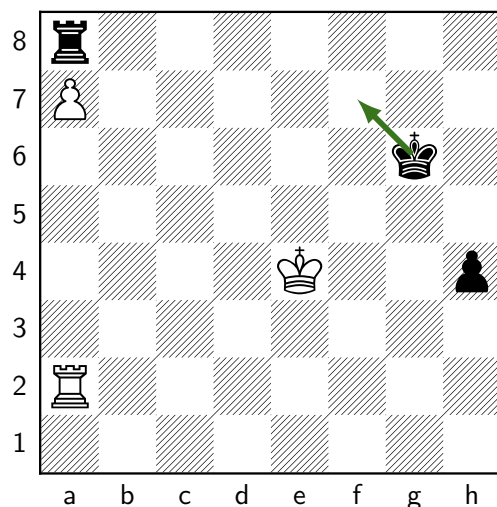
(a)



(b)

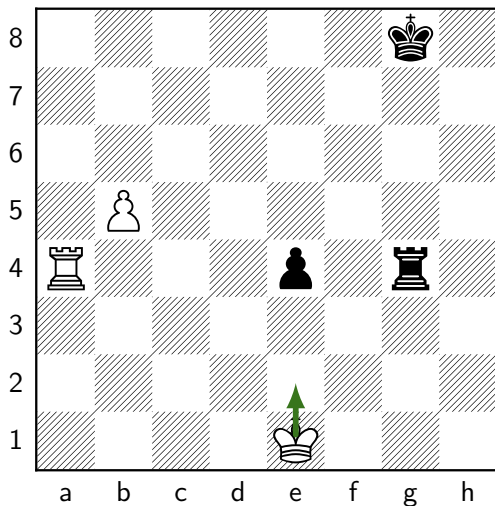


(c)

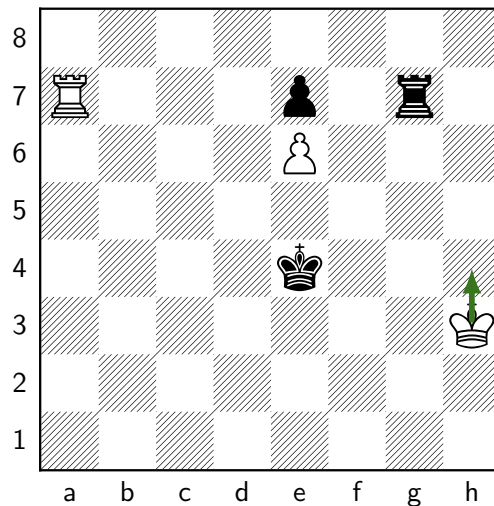


(d)

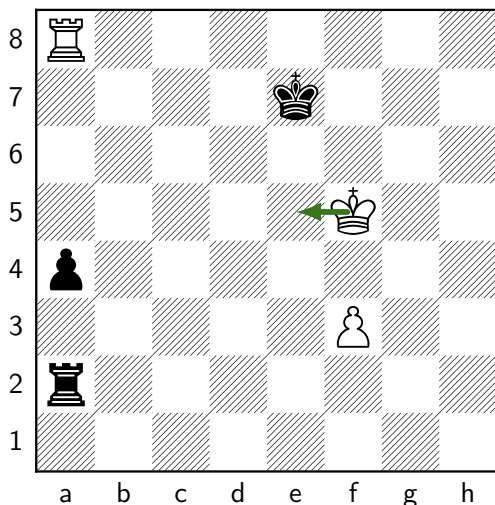
Figure 52: **Move 58, White: Ke1.** (a) Our position; Melanie plays **Ke1** (**e2e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2066` (ply 114) [structural distance $d=4$, QAP obj=45]: the master played **Kh4**; under μ our *Ke2* plays the role of that game's *Kh3*; transferred move **e2e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#513` (ply 104) [structural distance $d=1$, QAP obj=33]: the master played **Kh2**; under μ our *Ke2* plays the role of that game's *Kg3*; transferred move **e2e1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1460` (ply 117) [structural distance $d=2$, QAP obj=14]: the master played **Kf7**; under μ our *Ke2* plays the role of that game's *Kg3*; transferred move **e2e1** (**endorses this move**). Synthesis: 19 retrieved analogs transferred to **Ke1** (summed QAP objective 274, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



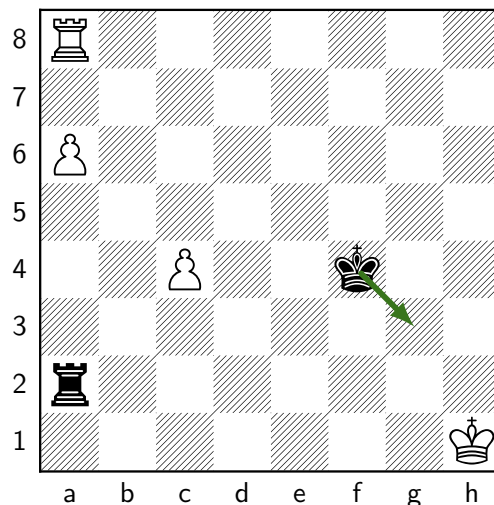
(a)



(b)

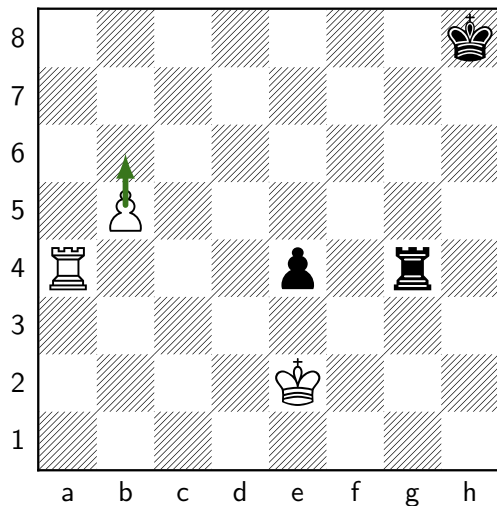


(c)

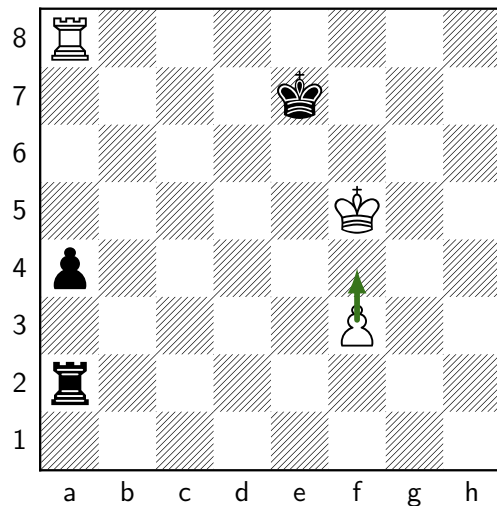


(d)

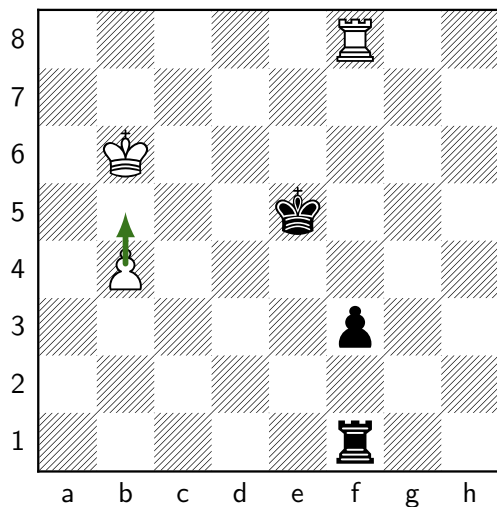
Figure 53: **Move 59, White: Ke2.** (a) Our position; Melanie plays **Ke2 (e1e2)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2066` (ply 114) [structural distance $d=8$, QAP obj=43]: the master played **Kh4**; under μ our *Ke1* plays the role of that game’s *Kh3*; transferred move **e1e2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#3937` (ply 124) [structural distance $d=5$, QAP obj=43]: the master played **Ke5**; under μ our *Ke1* plays the role of that game’s *Kf5*; transferred move **e1e2 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#2403` (ply 105) [structural distance $d=7$, QAP obj=43]: the master played **Kg3**; under μ our *Ke1* plays the role of that game’s *Kf5*; transferred move **e1e2 (endorses this move)**. Synthesis: 35 retrieved analogs transferred to **Ke2** (summed QAP objective 704, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



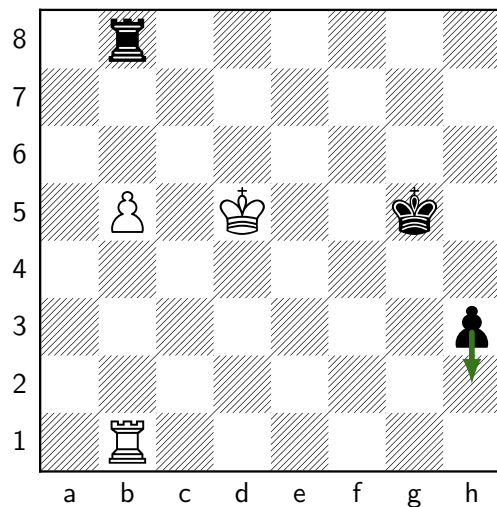
(a)



(b)

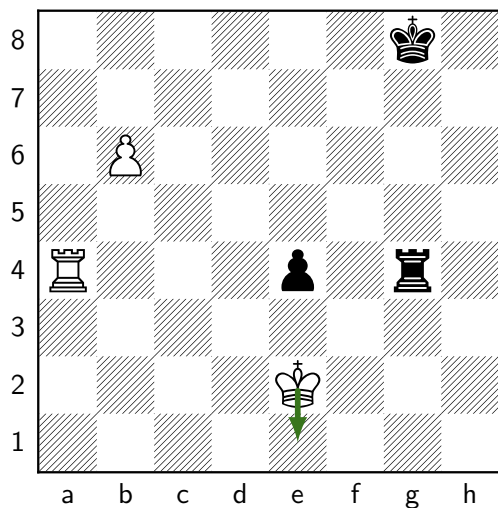


(c)

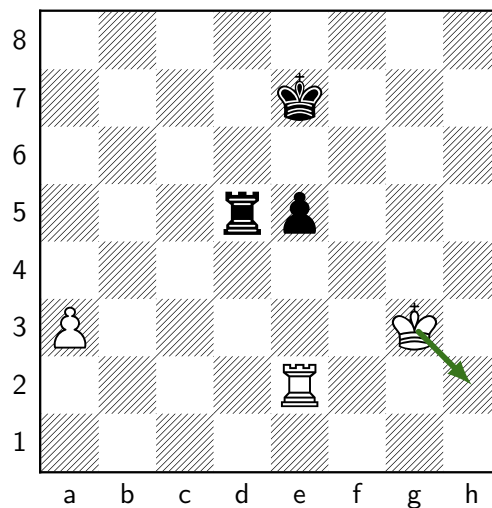


(d)

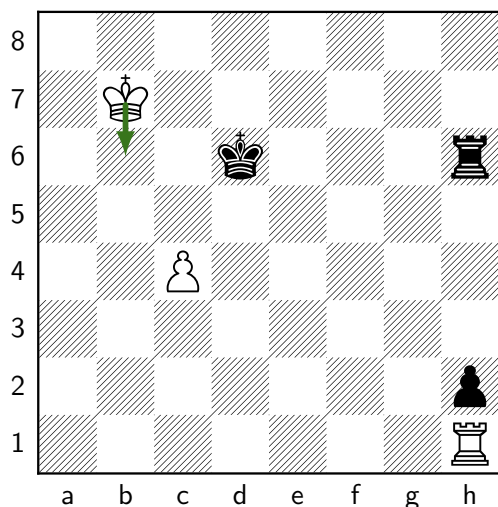
Figure 54: **Move 60, White: b6.** (a) Our position; Melanie plays b6 (b5b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3937` (ply 128) [structural distance $d=2$, QAP obj=44]: the master played f4; under μ our *Pb5* plays the role of that game’s *Pf3*; transferred move b5b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3508` (ply 124) [structural distance $d=3$, QAP obj=36]: the master played b5; under μ our *Pb5* plays the role of that game’s *Pb4*; transferred move b5b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#241` (ply 135) [structural distance $d=2$, QAP obj=33]: the master played h2; under μ our *Pb5* plays the role of that game’s *Ph6*; transferred move b5b6 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to b6 (summed QAP objective 294, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



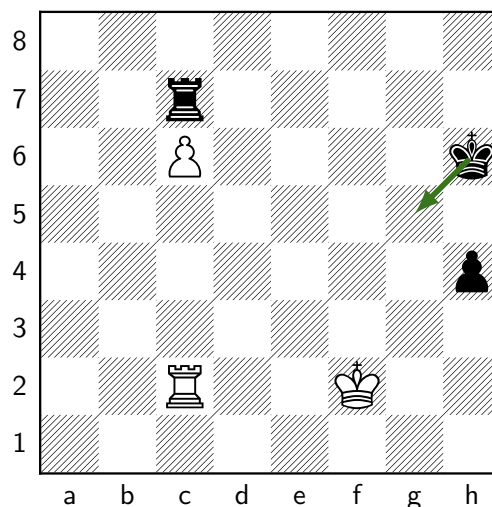
(a)



(b)

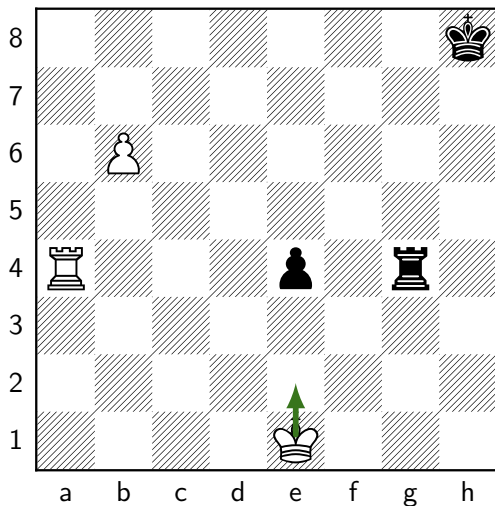


(c)

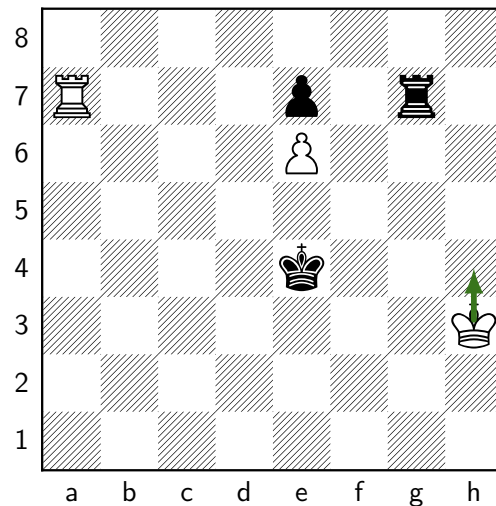


(d)

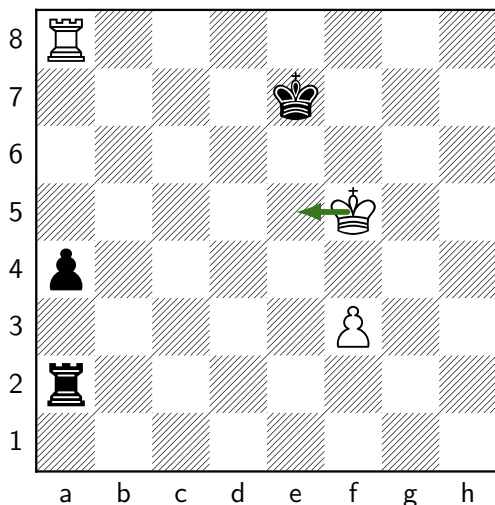
Figure 55: **Move 61, White: Ke1.** (a) Our position; Melanie plays **Ke1** (**e2e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#513` (ply 104) [structural distance $d=5$, QAP obj=32]: the master played **Kh2**; under μ our *Ke2* plays the role of that game's *Kg3*; transferred move **e2e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#136` (ply 108) [structural distance $d=3$, QAP obj=15]: the master played **Kb6**; under μ our *Ke2* plays the role of that game's *Kb7*; transferred move **e2e1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1663` (ply 113) [structural distance $d=2$, QAP obj=14]: the master played **Kg5**; under μ our *Ke2* plays the role of that game's *Kh3*; transferred move **e2e1** (**endorses this move**). Synthesis: 20 retrieved analogs transferred to **Ke1** (summed QAP objective 276, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



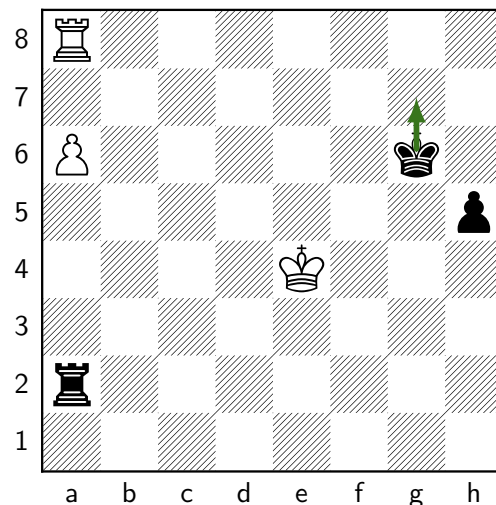
(a)



(b)

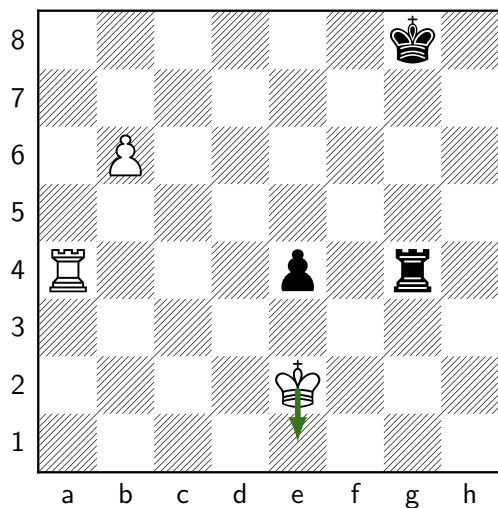


(c)

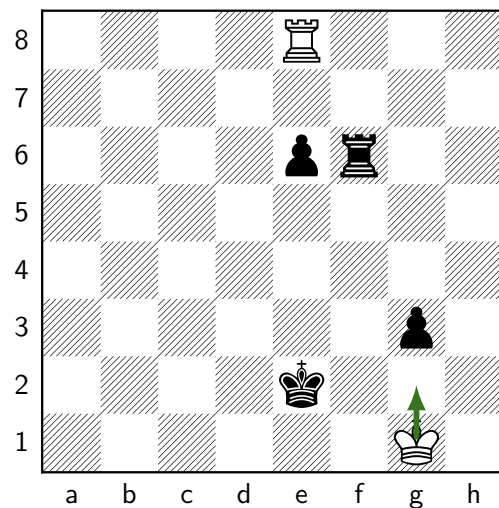


(d)

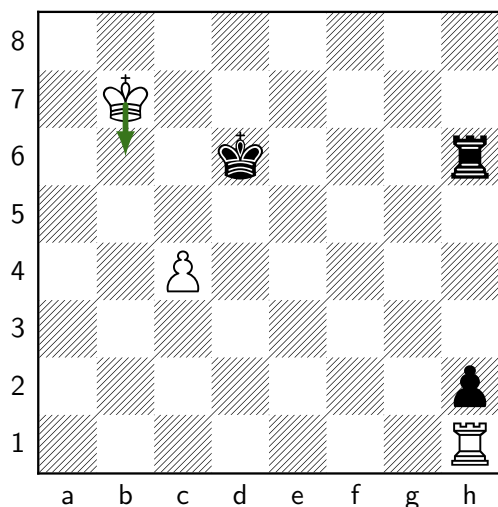
Figure 56: **Move 62, White: Ke2.** (a) Our position; Melanie plays **Ke2 (e1e2)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2066` (ply 114) [structural distance $d=4$, QAP obj=45]: the master played **Kh4**; under μ our *Ke1* plays the role of that game’s *Kh3*; transferred move **e1e2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#3937` (ply 124) [structural distance $d=1$, QAP obj=45]: the master played **Ke5**; under μ our *Ke1* plays the role of that game’s *Kf5*; transferred move **e1e2 (endorses this move)**. (d) Retrieved from `ccr1_top_p0003#1234` (ply 137) [structural distance $d=4$, QAP obj=44]: the master played **Kg7**; under μ our *Ke1* plays the role of that game’s *Kg3*; transferred move **e1e2 (endorses this move)**. Synthesis: 32 retrieved analogs transferred to **Ke2** (summed QAP objective 596, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



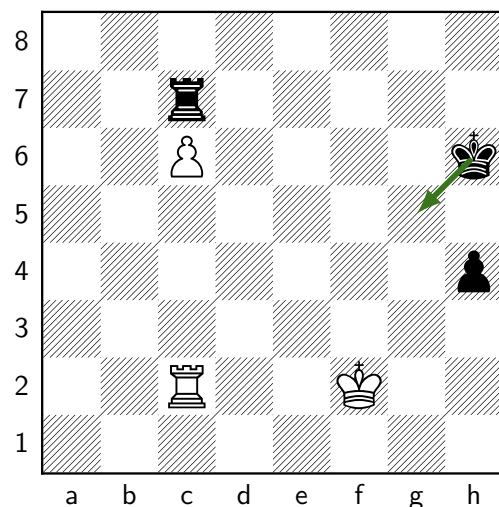
(a)



(b)



(c)



(d)

Figure 57: **Move 63, White: Ke1.** (a) Our position; Melanie plays **Ke1** (**e2e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#587` (ply 130) [structural distance $d=8$, QAP obj=21]: the master played **Kg2**; under μ our *Ke2* plays the role of that game’s *Kg1*; transferred move **e2e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#136` (ply 108) [structural distance $d=3$, QAP obj=15]: the master played **Kb6**; under μ our *Ke2* plays the role of that game’s *Kb7*; transferred move **e2e1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1663` (ply 113) [structural distance $d=2$, QAP obj=14]: the master played **Kg5**; under μ our *Ke2* plays the role of that game’s *Kh3*; transferred move **e2e1** (**endorses this move**). Synthesis: 21 retrieved analogs transferred to **Ke1** (summed QAP objective 274, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.